Fundamental Principles of Modern Portfolio Theory and its Application in R

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Chapter 1: Basic Concepts for R

1. What is R? Why Choose R? What R can do for us in Finance?

- **What is R?**

➢ The base code of R was developed by two academics, Ross Ihaka and Robert Gentleman, resulting in the programming platform we have today. For anyone curious about the name, **the letter R was chosen due to the common first letter of the name of their creators**.

- **Why Choose R?**

In the digital era, information is abundant and cheap. You should be able to disentangle useful and valuable information from the big data. R is a programming language aimed to **solve computational problems involving data analysis**. The reasons we choose R are presented as follows.

➢ **R is a mature, stable platform, continuously supported and intensively used in the industry. If you are a student**, learning R will create more options for your future career.

➢ **A lot of packages can be used to support** the computation and implementation for **the portfolio management**.

➢ The engine of R and the interface of RStudio (Integrated Development Environment for the engine of R) creates a highly productive environment. By combining both, the user can increase productivity in data analysis.

➢ R is **compatible with different operating systems** and it can interface with different programming languages. If you need to use a code in other programming language, such as C++, Python, Julia, it is easy to integrate it with R.

➢ **R is free**! The main software and all its packages are free to use. For most packages, the user's license gives you freedom to use and modify the code freely in your work.

- **What Can You Do with R and RStudio?**

With finance, I highlight the following possibilities in application with R:

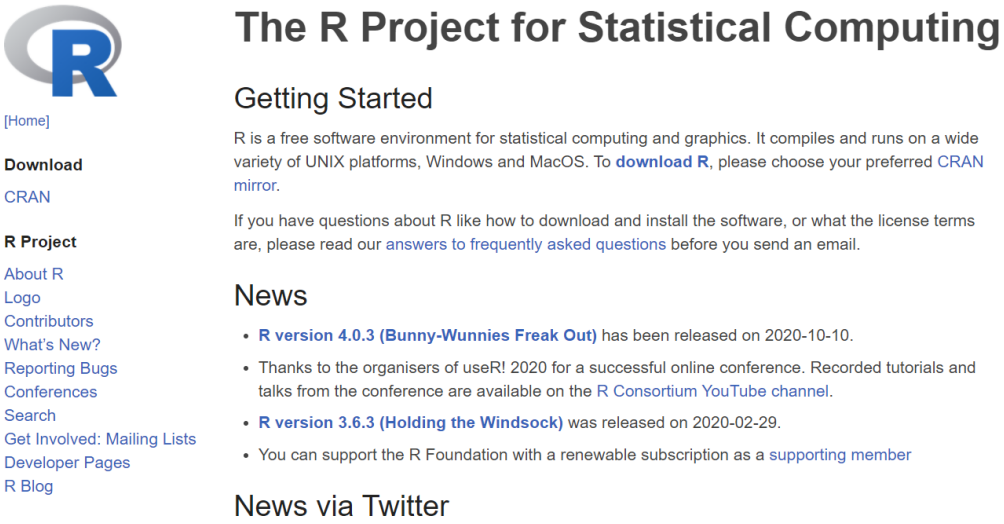
- Import, export, process, and store financial data based on **local files** or **the internet**.
- Develop routines for managing and controlling **investment portfolios** and executing financial orders.
- Implementation of various possibilities of **empirical research** through statistical tools, such as econometric models and hypothesis testing.
- Create dynamic websites with the Shiny package, allowing anyone in the world to use a financial tool created by you. Please see: https://hclee.shinyapps.io/treefore/.

Besides the previously highlighted uses, public access to packages developed by users further expands these capabilities. **The CRAN website offers a Task Views panel for the topic of Finance**. On this page, you can find the main packages available to perform specific operations in Finance. This includes importing financial data from the internet, estimating econometric model, calculation of different risk estimates, among many other possibilities.

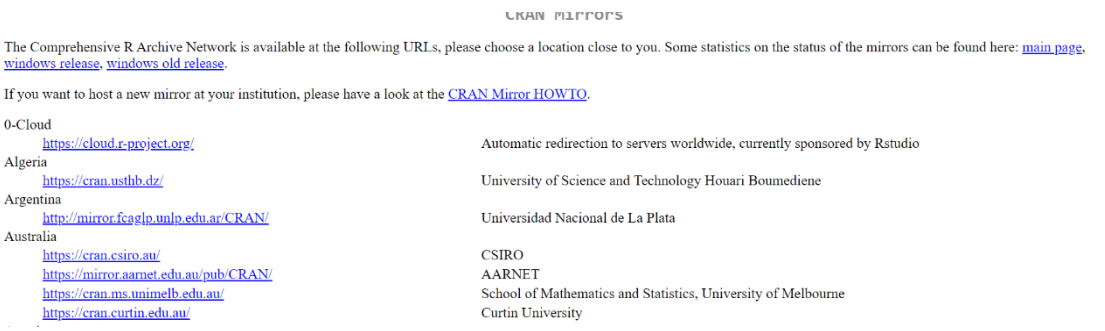
2. Installing R and R Studio

2.1. Install R

- **Go R website: https://www.r-project.org/**

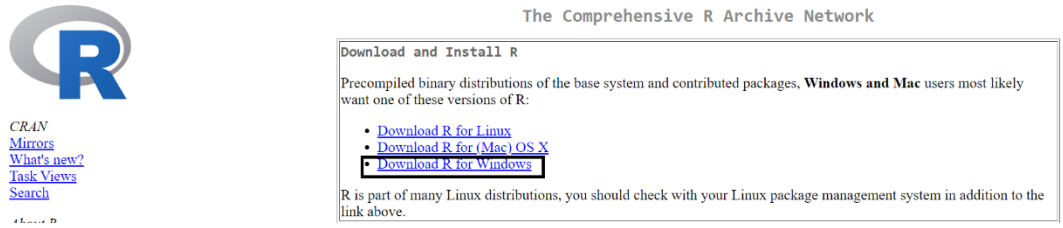


- **Choose CRAN repository (Comprehensive R Archive Network) and enter the website as follows:**

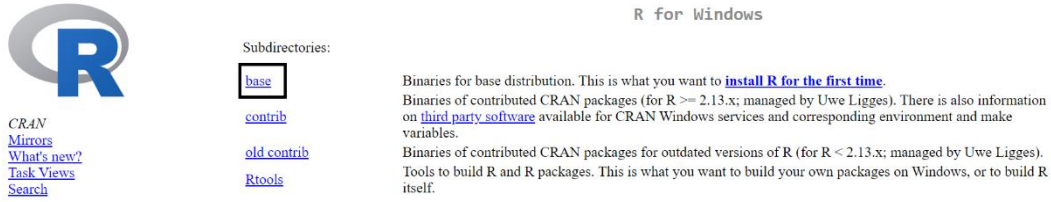


- **The Comprehensive R Archive Network is available at the following URLs, please choose a location close to you.**

- **The next step involves selecting your operating system. This is likely to be Windows. If you use Mac or Linux system, R also provides solutions for users.**



- **Among the download options, the first (base), should be selected. It contains the basic installation of R in Windows.**

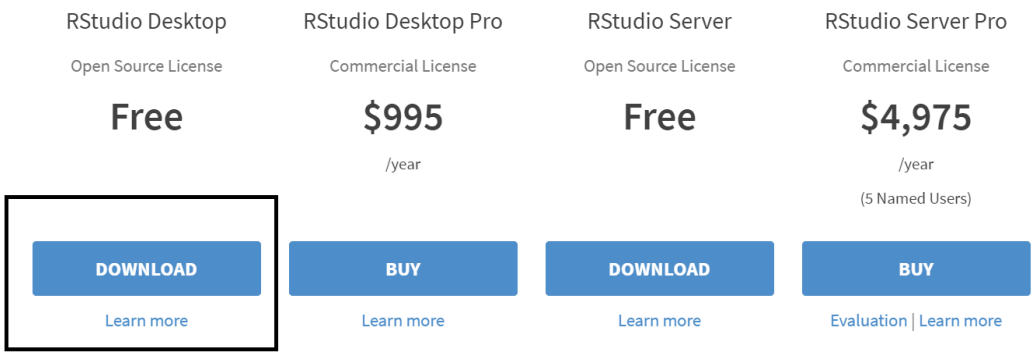


If the user is interested in creating and distributing their own R packages, it is necessary to install RTools. For most users, however, this should not be the case, so I suggest ignoring this program. The links to contrib and old.contrib relate to files for the current and old releases of R packages.

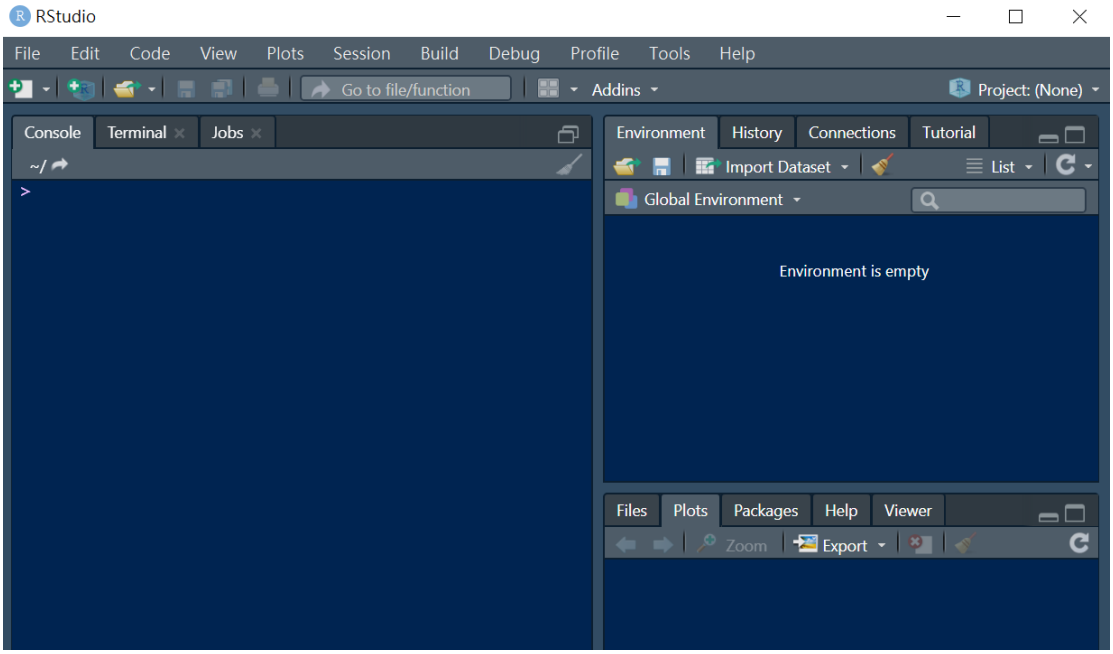
- **After clicking the link base, the next screen will show the link to the download of the R installation file. After downloading the file, open it and follow the steps in the installation screen.**

2.2. Install RStudio

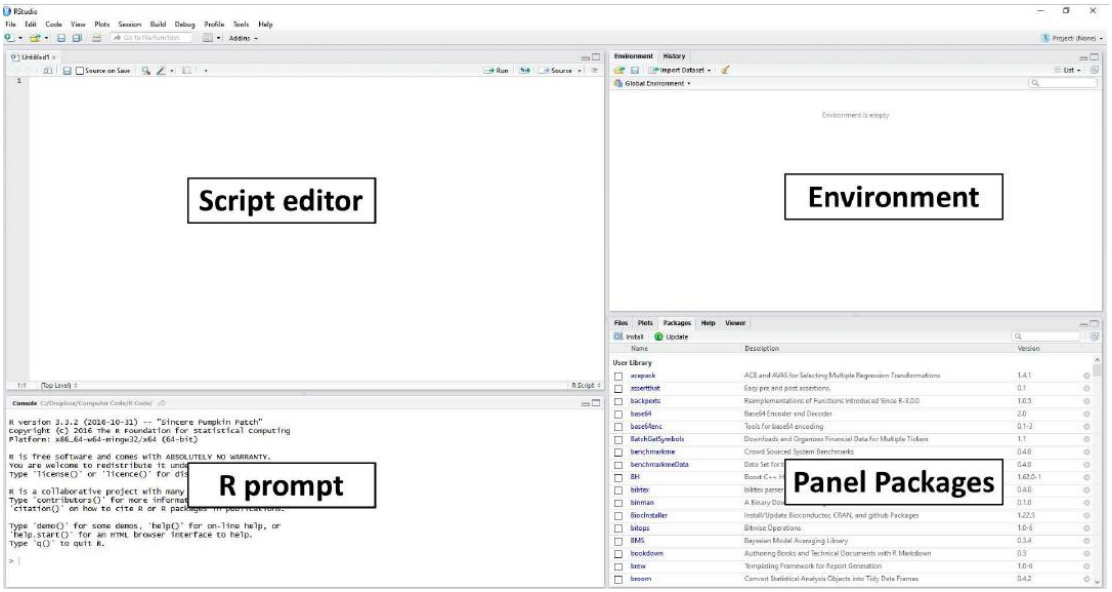
RStudio is an **Integrated Development Environment** (IDE) for R users. The RStudio IDE is a set of integrated tools designed to help you be more productive with R and Python. It includes a console, syntax-highlighting editor that supports direct code execution, and a variety of robust tools for plotting, viewing history, debugging and managing your workspace. The installation of RStudio is simpler than that of R. The files are available in RStudio website: https://rstudio.com/products/rstudio/download/.



- After opening RStudio, the resulting window should look like:



- Open script, the resulting screen should look like the following:



- **Script Editor**: located on the left side and above the screen. This panel is used to write scripts and functions.
- **R prompt**: located on the left side and below the script editor. It displays the prompt of R, which can also be used to give commands to R. The main function of the prompt is to test code and display the results of the commands entered in the script editor.
- **Environment**: located on the top-right of the screen. Shows all objects, including variables and functions currently available to the user. Also note a History panel, which shows the history of the commands previously executed by the user.
- **Panel Packages**: shows the packages installed and loaded by R. Here you have four tabs: Files, to load and view system files; Plots, to view pictures; Help to access the help system and Viewer to display dynamic and interactive results, such as a web page.

3. Background on R

The following operations are basic for R users, particular for beginners. We don't have enough time to introduce all the operations in R and students have to preview and review by themselves. See the PDF file for R beginners as follows:

https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf

3.1. Operations in R

3.1.1. Arithmetic Operators

These operators are used to carry out mathematical operations like addition and multiplication. Here is a list of arithmetic operators available in R.

Operator	Description
+	Addition
–	Subtraction
*	Multiplication
/	Division
^	Exponent
%%	Modulus (Remainder from division)
%/%	Integer Division

```
x <- 5
y <- 16
x+y
x-y
x*y
y/x
y%/%x
y%%x
y^x
```

3.1.2 Relational Operators

Relational operators are used to compare between values. Here is a list of relational operators available in R.

Operator	Description
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
==	Equal to
!=	Not equal to

```
x <- 5
y <- 16
x<y
x>y
x<=5
y>=20
y == 16
x != 5
```

3.1.3 Logical Operators

Logical operators are used to carry out Boolean operations like AND, OR etc.

Operator	Description
!	Logical NOT
&	Element-wise logical AND
&&	Logical AND
|	Element-wise logical OR
||	Logical OR

```
x <- c(TRUE,FALSE,0,6)
y <- c(FALSE,TRUE,FALSE,TRUE)
!x
x&y
x&&y
x|y
x||y
```

3.1.4. Assignment Operators

Operator	Description
<-, <<-, =	Leftwards assignment
->, ->>	Rightwards assignment

The operators <- and = can be used, almost interchangeably, to assign to variable in the same environment.

The <<- operator is used for assigning to variables in the parent environments (more like global assignments). The rightward assignments, although available are rarely used.

```
x <- 5
x

x = 9
x

10 -> x
x
```

3.2. Data Structure

The vector, matrix, dataframe, and list are the basic data structure in R.

Again, we don't have enough time to introduce all the operations in R and students have to preview and review by themselves. See the PDF file for R beginners as follows: https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf

3.2.1. Operation on vectors

The above-mentioned operators work on vectors. The variables used above were in fact single element vectors. We can use the function c() (as in *concatenate*) to make vectors in R. All operations are carried out in element-wise fashion. Here is an example.

```
x <- c(2,8,3)
y <- c(6,4,1)
x+y
x>y
x*y
x/y
length(x)
z=seq(1,10,1)
z=seq(1,10,2)
z=seq(10,1,-1)
rep(5,30)
```

3.2.2 Operation on Matrix

```
# R program to add two matrices by element-wise

# Creating 1st Matrix

B = matrix(c(1, 2, 3, 4, 5, 6), nrow = 2, ncol = 3)

# Creating 2nd Matrix

C = matrix(c(7, 8, 9, 10, 11, 12), nrow = 2, ncol = 3)

# Getting number of rows and columns

num_of_rows = nrow(B)

num_of_cols = ncol(B)

# Creating matrix to store results

sum = matrix(0, nrow = num_of_rows, ncol = num_of_cols)

# Printing Original matrices

print(B)

print(C)

print(B+C)
```

```
# Matrices Subtraction by element-wise
# Creating 1st Matrix
B = matrix(c(1, 2, 3, 4, 5, 6), nrow = 2, ncol = 3)
# Creating 2nd Matrix
C = matrix(c(7, 8, 9, 10, 11, 12), nrow = 2, ncol = 3)
# Getting number of rows and columns
num_of_rows = nrow(B)
num_of_cols = ncol(B)
# Creating matrix to store results
diff = matrix(, nrow = num_of_rows, ncol = num_of_cols)
# Printing Original matrices
print(B)
print(C)

# Calculating diff of matrices
for(row in 1:num_of_rows)
{
	for(col in 1:num_of_cols)
	{
		diff[row, col] <- B[row, col] - C[row, col]
	}
}

# Printing resultant matrix
print(diff)
print(B-C)
```

```
# R program to multiply two matrices by element-wise

# Creating 1st Matrix
B = matrix(c(1, 2, 3, 4, 5, 6), nrow = 2, ncol = 3)

# Creating 2nd Matrix
C = matrix(c(7, 8, 9, 10, 11, 12), nrow = 2, ncol = 3)

# Getting number of rows and columns
num_of_rows = nrow(B)
num_of_cols = ncol(B)
# Creating matrix to store results
prod = matrix(, nrow = num_of_rows, ncol = num_of_cols)

# Printing Original matrices
print(B)
print(C)

# Calculating product of matrices
for(row in 1:num_of_rows)
{
	for(col in 1:num_of_cols)
	{
		prod[row, col] <- B[row, col] * C[row, col]
	}
}

# Printing resultant matrix
print(prod)
print(B*C)
# %*% is matrix multiplication.
# For matrix multiplication, you need an m x n matrix times an n x p matrix
```

```
# R program to divide two matrices by element-wise

# Creating 1st Matrix
B = matrix(c(1, 2, 3, 4, 5, 6), nrow = 2, ncol = 3)

# Creating 2nd Matrix
C = matrix(c(7, 8, 9, 10, 11, 12), nrow = 2, ncol = 3)

# Getting number of rows and columns
num_of_rows = nrow(B)
num_of_cols = ncol(B)

# Creating matrix to store results
div = matrix(, nrow = num_of_rows, ncol = num_of_cols)

# Printing Original matrices
print(B)
print(C)

# Calculating product of matrices
for(row in 1:num_of_rows)
{
   for(col in 1:num_of_cols)
   {
      div[row, col] <- B[row, col] / C[row, col]
   }
}

# Printing resultant matrix
print(div)
print(B/C)
```

- **rbind** - Returns the merger (bind) of matrices, with row orientation.
- **cbind** - Returns the merger (bind) of matrices, with column orientation.
- **%*%** - matrix multiplication. For matrix multiplication, you need an m x n matrix times an n x p matrix.
- **t** - Returns a transposed matrix.

3.2.3. Data Frame

In short, dataframe is a table with rows and columns. Its big difference from the matrix type is it allows for each column to have a different class.

Another positive aspect of using dataframes in R is that several functions expect a dataframe as input. For example, the data manipulation package “dplyr” and the graphical package “ggplot2” work from a dataframe only.

```
# set ticker symbols as a vector
# The use of double quotes (" ") or single quotes (' ') defines objects of the class character.
ticker <- c(rep('AAP',5), rep('COG', 5), rep('BLK', 5), rep('CAM',5))

# set a date vector
date <- as.Date(rep(c("2010-01-04", "2010-01-05", "2010-01-06",
                      "2010-01-07", "2010-01-08"), 4) )

# set prices
prices <- c(40.38,   40.14,   40.49,   40.48,   40.64,
            46.23,   46.17,   45.97,   45.56,   45.46,
            238.58, 239.61, 234.67, 237.25, 238.92,
            43.43,   43.96,   44.26,   44.5,     44.86)

# create dataframe
my.df <- data.frame(ticker = ticker,
                    date = date,
                    prices = prices)

# print result
print(my.df)

# get names of columns with names
names(my.df)

# get names of columns with colnames
colnames(my.df)
```

```
# get column ticker from my.df
my.ticker <- my.df$ticker

# get column price from my.df
my.prices <- my.df['prices']

# get second column from my.df
my.date <- my.df[ ,2]

# print the results
print(my.ticker)
print(my.prices)
print(my.date)
```

3.2.4. List

A list is a flexible object that can *hold different types of objects*. In the same list object, we can group numeric objects, character objects, factor with Dates, and even lists within lists. Each element of a list need not have the same length or dimension as the others.

- A list can be created with the list command:

```
# create a list with three elements
my.l <- list(1, c(1,2,3), c('a', 'b'))
# print result
print(my.l)
```

- Notice how a list type object is printed differently than atomic vectors. The elements of the list are separated vertically and the content appears within **double brackets ([[ ]])**.

- The elements of a list can also be named.

```
# create named list
my.named.l <- list(ticker = 'ABC',
                   name.company = 'Company ABC',
                   price = c(1,1.5,2,2.3),
                   market = 'NYSE',
                   date.price = as.Date('2016-01-01')+0:3)
# print list
print(my.named.l)
```

- As mentioned, the individual elements of a list can be accessed with double brackets ([[ ]]), as in:

```
# set list
my.l <- list(2, 1:5, c('a', 'b'))
# print second element of my.l
print(my.l[[2]])
```

4. Install Package: Example for Data Collection using Package "quantmod" and "xts"

- One of the greatest benefits of using R is its package collection. A package is nothing aimed at solving a particular computational problem. R has a collaborative philosophy. Users provide their codes for others to use. And, most importantly, all packages are free.
- R packages can be accessed and installed from different sources. The main being **CRAN** (The Comprehensive R Archive network), **R-Forge** and **Github**. In practice, it is very common for developers to **maintain a development version on Github or R-Forge** and the official version in CRAN. When the development version reaches a certain stage of maturity, it is then sent to CRAN.
- CRAN is the official repository of R and it is built by the community. Anyone can send a package. However, there is an evaluation process to ensure that the certain strict rules about code format are respected.
- An important source for finding packages is CRAN Task Views. There you can find the most important packages for a given area of expertise. The website is https://cran.r-project.org/web/views/.
- To find out the current amount of packages on CRAN, type and execute the following commands in the prompt:

```
# get matrix with available packages
df.cran.pkgs <- available.packages()
# find the number of packages
n.cran.packages <- nrow(df.cran.pkgs)
# print it
print(n.cran.packages)
```

- For the convivence without API permission to import data for the use in the class, the package "quantmod" is employed and the syntax for installation is:

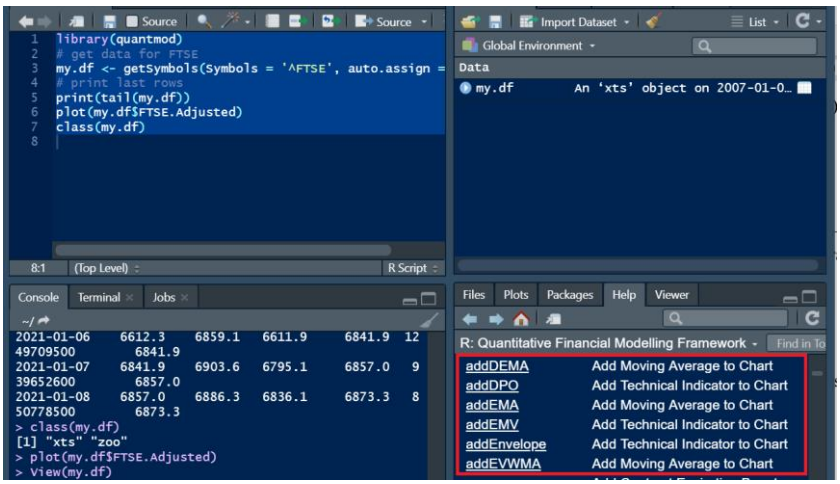
```
install.packages("quantmod")
```

- After executing this simple command, package quantmod and all of its dependencies will be installed and the functions related to the package will be ready for use once the package is loaded in a script. In other words, if the installed package is dependent on another package, R detects this dependency and automatically installs the missing packages.
- Note that we defined the package name in the installation as if it were text with the use of quotation marks (" ").
- The syntax for importing data from the website, e.g., Yahoo Finance, by the use of quantmod is presented as follows:

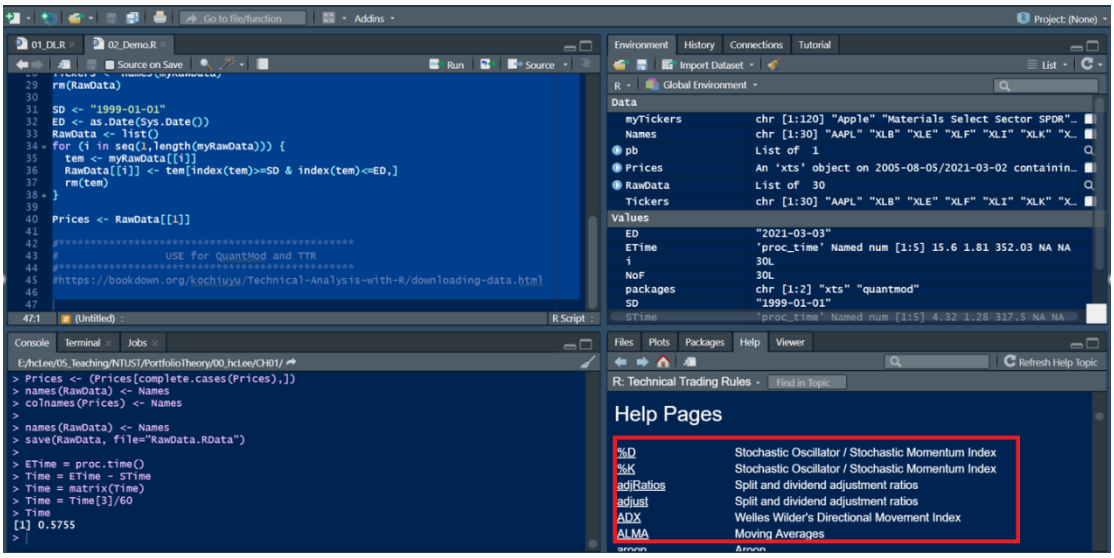
```
packages=c("quantmod")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}

# get data for FTSE
my.df <- getSymbols(Symbols = '^FTSE', auto.assign = FALSE)
# print last rows
print(tail(my.df))
plot(my.df$FTSE.Adjusted)
class(my.df)
```

- A "quantmod" package is also useful for developing trading strategy by using technical indicators in this package.



- For the convenience for developing trading strategy, R users can employ "quantmod" and "TTR" packages to calculate technical indicators and figures.



- Notice that the class for the data importing by using "quantmod" is "xts". A package "xts" is common in providing researchers to work with numeric data indexed by time. We can store this dataset in a matrix format, so each line represents dates and each column represents a variable. With this format, time operations, such as period aggregations, are easier to perform. This is the main idea of package xts (Ryan and Ulrich 2014). The great benefit of this alternative dataframe is that several functions for time aggregation and manipulation are available.

```
# load pkg
packages=c("xts")
for(i in packages){
    if(!require(i,character.only = TRUE)) install.packages(i)
    library(i,character.only = TRUE)
}
# set ticker symbols as a vector
ticker <- c('AAP', 'COG', 'BLK', 'CAM')
# set a date vector
date <- as.Date(c("2010-01-04", "2010-01-05", "2010-01-06",
"2010-01-07", "2010-01-08"))
```

```
# set prices as matrix
price.mat <- matrix(c(40.38, 40.13, 40.49, 40.48, 40.63,
46.23, 46.16, 45.97, 45.56, 45.45,
238.58, 239.61, 234.66, 237.25, 238.91,
43.43, 43.95, 44.25, 44.5, 44.86),
nrow = length(date))
# set xts object
my.xts <- xts(price.mat, order.by = date)
# set colnames
colnames(my.xts) <- ticker
# print it
print(my.xts)
# show its class
class(my.xts)
```

- In creating the xts object, notice how the time index is explicitly defined using argument "order.by". This is a necessary step in creating every xts object.

Chapter 2: Basic Concepts for Portfolio Selection and Capital Market Line

1. Reviewing Market History with Prices and Returns

To review the history of price and returns, the data for ETFs are imported from "Yahoo" by using package "quantmod". In the document "CH02", the R script named as "01_DL" is used to download data from "Yahoo" and the data is saved and named as "RawSTK.RData".

- Explain the syntax "01_DL"
- Code for the price figure by using the data collecting from script "01_DL" is presented as follows:

```
Sys.setlocale("LC_TIME","english")
packages=c("xts")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
load("RawSTK.RData")
Tickers <- names(RawSTK)
CP <- NULL
for (i in seq(1,length(RawSTK))) {
   CP <- cbind(CP,RawSTK[[i]][,6])
}
names(CP) <- Tickers
CP <- na.locf(CP)

# Available only on Windows. A graphics device is opened.
# For windows, win.graph, x11 and X11 this is a window on the current Windows
display.
windows()
plot(CP[,c(3,6)],
        main="",
        col=c("black", "blue"),
        grid.col = NA)
addLegend("topleft",
                legend.names = names(CP[,c(3,6)]),
```

```
                lty=c(1, 1), lwd=c(2, 1),
                col=c("black", "blue"))
```

- Asset return can be calculated as the log difference as follows:

$$R_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1})$$

- Asset return also can be calculated as the discrete approach

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

- Asset returns can also be compounded, which tell how much return over time one would get by keeping the investment. Compounding asset returns are given by this formula:

$$cumR_{i,t} = \prod (1 + R_{i,t})$$

- The code for the compounding asset returns can be shown as follows:

```
Sys.setlocale("LC_TIME","english")
packages=c("xts")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
load("RawSTK.RData")
Tickers <- names(RawSTK)
CP <- NULL
for (i in seq(1,length(RawSTK))) {
   CP <- cbind(CP,RawSTK[[i]][,6])
}
names(CP) <- Tickers
CP <- na.locf(CP)
RET <- diff.xts(log(CP),1,na.pad=T)

windows()
plot(RET[,c(3,6)],
        main="",
        col=c("black", "blue"),
        grid.col = NA)
addLegend("topleft",
```

```
                legend.names = names(cumRET[,c(3,6)]),
                lty=c(1, 1), lwd=c(2, 1),
                col=c("black", "blue"))

cumRET <- NULL
for(i in seq(1,ncol(RET))) {
   tem <- cumprod(1+RET[!is.na(RET[,i]),i])
   date <- as.Date(index(tem))
   tem <- c(1,tem)
   date <- c(date[1]-1,date)
   tem <- xts(tem,order.by = as.Date(date))
   cumRET <- cbind(cumRET,tem)
   rm(tem)
}
colnames(cumRET) <- colnames(RET)

windows()
plot(cumRET[,c(3,6)],
        main="",
        col=c("black", "blue"),
        grid.col = NA)
addLegend("topleft",
                legend.names = names(cumRET[,c(3,6)]),
                lty=c(1, 1), lwd=c(2, 1),
                col=c("black", "blue"))
```

- Annualized asset returns can be calculated as:

$$annR_{i,t} = R_{i,t} \times nFreq$$

If R_t is the daily data (or daily frequency), $nFreq = 256\ (or\ 365)$.

If R_t is the weekly data (or weekly frequency), $nFreq = 52$.

If R_t is the monthly data (or monthly frequency), $nFreq = 12$.

If R_t is the quarterly data (or quarterly frequency), $nFreq = 4$.

2. Risk Measures

2.1. Volatility, Covariance, and Correlation

- The uncertainty of returns is so-called the risks for assets. Usually, the standard deviation is used to measure the risk for the asset.

$$Vol_i = \sqrt{\frac{\sum(R_{i,t} - \bar{R}_i)^2}{n-1}}$$

- The co-movement for asset returns is measured by the covariance and correlation.

$$Cov_{i,j} = \frac{\sum(R_{i,t} - \bar{R}_i)\,(R_{j,t} - \bar{R}_j)}{n-1} \quad and \quad Cor_{i,j} = \frac{Cov_{i,j}}{Vol_i Vol_j}$$

- There are three risky assets, A, B, and C. The time periods for A, B, and C are 2002-2020, 2006-2020, 2019-2020. The question here is which period will you use to calculate returns, volatility, and covariance.

- Annualized asset return volatility can be calculated as:

$$annVol_i = Vol_i \times \sqrt{nFreq}$$

If Vol_i is computed by the daily data, $nFreq = 256\ (or\ 365)$.

If Vol_i is computed by the weekly data, $nFreq = 52$.

If Vol_i is computed by the monthly data, $nFreq = 12$.

If Vol_i is computed by the quarterly data, $nFreq = 4$.

- The code for calculating return, volatility, covariance, and correlation are displayed as follows:

```
Sys.setlocale("LC_TIME","english")
packages=c("xts")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
nFreq <- 256
load("RawSTK.RData")
Tickers <- names(RawSTK)
```

```
CP <- NULL
for (i in seq(1,length(RawSTK))) {
   CP <- cbind(CP,RawSTK[[i]][,6])
}
names(CP) <- Tickers
CP <- na.locf(CP)

RET <- diff.xts(log(CP),1,na.pad=T)
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq
corr <- cor(RET,use = "pairwise.complete.obs")

stdev
sqrt(diag(covmat))
```

- The outliers or extreme values for asset returns can affect the average asset returns and standard deviation. The effects of the outliers or extreme values on computing asset returns and standard deviation can be removed by using the syntax, Return.clean and StdDev, from package "PerformanceAnalytics". The following code helps us understanding the use of Return.clean and StdDev:

```
packages=c("PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}

?Return.clean
?StdDev
```

2.2.Sharpe Ratio

In order to compare the performance among assets or portfolios, we need to adjust the returns for risks. Sharpe (1966) develops a Sharpe ratio for this purpose. The Sharpe ratio is defined as follow:

$$Sharpe\ Ratio = \frac{R_i - R_f}{\sigma_i}$$

where R_i and σ_i are the returns and volatility for asset i. R_f is the risk-free rate.

- Higher Sharpe ratio reflects higher risk-adjusted returns and so a higher Sharpe ratio is reflective of better (superior) results.
- The Sharpe ratio is measured the reward-to-volatility with considering both up- and down-risk.
- Function for "SharpeRatio.annualized" from the package "PerformanceAnalytics" is employed to calculate annualized Sharpe ratio. The following code helps us understanding the use of SharpeRatio.annualized:

```
packages=c("PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
? SharpeRatio.annualized
```

- Example for calculating the annualized Sharpe ratio is presented as follows:

```
Sys.setlocale("LC_TIME","english")
packages=c("xts","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
rf <- 0.03
nFreq <- 256
load("RawSTK.RData")
Tickers <- names(RawSTK)
CP <- NULL
```

```
for (i in seq(1,length(RawSTK))) {
   CP <- cbind(CP,RawSTK[[i]][,6])
}
names(CP) <- Tickers
CP <- na.locf(CP)

RET <- diff.xts(log(CP),1,na.pad=T)
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq
corr <- cor(RET,use = "pairwise.complete.obs")

SharpeRatio.annualized(RET,
                       rf/nFreq,scale=nFreq,
                       geometric=T)

SharpeRatio.annualized(RET,
                       rf/nFreq,scale=nFreq,
                       geometric=F)
(avgRET-(rf))/stdev
```

2.3.Sortino Ratio

In practice, e.g., a real world of the trading or investment, traders and investors only care about the loss of money. As such, the Sortino ratio is proposed, and it only focuses on the downside deviation of returns for measuring risks.

$$Sortino\ Ratio = \frac{R_i - MAR}{\sigma_D}$$

where R_i and σ_i are the returns and downside volatility for asset i. MAR is the minimum acceptable return (threshold).

- A higher Sortino ratio indicates superior risk-adjusted results.
- Example for calculating the Sortino ratio is presented as follows:

```
Sys.setlocale("LC_TIME","english")
packages=c("xts","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
rf <- 0.03 ; nFreq <- 256
load("RawSTK.RData")
Tickers <- names(RawSTK)
CP <- NULL
for (i in seq(1,length(RawSTK))) {
   CP <- cbind(CP,RawSTK[[i]][,6])
}
names(CP) <- Tickers
CP <- na.locf(CP)
RET <- diff.xts(log(CP),1,na.pad=T)
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq
corr <- cor(RET,use = "pairwise.complete.obs")
SortinoRatio(RET[,-ncol(RET)])
SortinoRatio(RET[,-ncol(RET)],
                    MAR=colMeans(RET[,ncol(RET)] ,na.rm = T),
                    threshold=c("MAR"))
```

3. Asset Allocation Theory

- The portfolio or asset allocation problem is to answer the problem: **How much of your wealth should you invest in each security?**
- Mean variance portfolio analysis developed by Harry Markowitz (who won the 1990 Nobel Prize in Economics) in the 1952 is used to answer this question. His paper entitled as "Portfolio Selection" was published in 1952 in the *Journal of Finance*. Since then, the finance theory enters into a new era (chapter) "Modern Portfolio Theory (MPT)."
- In its simplest form, mean-variance analysis provides a framework to construct and select portfolios, based on the **expected performance of the investments** and the **risk appetite of the investor**.
- The basic concept for the portfolio selection is based the benefit of **diversification**. Conventional wisdom has always dictated "not putting all your eggs into one basket."
- Markowitz quantified the concept of diversification through the statistical notion of covariance between individual securities.
  - Suppose N $(N = 3)$ asset returns have expected returns and covariance as follows:

$$\boldsymbol{R}_{3\times 1} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} \text{ and } \boldsymbol{\Sigma}_{3\times 3} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 \end{bmatrix}$$

  - The weights for N $(N = 3)$ is:

$$\mathbf{W}_{3\times 1} = \begin{bmatrix} W_1 \\ W_2 \\ W_3 \end{bmatrix}$$

- The portfolio return and variance is calculated as follows:

$$R_P = \mathbf{W}'\boldsymbol{R} \text{ and } \boldsymbol{\sigma}_P^2 = \mathbf{W}'\boldsymbol{\Sigma}\mathbf{W}$$

- The global minimum variance portfolio (GMV) can be computed directly by solving the optimization problem:

$$\min_{W} \mathbf{W}'\boldsymbol{\Sigma}\mathbf{W}$$

$$\text{s.t. } \mathbf{W}'\boldsymbol{\iota} = \mathbf{1}, \ \boldsymbol{\iota}' = [\mathbf{1}, \mathbf{1}, \mathbf{1}] \ (\boldsymbol{\iota}, \text{Iota})$$

- This problem can also be solved by the method of Lagrange multipliers, which has the solution:

$$\mathbf{W} = \frac{\mathbf{1}}{\boldsymbol{\iota}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\iota}}\boldsymbol{\Sigma}^{-1}\boldsymbol{\iota}$$

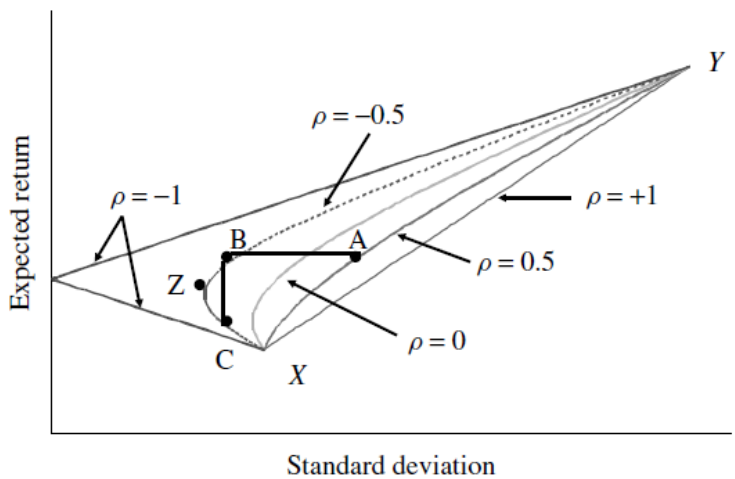
- Furthermore, following Markowitz, the investor's problem for the portfolio selection with a given expected return ($\hat{R}_P$) is a constrained minimization problem in the sense that the investor must seek:

$$\min_{W} \mathbf{W}'\boldsymbol{\Sigma}\mathbf{W}$$

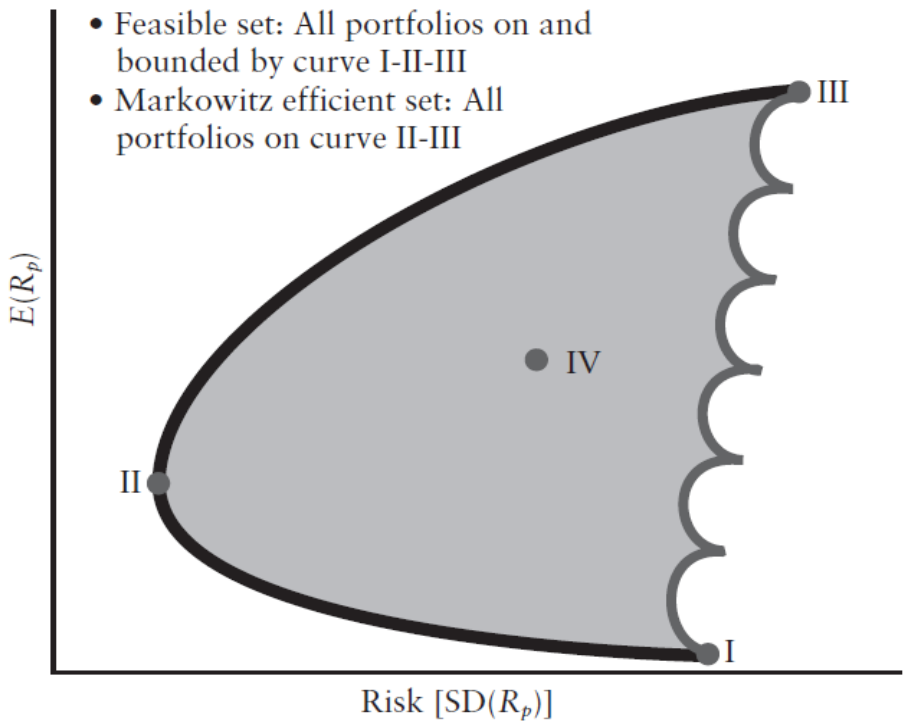
$$\text{s.t. } \mathbf{W}'\boldsymbol{R} = \hat{R}_P$$

$$\text{s.t. } \mathbf{W}'\boldsymbol{\iota} = \mathbf{1}, \ \boldsymbol{\iota}' = [\mathbf{1}, \mathbf{1}, \mathbf{1}] \ (\boldsymbol{\iota}, \text{Iota})$$

- Theoretically, the solutions can be shown as the following figure under the different asset correlations:

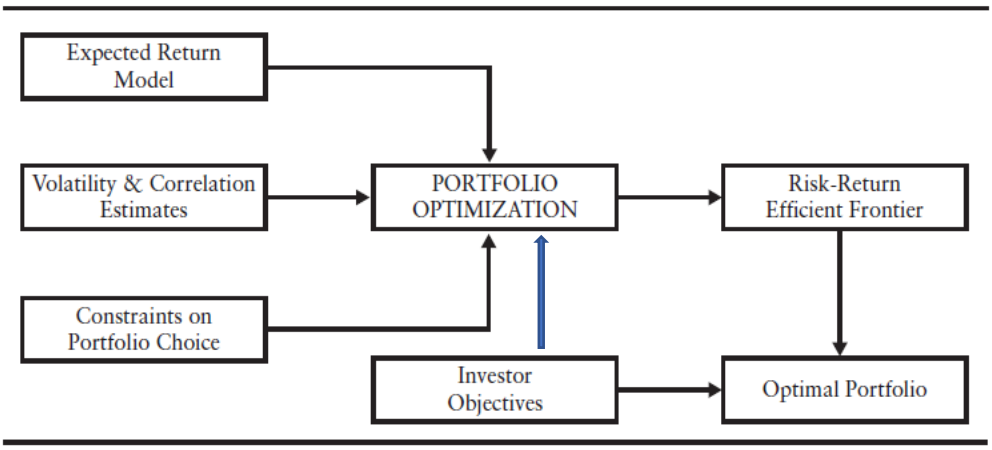


- In practice, an optimization software package is then used to solve a series of optimization problems in order to generate the efficient frontier.
- The software R and package "quadprog" will be used to solve the optimization problem. The solution of optimization problem can be shown as following figures since the correlations among assets are from -1 to 1:



- The set of all possible portfolios that can be constructed is called the feasible set. Notice that the feasible set is bounded by the curve I-II-III
- II is so-called the global minimum variance (GMV) portfolio.
- All portfolios on the curve II-III are efficient portfolios for different levels of risk. II-III is called the efficient frontier.

- The following figure shows a schematic view of the investment process as seen from the perspective of modern portfolio theory. This process is often also referred to as mean-variance optimization or theory of portfolio selection. The inputs to the process are estimates of the expected returns, volatilities, and correlations of all the assets together with various portfolio constraints.



➢ The effect of diversification and decrease portfolio risk (and keeping the expected portfolio return constant) by increasing the asset universe.

$$\boldsymbol{\sigma_P^2} = \mathbf{W'\Sigma W}$$

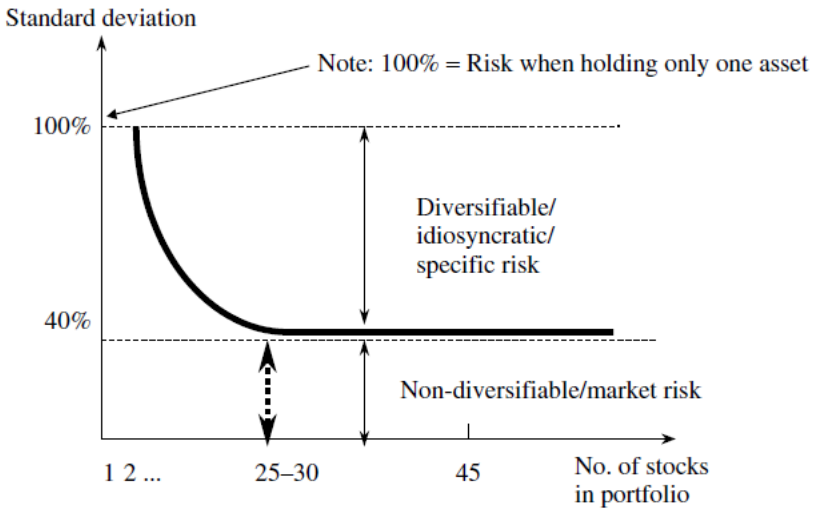
$$= \frac{\mathbf{1}}{N^2}\sum_{i=1}^{n}\boldsymbol{\sigma_i^2} + \frac{\mathbf{1}}{N^2}\sum_{i\neq j}^{N}\boldsymbol{\sigma_i\,\sigma_j}$$

$$= \frac{\mathbf{1}}{N^2}N\bar{\boldsymbol{\sigma}} + \frac{\mathbf{1}}{N^2}N(N-1)\overline{COV}_{i\neq j}$$

$$= \frac{\bar{\boldsymbol{\sigma}}}{N} + \frac{(N-1)}{N}\overline{COV}_{i\neq j}$$

$$where\ \ \overline{COV}_{i\neq j} = \frac{\mathbf{1}}{N(N-1)}\sum_{i\neq j}^{N}\boldsymbol{\sigma_i\,\sigma_j}$$

- This implies that the portfolio variance approaches average covariance as the number of assets becomes large. Therefore, we see that, in general, the benefits of diversification are limited up to a point and that we cannot expect to be able to completely eliminate portfolio risk. The following figure show this finding:



4. Capital Market Line

As demonstrated by previous research, the performance of the efficient set of portfolios with including a risk-free asset is better than those of without considering a risk-free asset. We present this formulation in this section.

- Assume that there is a risk-free asset, with a risk-free return denoted by R_f and that the investor is able to borrow and lend at this rate.
- The investor has to choose a combination of the N risky assets plus the risk-free asset. The weights $W_R' = (W_{R1} \cdots W_{R1})$ do not have to sum to 1 as the remaining part $(1 - W_R'\iota)$ can be invested in the risk-free asset. Note also that this portion of the investment can be positive or negative if we allow risk-free borrowing and lending.
- In this case, the portfolio's expected return and variance are:

$$u_P = W_R'\boldsymbol{R} + (1 - W_R'\iota)\, R_f$$

$$\sigma_P^2 = W_R'\Sigma W_R$$

- Because the risk-free asset has zero variance and is uncorrelated with the risky assets, than portfolio variance is no associated with a risk-free rate and is shown as $\sigma_P^2 = W_R'\Sigma W_R$
- The investor's objective is for a targeted level of expected portfolio return, u_0, to choose allocations by solving a quadratic optimization problem:

$$\min_{W_R} W_R'\Sigma W_R$$

$$\text{s.t. } W_R'\boldsymbol{R} + (1 - W_R'\iota)R_f = u_0$$

- Again, this problem can also be solved by the method of Lagrange multipliers, which has the solution:

$$W_R = C\Sigma^{-1}(\boldsymbol{R} - R_f\iota),$$

where

$$C = \frac{u_0 - R_f}{(\boldsymbol{R} - R_f\iota)'\Sigma(\boldsymbol{R} - R_f\iota)}$$

- The above formula shows that the weights of the risky assets of any minimum variance portfolio are proportional to the vector $\Sigma^{-1}(\boldsymbol{R} - R_f\iota)$, with the proportionality constant C, defined above.
- Therefore, with a risk-free asset, all minimum variance portfolios are a combination of the risk-free asset and a given risky portfolio. The line for the combination of the risk-free asset and a given risky portfolio is the CML and this risky portfolio is called the *tangency portfolio*.
- Fama (1970) demonstrated that **under certain assumptions** the tangency portfolio must consist of all assets available to investors, and each asset must be held in proportion to its market value relative to the total market value of all assets. Therefore, the tangency portfolio is often referred to as the "**market portfolio**," or simply the "market."[1]
- Notice that it is also to verify that the market portfolio can be calculated directly from the maximal Sharpe ratio optimization problem:

$$\max_{W} \frac{W'R - R_f}{\sqrt{W'\Sigma W}}$$

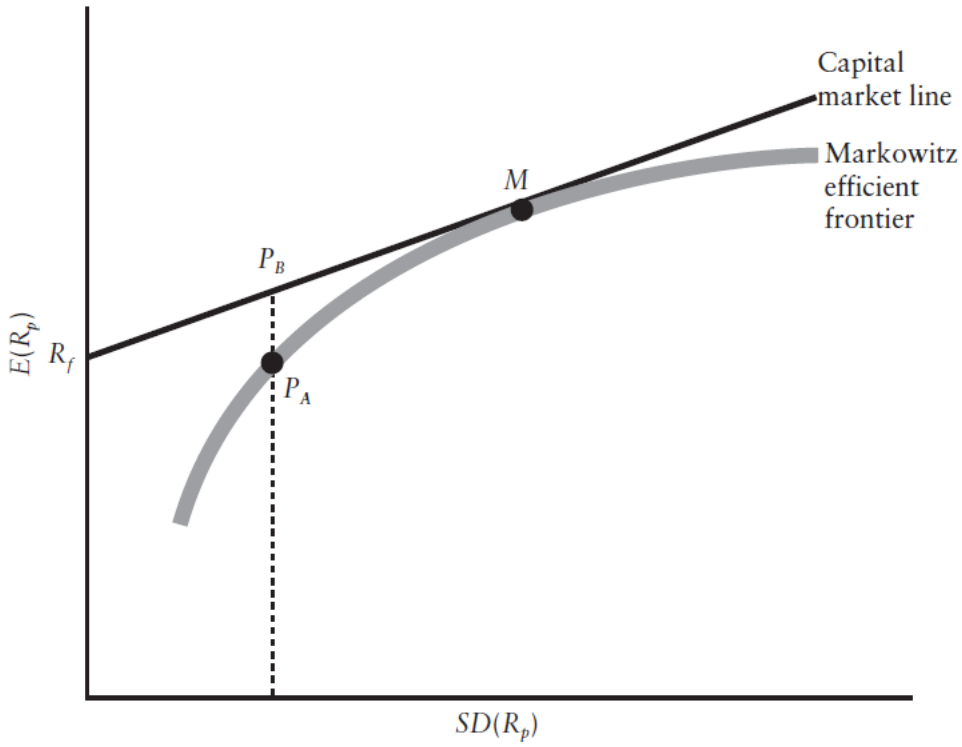
$$\text{s.t. } W'\iota = 1$$

- As such, the tangency portfolio = market portfolio = maximal Sharpe ratio portfolio. We don't spend time on proving it by math but instead showing it by R application with real data in the next chapter.
- In the following figure, every combination of the risk-free asset and the market portfolio M is shown on the line drawn from the vertical axis at the risk-free rate tangent to the Markowitz efficient frontier. All the portfolios on the line are feasible for the investor to construct. The line from the risk-free rate that is tangent

[1] Fama, E.F., 1970. Efficient capital markets: A review of theory and empirical work 25, pp. 383–417.

to the efficient frontier of risky assets is called the Capital Market Line (CML).

Capital Market Line and the Markowitz Efficient Frontier



- With the introduction of the risk-free asset, we can now say that an investor will select a portfolio on the CML that represents a combination of borrowing or lending at the risk-free rate and the market portfolio. This important property is called separation theory.
- Portfolios to the left of the market portfolio represent combinations of risky assets and the risk-free asset.
- Portfolios to the right of the market portfolio include purchases of risky assets made with funds borrowed at the risk-free rate. Such a portfolio is called a **leveraged portfolio** because it involves the use of borrowed funds.
- The separation property also has important implications in practice. Specifically, practical portfolio construction is normally broken down into at least the following two steps:

(1) Asset allocation: Decide how to allocate the investor's wealth between the risk-

free security and the set of risky securities.

(2) Risky portfolio construction: Decide how to distribute the risky portion of the investment among the set of risky securities.

- The first point is closely linked to an investor's strategic goals and general risk profile as well as his liquidity requirements.

(3) The focus of individual investors is more on the first point.

(4) The focus of investment management is more on the second point. In later chapters we will discuss various kinds of forecasting techniques that can be used in order to maximize different investment objectives and controlling the risk of the risky portion of the portfolio.

5. Deriving the Capital Market Line

- As we have seen above, in the absence of a risk-free asset, Markowitz efficient portfolios can be constructed by solving a constrained optimization problem that finds an optimal trade-off between return and variance, with the optimal portfolio being the one portfolio selected based on the investor's preference.
- However, the efficient frontier changes once a risk-free asset is introduced and we assume that investors can borrow and lend at the risk-free rate. As such, in the presence of a risk-free asset, the efficient frontier will be extended to the Capital Market Line (CML).
- We can derive a formula for the CML as follows. Based on the assumption of homogeneous expectations regarding the inputs in the portfolio construction process, all investors can create an efficient portfolio consisting of w_f placed in the risk-free asset and w_M in the market portfolio, where w represents the corresponding percentage (weight) of the portfolio allocated to each asset. Thus, $w_f + w_M = 1$.
- As the expected return of the portfolio, $E(R_P)$, is equal to the weighted average of the expected returns of the two assets, we have:

$$E(R_P) = w_f R_f + w_M E(R_M)$$

- Since we know that $w_f = 1 - w_M$, we can rewrite $E(R_P)$ as

$$E(R_P) = (1 - w_M)R_f + w_M E(R_M)$$

$$\rightarrow E(R_P) = R_f + w_M\left[E(R_M) - R_f\right]$$

- Since the return of the risk-free asset and the return of the market portfolio are uncorrelated and the variance of the risk-free asset is equal to zero, the variance of the portfolio consisting of the risk-free asset and the market portfolio is given by

$$\sigma_P^2 = var(R_P) = w_f^2\sigma_f^2 + w_M^2\sigma_M^2 + 2w_f w_M \sigma_{f,M}$$

$$\rightarrow \quad \sigma_P^2 = var(R_P) = w_M^2 \sigma_M^2 \ \text{(Since } \sigma_f^2 = 0 \ and \ \sigma_{f,M} = 0)$$

$$\rightarrow w_M = \frac{\sigma_P}{\sigma_M}$$

- If we substitute the $w_M = \frac{\sigma_P}{\sigma_M}$ into $E(R_P) = R_f + w_M[E(R_M) - R_f]$ and rearrange terms, we get the explicit expression for the CML:

$$E(R_P) = R_f + \left[\frac{E(R_M) - R_f}{\sigma_M}\right]\sigma_P$$

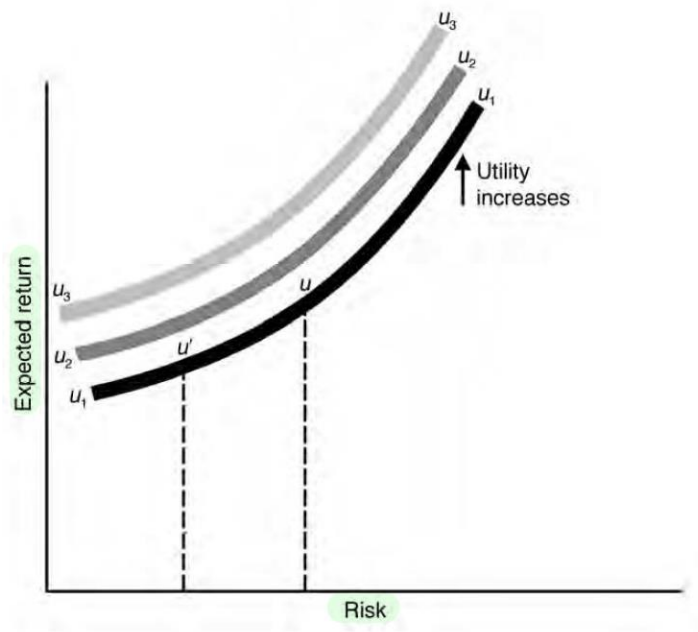
- The bracketed portion of the second term in the equation for the CML, $\left[\frac{E(R_M)-R_f}{\sigma_M}\right]$, is often referred to as the risk premium. Specifically, $\left[\frac{E(R_M)-R_f}{\sigma_M}\right]$ is the slope of the CML and measures the reward per unit of market risk.
- Since the CML represents the return offered to compensate for a perceived level of risk, each point on the CML is a balanced market condition, or equilibrium.
- The slope of the CML is also referred to as the equilibrium market price of risk because it (the slope of the CML) determines the additional return needed to compensate for a unit change in (total market) risk.
- In other words, the CML says that the expected return on a portfolio is equal to the risk-free rate plus a risk premium, where **the risk premium is equal to the market price of risk** (as measured by the reward per unit of market risk) times the quantity of risk for the portfolio (as measured by the standard deviation of the portfolio). Summarizing, we can write:

$$E(R_P) = R_f + Market\ Price\ of\ Risk \times Quantity\ of\ risk$$

6. Utility Functions, Indifference Curves, and Optimal Portfolio

In portfolio theory, entities are faced with a set of choices. Different portfolios have different levels of expected return and risk—the higher the level of expected return, the larger the risk. Entities are faced with the decision of choosing a portfolio from the set of all possible risk/return combinations. Whereas they like return, they dislike risk. Therefore, entities obtain different levels of utility from different risk/return combinations. The utility obtained from any possible risk/return combination is expressed by the utility function, expressing the preferences of entities over perceived risk and expected return combinations.

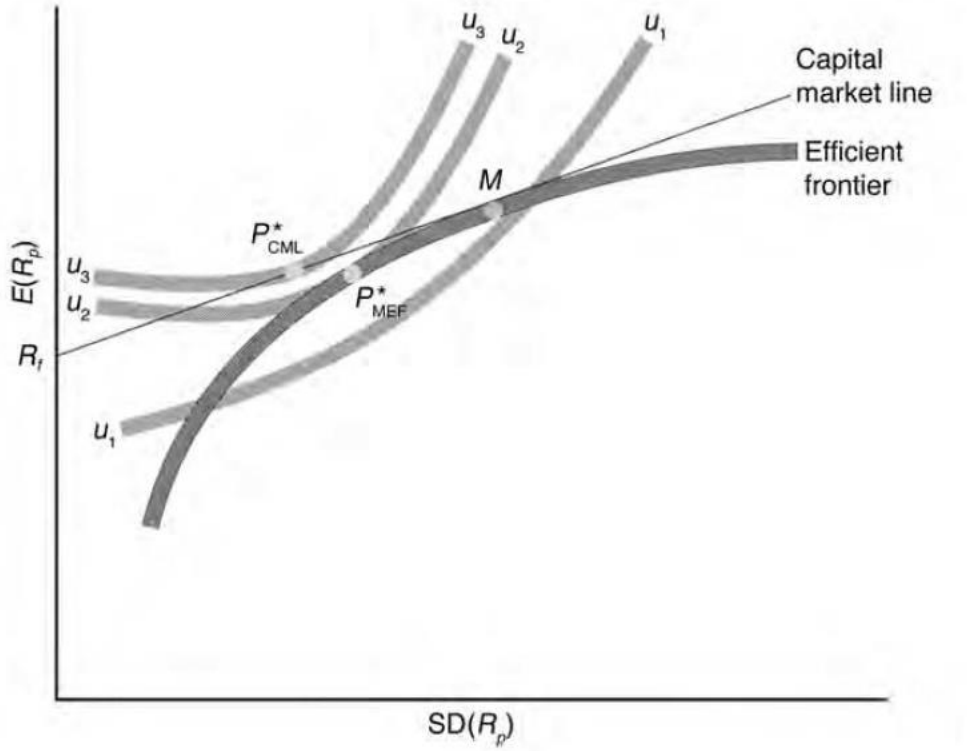
- A utility function can be presented in graphical form by a set of indifference curves. Each curve represents a set of portfolios with different combinations of risk and return. All the points on a given indifference curve indicate combinations of risk and expected return that will give the same level of utility to a given investor.



- All the points on a given indifference curve indicate combinations of risk and expected return that will give the same level of utility to a given investor. For example, on utility curve u_1 there are two points u and u', with u having a

higher expected return than u', but also having a higher risk. Because the two points lie on the same indifference curve, the investor has an equal preference for (or is indifferent between) the two points, or, for that matter, any point on the curve. The positive slope of an indifference curve reflects that, to obtain the same level of utility, the investor requires a higher expected return in order to accept higher risk.

- A reasonable assumption is that investors are risk averse. A risk averse investor is an investor who, when faced with choosing between two investments with the same expected return but two different risks, prefers the one with the lower risk.
- In selecting portfolios, an investor seeks to maximize the expected portfolio return given his tolerance for risk. Given a choice from the set of efficient portfolios, the optimal portfolio is the one that is preferred by the investor. In terms of utility functions, the optimal portfolio is the efficient portfolio that has the maximum utility.
- According to the following Figure, the particular efficient portfolio that the investor will select will depend on the investor's risk preference. The investor will select the portfolio P^*_{CML} on the CML that is tangent to the highest indifference curve, u_1 in the Figure. Notice that without the risk-free asset, an investor could only get to u_2, which is the indifference curve that is tangent to the Markowitz efficient frontier. This portfolio is denoted by P^*_{MEF} in the Figure.



u_1, u_2, u_3 = Indifference curves with $u_1 < u_2 < u_3$
M = Market portfolio
R_f = Risk-free rate
P^*_{CML} = Optimal portfolio on capital market line
P^*_{MEF} = Optimal portfolio on efficient frontier

- Five Phases of the Business Cycle

EXHIBIT 4-15 Five Phases of the Business Cycle

Phase	Economy	Fiscal and Monetary Policy	Confidence	Capital Markets
1. Initial recovery	Inflation still declining	Stimulatory fiscal policies	Confidence starts to rebound	Short rates low or declining; bond yields bottoming; stock prices strongly rising
2. Early upswing	Healthy economic growth; inflation remains low		Increasing confidence	Short rates moving up; bond yields stable to up slightly; stock prices trending upward
3. Late upswing	Inflation gradually picks up	Policy becomes restrictive	Boom mentality	Short rates rising; bond yields rising; stocks topping out, often volatile
4. Slowdown	Inflation continues to accelerate; inventory correction begins		Confidence drops	Short-term interest rates peaking; bond yields topping out and starting to decline; stocks declining
5. Recession	Production declines; inflation peaks		Confidence weak	Short rates declining; bond yields dropping; stocks bottoming and then starting to rise

- Inflation/Deflation Effects on Asset Classes[2,3]

EXHIBIT 4-18 Inflation/Deflation Effects on Asset Classes

	Cash	Bonds	Equity	Real Estate/ Other Real Assets
Inflation at or below expectations	Short-term yields steady or declining. **[Neutral]**	Yield levels maintained; market in equilibrium. **[Neutral]**	Bullish while market in equilibrium state. **[Positive]**	Cash flow steady to rising slightly. Returns equate to long-term average. Market in general equilibrium. **[Neutral]**
Inflation above expectations	Bias toward rising rates. **[Positive]**	Bias toward higher yields due to a higher inflation premium. **[Negative]**	High inflation a negative for financial assets. Less negative for compa-nies/industries able to pass on inflated costs. **[Negative]**	Asset values increasing; increased cash flows and higher expected returns. **[Positive]**
Deflation	Bias toward 0% short-term rates. **[Negative]**	Purchasing power increasing. Bias toward steady to lower rates (may be offset by increased risk of potential defaults due to falling asset prices). **[Positive]**	Negative wealth effect slows demand. Especially affects asset-intensive, commodity-producing (as opposed to commodity-using), and highly levered companies. **[Negative]**	Cash flows steady to falling. Asset prices face downward pressure. **[Negative]**

- Chih-Hsiang Hsu*, **Hsiu-Chuan Lee**, and Donald Lien**.** (2020). Stock market uncertainty, volatility connectedness of financial institutions, and stock-bond return correlations. *International Review of Economics and Finance,* 70, 600-621. (MOST: A- , SSCI,*Correspondence author*)

[2] Campbell and Ammer (1993) propose a traditional fundamental approach that involves several counteracting forces behind the return correlation between bonds and stocks. First, the variation in the interest rate produces a positive return correlation between stock and bond prices, because both are negatively related to interest rates. Second, the variation in expected inflation leads to a negative return correlation because an increase in expected inflation negatively affects bond returns but positively affects stock returns.

[3] Campbell, J.Y., Ammer, J., 1993. What moves the stock and bond markets? A variance decomposition for long-term asset returns. Journal of Finance 48, 3–37.

Chapter 3: Asset Allocation by using Optimization Package "quadprog"

1. solve.QP in package "quadprog"

In solve.QP of the package "quadprog", the quadratic programming problems of the following general type:

$$\min_{X} \left[\frac{1}{2} X^T D X - d^T X\right]$$

$$\text{s.t. } A^T X \geq b$$

$$(\ decompose\ into$$

$$\text{s.t. } A^T X = b$$

$$\text{and}$$

$$\text{s.t. } A^T X > b)$$

where X is $n \times 1$ vector, d is $n \times 1$ vector, and D is $n \times n$ matrix for the object objective equation. A^T is $k \times n$ matrix and b is $k \times 1$ vector for the constraint equation.

In solve.QP, the parameters are displayed as follows:

function (Dmat, dvec, Amat, bvec, meq = 0, factorized = FALSE)

Dmat: the matrix D in the above equation

dvec: the vector d in the above equation

Amat: the matrix A in the above equation

bvec: the vector b in the above equation

meq: the first meq constraints are treated as equality constraints, and the remainder are inequality constraints.

factorized: factorized logical flag: if TRUE, then we are passing R^{-1} where $D = R^T R$, instead of D in the argument Dmat

2. Portfolio Optimization in Four Case by using solve.QP

Case 1: Minimum Variance Portfolio with Short Selling

$$\min_{X} \left[\frac{1}{2} W^T \Sigma W\right]$$

$$\text{s.t. } A^T W = 1$$

So, the notation mapping can be presented as follows:

$d \rightarrow 0 = n \times 1$ vector of 0 $\rightarrow d =$ dvec

$D \rightarrow \sum = n \times n$ return covariance matrix $\rightarrow \sum$ = Dmat

$X \rightarrow W = n \times 1$ vector of portfolios weights

A is a $n \times 1$ matrix of ones $\rightarrow A$ = Amat

b is a $n \times 1$ vector for the constraints $\rightarrow b =$ bvec = 1

meq: meq is 1 because we only have "one" equality constraint

```
packages=c("quadprog","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}

data(managers)
RET <- managers[,5:10]

# Ann Ret and Vol
nFreq <- 12
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq

# Parms for solve.QP
cov.mat <- covmat
n <- ncol(cov.mat)
dvec.set <- array(0, dim=c(n,1))
Amat.set <- matrix (1, nrow=n)
bvec.set <- 1
```

```
meq.set    <- 1

# solve by solve.QP and obtain the optimal weights
port <-solve.QP(Dmat=cov.mat, dvec=dvec.set, Amat=Amat.set,
                        bvec=bvec.set, meq=meq.set)
port.wgt <- round(as.matrix(port$solution)* 100, digits=2)
rownames(port.wgt) <-c(names(RET))

# Optimal portfolio returns and vol by using the optimal weights
port.wgt <- port.wgt/100
PortRET <- t(port.wgt)%*%avgRET
PortVol <- sqrt(t(port.wgt)%*%covmat%*%port.wgt)

Ans <- cbind(t(port.wgt),PortRET,PortVol, PortRET/PortVol)
colnames(Ans) <- c(paste0("W:",names(RET)),
                           "PortR","PortV","R/V")
round(Ans, digits=5)
```

Case 2: Minimum Variance Portfolio with Target Return allowing Short Selling

$$\min_{X}\left[\frac{1}{2}W^T\Sigma W\right]$$

$$\text{s.t. } A^T W = 1$$

$$\text{s.t. } R^T W = \hat{R}$$

In this case, $A^T X = b$ should be presented as follows:

$$A = \begin{bmatrix} 1 & R_1 \\ 1 & R_2 \\ 1 & R_3 \\ 1 & R_4 \\ 1 & R_5 \\ 1 & R_6 \end{bmatrix}$$

And

R is a $n \times 1$ matrix of returns

b is a $n \times 1$ vector for the constraints $\rightarrow$ $b =$ bvec = $\begin{bmatrix} 1 \\ \hat{R} \end{bmatrix}$

meq: meq is 2 because we only have "two" equality constraint

```
packages=c("quadprog","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
data(managers)
RET <- managers[,5:10]
selTR <- 150    # select # target reutns from mu.P 1~300

# Ann Ret and Vol
nFreq <- 12
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq

# Parms for solve.QP
mean.R <- avgRET
cov.mat <- covmat
n <- ncol(cov.mat)
```

```
dvec.set <- array(0, dim=c(n,1))
Amat.set <- cbind(rep(1, n), mean.R)
mu.P <- seq(min(mean.R - 5e-04), max(mean.R + 5e-04), length = 300) ## set of 300
possible target portfolio returns
bvec.set <- c(1, mu.P[selTR])
meq.set    <- 2

# solve by solve.QP and obtain the optimal weights
port <-solve.QP(Dmat=cov.mat, dvec=dvec.set, Amat=Amat.set,
                        bvec=bvec.set, meq=meq.set)
port.wgt <- round(as.matrix(port$solution)* 100, digits=2)
rownames(port.wgt) <-c(names(RET))

# Optimal portfolio returns and vol by using the optimal weights
port.wgt <- port.wgt/100
PortRET <- t(port.wgt)%*%avgRET
PortVol <- sqrt(t(port.wgt)%*%covmat%*%port.wgt)
Ans <- cbind(t(port.wgt),PortRET,PortVol, PortRET/PortVol)
colnames(Ans) <- c(paste0("W:",names(RET)),
                        "PortR","PortV","R/V")
round(Ans, digits=5)
mu.P[selTR]
```

Case 3: Minimum Variance Portfolio with Target Return without Short Selling

$$\min_{X}\left[\frac{1}{2}W^T\Sigma W\right]$$

$$\text{s.t. } A^T W = 1$$

$$\text{s.t. } R^T W = \hat{R}$$

$$\text{s.t. } W > 0$$

In this case, $A^T X = b$ and $A^T X > b$ should be presented as follows:

$$A = \begin{bmatrix} 1 & R_1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & R_2 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & R_3 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & R_4 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & R_5 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & R_6 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

And

b is a $n \times 1$ vector for the constraints $\rightarrow$ $b =$ bvec $= \begin{bmatrix} 1 \\ \hat{R} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

meq: meq is 2 because we only have “two” equality constraint

```
packages=c("quadprog","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
data(managers)
RET <- managers[,5:10]
selTR <- 150    # select # target reutns from mu.P 1~300

# Ann Ret and Vol
nFreq <- 12
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq
```

```
# Parms for solve.QP
mean.R <- avgRET
cov.mat <- covmat
n <- ncol(cov.mat)
dvec.set <- array(0, dim=c(n,1))
Amat.set <- cbind(rep(1, n), mean.R, diag(n))
mu.P <- seq(min(mean.R - 5e-04), max(mean.R + 5e-04), length = 300) ## set of 300
possible target portfolio returns
bvec.set <- c(1, mu.P[selTR], rep(0, n))
meq.set    <- 2

# solve by solve.QP and obtain the optimal weights
port <-solve.QP(Dmat=cov.mat, dvec=dvec.set, Amat=Amat.set,
                        bvec=bvec.set, meq=meq.set)
port.wgt <- round(as.matrix(port$solution)* 100, digits=2)
rownames(port.wgt) <-c(names(RET))

# Optimal portfolio returns and vol by using the optimal weights
port.wgt <- port.wgt/100
PortRET <- t(port.wgt)%*%avgRET
PortVol <- sqrt(t(port.wgt)%*%covmat%*%port.wgt)
Ans <- cbind(t(port.wgt),PortRET,PortVol, PortRET/PortVol)
colnames(Ans) <- c(paste0("W:",names(RET)),
                          "PortR","PortV","R/V")
round(Ans, digits=5)
mu.P[selTR]
```

Case 4: Minimum Variance Portfolio with Target Return and Bounds of Min and Max Weights without Short Selling

$$\min_{X}\left[\frac{1}{2}W^{T}\Sigma W\right]$$

$$\text{s.t. } A^{T}W = 1$$

$$\text{s.t. } R^{T}W = \hat{R}$$

$$\text{s.t. } W > 0$$

$$\text{s.t. } minW \leq W \leq maxW$$

In this case, $A^{T}X = b$ and $A^{T}X > b$ should be presented as follows:

$$A = \begin{bmatrix} 1 & R_1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \mathbf{-1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1 & R_2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \mathbf{0} & \mathbf{-1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1 & R_3 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1 & R_4 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{0} & \mathbf{0} \\ 1 & R_5 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} & \mathbf{0} \\ 1 & R_6 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-1} \end{bmatrix}$$

(why the bold red numbers are negative? Hint $A^{T}X \geq b$)

And

b is a $n \times 1$ vector for the constraints $\rightarrow$ $b =$ bvec $= \begin{bmatrix} 1 \\ \hat{R} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.15 \\ 0.15 \\ 0.15 \\ 0.15 \\ 0.15 \\ 0.15 \\ \mathbf{-0.50} \\ \mathbf{-0.50} \\ \mathbf{-0.50} \\ \mathbf{-0.50} \\ \mathbf{-0.50} \\ \mathbf{-0.50} \end{bmatrix}$

meq: meq is 2 because we only have “two” equality constraint

```
packages=c("quadprog","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
data(managers)
RET <- managers[,5:10]
selTR <- 150    # select # target reutns from mu.P 1~300

# Ann Ret and Vol
nFreq <- 12
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq

# Parms for solve.QP
minW <- 0.15    # Min Weight
maxW <- 0.50    # Max Weight
mean.R <- avgRET
cov.mat <- covmat
n <- ncol(cov.mat)
dvec.set <- array(0, dim=c(n,1))
Amat.set <- cbind(rep(1, 3), mean.R, diag(n), 1*diag(n), -1*diag(n))
mu.P <- seq(min(mean.R - 5e-04), max(mean.R + 5e-04), length = 300) ## set of 300
possible target portfolio returns
min.wgt <-rep(minW , n)
max.wgt <-rep(maxW, n)
bvec.set <- c(1, mu.P[selTR], rep(0, n), min.wgt, -max.wgt)
meq.set    <- 2

# solve by solve.QP and obtain the optimal weights
port <-solve.QP(Dmat=cov.mat, dvec=dvec.set, Amat=Amat.set,
                        bvec=bvec.set, meq=meq.set)
port.wgt <- round(as.matrix(port$solution)* 100, digits=2)
rownames(port.wgt) <-c(names(RET))
```

```
# Optimal portfolio returns and vol by using the optimal weights
port.wgt <- port.wgt/100
PortRET <- t(port.wgt)%*%avgRET
PortVol <- sqrt(t(port.wgt)%*%covmat%*%port.wgt)
Ans <- cbind(t(port.wgt),PortRET,PortVol, PortRET/PortVol)
colnames(Ans) <- c(paste0("W:",names(RET)), "PortR","PortV","R/V")
round(Ans, digits=5)
mu.P[selTR]
```

- **Investment set and efficient frontier**

```
packages=c("quadprog","PerformanceAnalytics")
for(i in packages){
   if(!require(i,character.only = TRUE)) install.packages(i)
   library(i,character.only = TRUE)
}
data(managers)
RET <- managers[,1:3]

# Ann Ret and Vol
nFreq <- 12
avgRET <- colMeans(RET ,na.rm = T)*nFreq
stdev <- apply(RET, 2, sd,na.rm = T)*nFreq^0.5
covmat <- cov(RET ,use = "pairwise.complete.obs")*nFreq

# Parms for solve.QP
minW <- 0.00    # Min Weight
maxW <- 1.00    # Max Weight
mean.R <- avgRET
cov.mat <- covmat
mean.R <- avgRET
cov.mat <- covmat
mu.P <- seq(min(mean.R),
                    max(mean.R), length = 300) ## set of 300 possible target portfolio
returns

# solve by solve.QP and obtain the optimal weights
Weight <- NULL
PortR <- NULL
PortV <- NULL
for (i in 1:length(mu.P)) {
   # Parms for solve.QP
   n <- ncol(cov.mat)
   dvec.set <- array(0, dim=c(n,1))
   Amat.set <- cbind(rep(1, 3), mean.R, diag(n), 1*diag(n), -1*diag(n))
   min.wgt <-rep(minW , n)
   max.wgt <-rep(maxW, n)
   bvec.set <- c(1, mu.P[i], rep(0, n), min.wgt, -max.wgt)
```

```
  meq.set    <- 2
  port <- solve.QP(Dmat=cov.mat, dvec=dvec.set, Amat=Amat.set,
                          bvec=bvec.set, meq=meq.set)
  Weight <- rbind(Weight,port$solution)
  PortR <- rbind(PortR,
                       (t(port$solution)%*%avgRET))
  PortV <- rbind(PortV,
                       sqrt(t(port$solution)%*%covmat%*%port$solution))
  rm(port)
}
# Port sort by indicators
SV <- 3;     # Sorting by indicators. 1=RET, 2=VOL, 3=RET/VOL
Port <- round(cbind(PortR,PortV,PortR/PortV,Weight),digits = 4)
colnames(Port) <- c("RET","SD","RET/SD",names(RET))
SPort <- Port[order(Port[,SV],decreasing = T), ]
View(SPort)

gmin.port <- Port[order(Port[,2],decreasing = F)[1],]
tan.port <- Port[order(Port[,SV],decreasing = T)[1],]

# Stat Fig
adj = 0.50
windows()
plot(PortR~PortV,
       xlim = c(min(stdev)-max(stdev)*adj,
                max(stdev)+max(stdev)*adj),
       ylim = c(min(avgRET),
                max(avgRET)),
       type = 'l', lty = 6, xlab = 'risk(sd)', ylab = 'return',
       main = 'Feasible Investment Set', col = 'navyblue')
cols <- ifelse(avgRET > 0, 'red','black')
points(stdev, avgRET, pch = 8, cex = 1.4, col = cols)
text(stdev, avgRET, colnames(RET),
       col = cols, cex = 0.70, adj = -.2)
points(gmin.port[1]~gmin.port[2], col="green", pch=16, cex=2)
points(tan.port[1]~tan.port[2], col="red", pch=16, cex=2)
text(gmin.port[1]~gmin.port[2], labels="GLOBAL MIN", pos=2,col="green")
text(tan.port[1]~tan.port[2], labels="TANGENCY", pos=1,col = "red")
```

3. Investment Opportunity Set, Minimum Variance Portfolio and Tangency Portfolio

For a given $\hat{R}$, the weights of optimal portfolio can be calculated as follows:

$$\min_{X} \left[\frac{1}{2} W^T \Sigma W\right]$$

$$\text{s.t. } A^T W = 1$$

$$\text{s.t. } W^T R = \hat{R}$$

$$\text{s.t. } W > 0$$

$$\text{s.t. } minW \leq W \leq maxW$$

(1) A feasible investment set (investment opportunity set)

We need the $\hat{R}_i$ from the minimum asset return to the maximum asset return and solve the objective and subjective equations. Using a loop for solving.

(2) Efficient frontier:

(a) Find a minimum variance from a feasible investment set.

(b) Find the returns given a minimum variance.

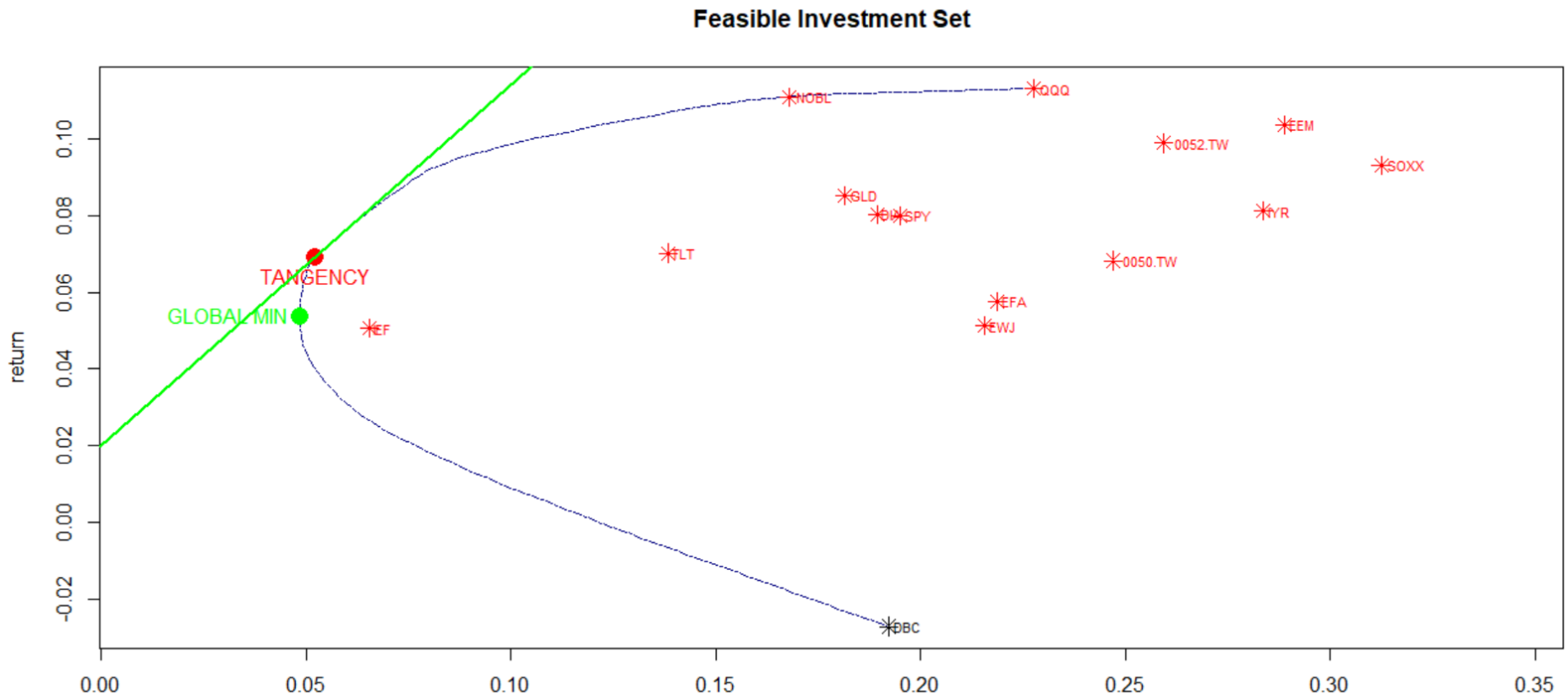
(c) Find the portfolio returns are higher than those of the returns given a minimum variance.

(3) Tangency Portfolio

(a) Find the portfolio with the maximum sharpe ratio from a feasible investment set (same for the efficient frontier).

(b) The portfolio with the maximum sharpe ratio is also the tangency portfolio.

(4) Code in the document “CH03”.


Feasible Investment Set
return
0.10
0.08
0.06
0.04
0.02
0.00
-0.02
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
TANGENCY
GLOBAL MIN
NOBL
QQQ
EEM
0052.TW
SOXX
GLD
SPY
IYR
TLT
0050.TW
EFA
EWJ
EF
DBC

4. Issue of Parameter Estimation and the Single-Index Model

- The Markowitz approach to mean–variance efficiency and its variations need two crucial inputs: The vector of expected returns μ and the covariance matrix Σ.
- The question to this end is that the accurate estimates may not be found for a large number of assets. For example, let us consider a universe of n = 500 firms. In this case, a total of 125,750 need to be estimated as follows:
  - 500 returns, $R_{i,t}$
  - 500 variances $\sigma_{ii} = \sigma_i^2$
  - $n \times (n-1)/2 = 124{,}750$ covariances, $\sigma_{ii}, i > j$.
  - A total of $2 \times n + n \times (n-1)/2 = 1{,}000 + 124{,}750 = 125{,}750$
- One way to simplify the task is to reduce dimensionality by spotting common risk factors, such as market return, inflation, economic growth, oil price, etc., which explain the return of a stock share together with firm-specific factors (or so-called individual firm factors). This simplification may also be useful for forecasting purposes, as it is certainly easier to predict a few key variables than a large array of returns.
- The simplest factor model consists of a linear relationship between the return of each asset and a single common risk factor related to market return. The market return may be proxied by a relevant index, such as S&P500, which is why such a model is referred to as **single-index model**.
- Define excess returns of each asset as $\hat{R}_{i,t} = \hat{r}_{i,t} - \hat{r}_{RF,t}$ and of the market index $\hat{R}_{M,t} = \hat{r}_{M,t} - \hat{r}_{RF,t}$. Then, the single-index model is given by the linear regression model:

$$\hat{R}_{i,t} = \alpha_i + \beta_i \hat{R}_{M,t} + \hat{\varepsilon}_{i,t}$$

  - The error term $\hat{\varepsilon}_{i,t}$ is interpreted as a specific risk factor of asset i.
  - β_i is explained as the systematic risk.

- We make the following assumptions for the single-index model:
  - The specific risks are random variables with expected value $E(\hat{\varepsilon}_{i,t}) = 0$. This makes sense, as any predictable component should be included in the intercept α_i.
  - The systematic and specific risks are uncorrelated. We know from the theory of linear regression that, after estimating the model using ordinary least-squares, we indeed have $COV(\hat{R}_{M,t}, \hat{\varepsilon}_{i,t}) = 0$.
  - Last but not least, a further important assumption is the lack of correlation among specific risks, $COV(\hat{\varepsilon}_{i,t}, \hat{\varepsilon}_{j,t}) = 0$ for $i \neq j$. This condition displays a diagonal model, since the covariance matrix for the errors is diagonal. Indeed, **if idiosyncratic factors are truly specific**, this assumption should hold.[4]
- According to the index model,

$$\hat{\pi}_{i,t} = E(\hat{R}_{i,t}) = \alpha_i + \beta_i E(\hat{R}_{M,t}) = \alpha_i + \beta_i \hat{\pi}_{M,t}$$

  - From the theory of linear regression by least-squares

$$\beta_i = \frac{COV(\hat{R}_{i,t}, \hat{R}_{M,t})}{VAR(\hat{R}_{M,t})} = \frac{\sigma_{i,M}}{\sigma_M^2}$$

  - According to $\hat{R}_{i,t} = \alpha_i + \beta_i \hat{R}_{M,t} + \hat{\varepsilon}_{i,t}$, the total risk of an asset is the sum of systematic and specific risks:

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\hat{\varepsilon}_{i,t}}^2$$

  - Given that $COV(\hat{\varepsilon}_{i,t}, \hat{\varepsilon}_{j,t}) = 0$ and $COV(\hat{R}_{M,t}, \hat{\varepsilon}_{i,t}) = 0$, the covariance between assets returns is the product of betas and the market risk:

$$COV(\hat{R}_{i,t}, \hat{R}_{j,t})$$
$$= COV(\alpha_i + \beta_i \hat{R}_{M,t} + \hat{\varepsilon}_{i,t}, \alpha_j + \beta_j \hat{R}_{M,t} + \hat{\varepsilon}_{j,t}) = \beta_i \beta_j \sigma_M^2$$

[4] However, this is a debatable condition, and it holds only if the single common factor really accounts for all of the common risk. When multiple common risk factors, possibly related to macroeconomic conditions, are at play, a single factor will not account for the whole correlation, and this may be reflected in correlated errors. Nevertheless, the diagonality assumption will prove to be quite convenient.

- As to the correlation, it is the product of correlations with the common risk factor:

$$\rho_{i,j} = \frac{\beta_i \beta_j \sigma_M^2}{\sigma_i \sigma_j} = \frac{(\beta_i \sigma_M^2)(\beta_j \sigma_M^2)}{(\sigma_i \sigma_M)(\sigma_j \sigma_M)} = \rho_{i,M} \rho_{j,M}$$

Hint: $\beta_i = \frac{\sigma_{i,M}}{\sigma_M^2} \rightarrow \beta_i \sigma_M^2 = \sigma_{i,M}$

- Hence, in order to feed a mean–variance optimization model, we need to estimate:
  - The $2 \times n$ parameters α_i and β_i, $i = 1 \ldots 500$.[5]
    - As such, we can obtain expected returns for each asset, $E(\hat{R}_{i,t})$,
  - The n variances $\sigma_{\hat{\varepsilon}_{i,t}}^2$ $(\hat{\sigma}_i)$ of idiosyncratic risk factors.[6]
    - In this case, we can obtain the variance of each asset for the diagonal of covariance matrix., e.g., $\sigma_{\hat{\varepsilon}_{i,t}}^2$.
  - The expected risk premium $\hat{\pi}_{M,t}$, *or* $E(\hat{R}_{M,t})$, and the σ_M^2 of the common risk factor.
    - After obtaining the σ_M^2, we can calculate the covariance between assets for the covariance matrix by using $\hat{\sigma}_{i,j} = COV(\hat{R}_{i,t}, \hat{R}_{j,t}) = \beta_i \beta_j \sigma_M^2$.
  - In summary, the total number of these parameters is $3 \times n + 2$, rather than $2 \times n + n \times (n-1)/2$.
  - These $3 \times n + 2$ parameters generate $E(\hat{R}_{i,t})$, $\hat{\sigma}_i$, and $\hat{\sigma}_{i,j}$. We get the parameters by estimating the index model for each of the n securities.

[5] The expected returns may be found from $\hat{R}_{i,t} = \alpha_i + \beta_i \hat{R}_{M,t} + \hat{\varepsilon}_{i,t}$, which requires estimates of the α_i and β_i.

[6] After estimating $\hat{R}_{i,t} = \alpha_i + \beta_i \hat{R}_{M,t} + \hat{\varepsilon}_{i,t}$, $\sigma_{\hat{\varepsilon}_{i,t}}^2$ (or $\hat{\sigma}_i$) can be calculated as the variance of $\hat{\varepsilon}_{i,t}$.

Chapter 4: CAPM and APT

1. Why Should We Need the Capital Asset Pricing Model (CAPM)?

- Since that the efficient frontier should be the same for all investors, they must have **homogeneous expectations** about the underlying market variables, R_i $(or\ u_i)$, σ_i, and $\rho_{i,j}$,
- Hence, with **homogeneous expectations**, all investors hold all the risky assets in the proportions given by point M, the market portfolio. The assumption of **homogeneous expectations** is crucial in producing a **market equilibrium** where all risky assets are willingly held in the optimal proportions w^*given by M or, in other words, in producing **market clearing**.
- Why should we need the CAPM when the relationship between the total risk of a portfolio (or an asset) and return is captured by the CML?
  - ✧ When the market is clearing, investors hold the well-diversified portfolio (the market portfolio M) and only care about the systematic risk because the specific risk for the assets are diversified in the market portfolio.
  - ✧ Hence, if we want to create an artificial portfolio by buying a stock (or a portfolio) with weight w_i and a market portfolio with weight $(1 - w_i)$, we need to justify whether the inclusion of a stock (or a portfolio) can reduce the systematic risk.
  - ✧ As such, a measure of the sensitivity of a stock or portfolio to the market (systematic) risk need to be measured.[7] The CAPM and its beta (β) help us in solving this question.

[7] In other words, the sensitivity of a stock's (or portfolio) returns to fluctuations in the market portfolio returns need to be measured.

2. Deriving the CAPM

- The Capital Asset Pricing Model is an equilibrium asset pricing model that was developed from mean-variance portfolio selection. The CAPM is an abstraction of the real-world capital markets based on the following assumptions:

✧ Investors make investment decisions based on the expected return and variance of returns.

✧ Investors are rational and risk-averse.

✧ Investors subscribe to the Markowitz method of portfolio diversification.

✧ Investors all invest for the same period of time.

✧ Investors have the same expectations about the expected return and variance of all assets.

✧ There is a risk-free asset and investors can borrow or lend any amount at the risk-free rate.

✧ Capital markets are (perfectly) competitive and frictionless.

- We create an artificial portfolio and this artificial portfolio (call it p) consists of two risky portfolios with proportions w_i in asset i and $(1 - w_i)$ in the portfolio at M. This portfolio p has expected return $E(R_p)$ and standard deviation σ_p:

$$u_p = E(R_p) = w_i E(R_i) + (1 - w_i) E(R_M)$$

$$\sigma_p = \left[w_i^2 \sigma_i^2 + (1 - w_i)^2 \sigma_M^2 + 2 w_i (1 - w_i) \sigma_{i,M}\right]^{1/2}$$

- When the market is in equilibrium, $w_i = 0$ can be observed. Hence, to find the slope of the efficient frontier at M, we require:

$$\left[\frac{\partial u_p}{\partial \sigma_p}\right]_{w_i=0} = \left[\frac{\partial u_p}{\partial w_i}\right]\left[\frac{\partial w_i}{\partial \sigma_p}\right] = \left[\frac{\partial u_p}{\partial w_i}\right]\left[\frac{\partial \sigma_p}{\partial w_i}\right]^{-1}$$

where all the derivatives are evaluated at $w_i = 0$

- Using the partial derivatives,

$$\left[\frac{\partial E(R_p)}{\partial w_i}\right] = \frac{\partial(w_i E(R_i) + (1-w_i)E(R_M))}{\partial w_i} = E(R_i) - E(R_M)$$

$$\left[\frac{\partial \sigma_p}{\partial w_i}\right] = \frac{\partial\left(\left[w_i^2\sigma_i^2 + (1-w_i)^2\sigma_M^2 + 2w_i(1-w_i)\sigma_{i,M}\right]^{\frac{1}{2}}\right)}{\partial w_i}$$

$$= \frac{1}{2}\left[w_i^2\sigma_i^2 + (1-w_i)^2\sigma_M^2 + 2w_i(1-w_i)\sigma_{i,M}\right]^{-\frac{1}{2}} \times$$

$$\left[2w_i\sigma_i^2 - 2\sigma_M^2 + 2w_i\sigma_M^2 + 2\sigma_{i,M} - 4w_i\sigma_{i,M}\right]$$

- At $w_i = 0$ (point M), hence $\left[\frac{\partial \sigma_p}{\partial w_i}\right]$ can be rewritten as:

$$\left[\frac{\partial \sigma_p}{\partial w_i}\right]_{w_i=0} = \frac{1}{2\sigma_M^2}\left[-2\sigma_M^2 + 2\sigma_{i,M}\right] = \frac{\sigma_{i,M} - \sigma_M^2}{\sigma_M}$$

- As such, $\left[\frac{\partial E(R_p)}{\partial \sigma_p}\right]_{w_i=0}$ can be presented as follows:

$$\left[\frac{\partial E(R_p)}{\partial \sigma_p}\right]_{w_i=0} = \left[\frac{\partial E(R_p)}{\partial w_i}\right]\left[\frac{\partial \sigma_p}{\partial w_i}\right]^{-1}$$

$$= [E(R_i) - E(R_M)]\left[\frac{\sigma_{i,M} - \sigma_M^2}{\sigma_M}\right]^{-1}$$

$$= \frac{[E(R_i) - E(R_M)]\sigma_M}{\sigma_{i,M} - \sigma_M^2}$$

- Since the tangent points for the CML and CAPM are the market portfolio M, the slope of the efficient frontier regarding the CAPM equals the slope of the CML:

$$\frac{[E(R_i) - E(R_M)]\sigma_M}{\sigma_{i,M} - \sigma_M^2} = \left[\frac{E(R_M) - R_f}{\sigma_M}\right]$$

$$[E(R_i) - E(R_M)]\sigma_M = \left[\frac{E(R_M) - R_f}{\sigma_M}\right]\left[\sigma_{i,M} - \sigma_M^2\right]$$

$$[E(R_i) - E(R_M)] = \left[\frac{E(R_M) - R_f}{\sigma_M}\right]\left[\sigma_{i,M} - \sigma_M^2\right]\frac{1}{\sigma_M}$$

$$[E(R_i) - E(R_M)] = \left[E(R_M) - R_f\right]\left[\frac{\sigma_{i,M} - \sigma_M^2}{\sigma_M^2}\right]$$

$$[E(R_i) - E(R_M)] = \left[E(R_M) - R_f\right]\left[\frac{\sigma_{i,M}}{\sigma_M^2} - 1\right]$$

$$[E(R_i) - E(R_M)] = \left[E(R_M) - R_f\right]\left[\frac{\sigma_{i,M}}{\sigma_M^2}\right] - \left[E(R_M) - R_f\right]$$

$$[E(R_i) - E(R_M)] = [E(R_M) - R_f]\left[\frac{\sigma_{i,M}}{\sigma_M^2}\right] - E(R_M) + R_f$$

$$E(R_i) = R_f + [E(R_M) - R_f]\left[\frac{\sigma_{i,M}}{\sigma_M^2}\right]$$

- Let $\beta_i = \left[\frac{\sigma_{i,M}}{\sigma_M^2}\right]$, $E(R_i) = R_f + [E(R_M) - R_f]\left[\frac{\sigma_{i,M}}{\sigma_M^2}\right]$ can be rewritten as:

$$E(R_i) = R_f + \beta_i[E(R_M) - R_f]$$

- The above equation is so-called the CAPM. The CAPM predicts that only the covariance of returns between asset i and the market portfolio influence the cross-section of excess returns, across assets. No additional variables such as the dividend–price ratio, the size of the firm or the earnings–price ratio should influence the cross-section of expected excess returns.
- This equation is also called the security market line (SML). **In equilibrium, the expected return of individual securities will lie on the SML and not on the CML. This is true because of the high degree of nonsystematic risk that remains in individual assets that can be diversified out of portfolios.** In equilibrium, only **efficient portfolios** (market portfolio, M) will lie on both the CML and the SML.
- The portfolio beta, β_p, is the linear combination of the individual stocks' beta and can be presented as:

$$\beta_p = \sum w_i \beta_i \text{ and } \sum w_i = 1$$

- To obtain the beta, the *characteristic line* can be estimated by the linear regression as follows: [8]

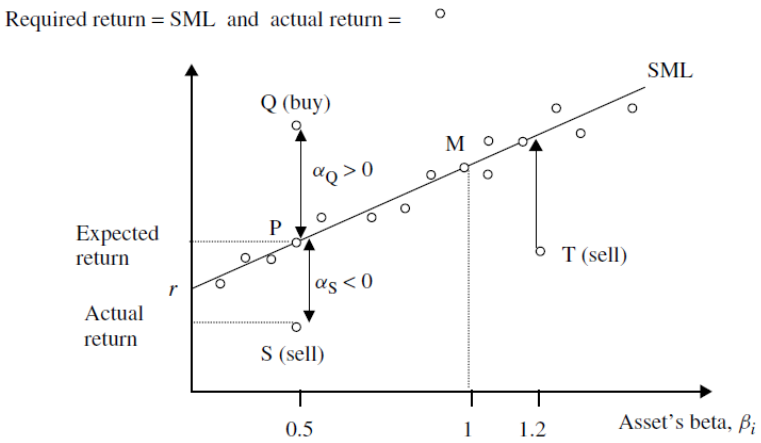
$$E(R_{i,t}) = R_{f,t} + \beta_i[E(R_{M,t}) - R_{f,t}] + \varepsilon_{i,t}$$

[8] The regression might use 60 or more months of time-series data on the (excess) market return and the excess return on the portfolio-i, adopted by the 'stock picker'.

- The *characteristic line* can be used to estimate Jensen's Alpha by the linear regression as follows:

$$E(R_{i,t}) - R_{f,t} = \alpha_i + \beta_i[E(R_{M,t}) - R_{f,t}] + \varepsilon_{i,t}$$

- The SML can be used to try to pick underpriced and overpriced stocks by using the Jensen's Alpha as follows:



Securities that lie above (below) the SML have a positive (negative) 'alpha', indicating a positive (negative) 'abnormal return', after correcting for 'beta risk'.

3. Empirically Testing the CAPM

A vast amount of research has been undertaken on empirically testing whether the CAPM is supported by real-world data on stocks and many different regression-based tests have been proposed. A simple Jensen's Alpha and the two-step regression-based suggested by Fama and MacBeth (1973)[9] are frequently used in the literature.

3.1. Jensen's Alpha

- Jensen's index ('alpha') assumes that investors hold an activity-managed portfolio-i, and performance is measured by the intercept α_i in the following regression:

$$E(R_{i,t}) - R_{f,t} = \alpha_i + \beta_i[E(R_{M,t}) - R_{f,t}] + \varepsilon_{i,t}$$

- The regression might use 60 or more months of time-series data on the (excess) market return and the excess return on the portfolio-i, adopted by the 'stock picker'.
- If α_i is zero, then we have the standard CAPM. Portfolio-i earns a return in excess of that given by the CAPM/SML if α_i is higher 0. For α_i < 0, the active portfolio manager has underperformed relative to beta risk of her portfolio.
- If α_i significantly differs from 0, the CAPM fails. By contrast, if α_i insignificantly differs from 0, the CAPM holds.

[9] Fama, E.F., MacBeth, J., 1973. Risk, return, and equilibrium: Empirical test. Journal of Political Economy 81, pp. 607–636.

3.2. Fama-MacBeth Approach

- A simple two-step regression-based empirical test of the CAPM suggested by Fama and MacBeth (1973) can be computed as follows:

1. Collect a sample of time series data on the returns for a sample of stocks (e.g., the monthly returns for N US stocks listed on the NYSE) over a particular time period (e.g., five years), and on a proxy for the market portfolio (e.g., the NYSE Composite index) and the risk-free rate (e.g., a US Treasury bill rate), and estimate the parameters of the security characteristic line (SCL) for each stock using ordinary least squares (OLS),

$$E(R_{i,t}) - R_{f,t} = \alpha_i + \beta_i[E(R_{M,t}) - R_{f,t}] + \varepsilon_{i,t}$$

   Assume that $\varepsilon_{i,t} \sim NID(0, \sigma^2_{\varepsilon_{i,t}})$. [10] These are referred to as the first-pass regressions.

2. Run a second cross-sectional regression of the sample mean excess return for the stocks $\bar{R}_i - \bar{R}_f$ on the estimated betas from the first-pass regressions,

$$\bar{R}_i - \bar{R}_f = \gamma_0 + \gamma_1 \hat{\beta}_i + \epsilon_i$$

   Assume that $\epsilon_i \sim NID(0, \sigma^2_{\epsilon_i})$. This is referred to as the second-pass regression. The CAPM implies that $H_0: \gamma_0 = 0$ should not be rejected. The CAPM also implies that $H_0: \gamma_1 = \bar{R}_M - \bar{R}_f$ should not be rejected. These hypotheses ($H_0: \gamma_0 = 0$ and $H_0: \gamma_1 = \bar{R}_M - \bar{R}_f$) can be tested separately using t-tests or jointly using x^2 statistics. A rejection implies that the CAPM expected return-beta relationship can be rejected.

3. With the rolling method for T-period, the null hypothesis, $H_0: \gamma_1 = 0$, can also be tested as:

[10] $NID(0, \sigma^2_{\varepsilon_{i,t}})$ indicates a normally independently distributed with an unconditional mean of zero and an unconditional variance $\sigma^2_{\varepsilon_{i,t}}$.

$$\frac{\overline{\gamma_1}}{\boldsymbol{\sigma}_{\gamma_1}/\sqrt{\boldsymbol{T}}}$$

where σ_{γ_1} is the standard deviation of the $\sigma_{\gamma_1,t}$ and $\overline{\gamma_1}$ is the mean value of $\gamma_{1,t}$ for the period of T-period.

4. Arbitrage Pricing Theory (APT)

- In well-functioning capital markets, an investor should be rewarded for accepting the various risks associated with investing in a security. In this, and in the following section, we consider the type of asset pricing models that practitioners refer to as factor models. These models are so-called because they attempt to model each exposure to risk as a separate factor. In this type of models, **risks are also commonly referred to as "risk factors" or just "factors."**
- In this section, we discuss the **theoretical foundation for factor models** laid by (*or* based on) the Arbitrage Pricing Theory (APT). In the next section, we present the factors models used by practitioners**.**
- We can express an asset pricing model in general terms as

$$E(R_i) = f(F_1, F_2, F_3, \dots F_N)$$

where $E(R_i)$, F_k, and N denote the expected return on asset i, the k-th risk factor, and the number of risk factors, respectively.

- By investing in an asset other than risk-free securities, investors will demand a premium over the risk-free rate. That is, the expected return that an investor will demand is:

$$E(R_i) = R_f + risk\ premium$$

where R_f is the risk-free rate.

- The "risk premium," or excess return expected over the risk-free rate, depends on the risk factors associated with investing in the asset. Thus, we can rewrite the general form of the asset pricing model given above as

$$E(R_i) = R_f + g(F_1, F_2, F_3, \dots F_N)$$

- Risk factors can be divided into two general categories:
  - ✧ Systematic, or non-diversifiable risk factors
  - ✧ Unsystematic, or diversifiable risk factors

- As an alternative to the capital asset pricing model just discussed, Stephen Ross derived an asset pricing model based purely on arbitrage arguments called the Arbitrage Pricing Theory (APT).[11]
- This approach (the APT) postulates that an asset's expected return is influenced by a variety of risk factors, as opposed to just the "market risk", as suggested by the CAPM. The APT states that the return on a security **is linearly related** to some K risk factors. However, the **APT does not specify what these risk factors are**, but it is assumed that the relationship between asset returns and the risk factors is **linear**. **Moreover, as in CAPM, the APT assumes that unsystematic risk can be eliminated so that an investor is only compensated for accepting the systematic risk factors**.
- This arbitrage relies on a fundamental principle of finance called the law of one price. The law of one price implies that if the payoff of an asset can be created by a portfolio of assets, the price of the portfolio, and the price of the asset whose payoff it replicates, must be equal. That is:

$$E(R_i) = E(R_p)$$

- Let us now suppose that there are N securities and that each have a return distribution according to the factor structure:

$$R_i = \alpha_i + \sum_{i=k}^{K} \beta_{i,k} F_k + \varepsilon_i$$

where we also assume that

$$E(\varepsilon_i) = E(F_k) = 0$$

$$E(\varepsilon_i \varepsilon_j) = E(\varepsilon_i F_j) = E(F_i F_j) = 0$$

$$E(\varepsilon_i^2) = \sigma^2$$

[11] Ross, S.A. 1976. The arbitrage theory of capital asset pricing. Journal of Economic Theory 13, 343–362.

for all $i \neq j$.[12] Here F_k, $k = 1, 2, \ldots, K$ are the K factors common to all the securities, $\beta_{i,k}$ is the sensitivity of the i-th security to the k-th factor, and ε_i is the nonsystematic (idiosyncratic) return for the i-th security.

- In vector form, we can write the above relationship as:

$$\boldsymbol{R} = \boldsymbol{\alpha} + \boldsymbol{BF} + \boldsymbol{\varepsilon}$$

$$\boldsymbol{R} = \begin{bmatrix} R_1 \\ \vdots \\ R_N \end{bmatrix}, \ \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_N \end{bmatrix}, \ \boldsymbol{F} = \begin{bmatrix} F_1 \\ \vdots \\ F_N \end{bmatrix}, \ \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_N \end{bmatrix}, \ \boldsymbol{B} = \begin{bmatrix} \beta_{11} & \cdots & \beta_{1N} \\ \vdots & \ddots & \vdots \\ \beta_{N1} & \cdots & \beta_{NN} \end{bmatrix}$$

- Ross (1976) showed that in the absence of arbitrage, the following relationship holds:

$$E(R_i) = R_f + \sum_{i=k}^{K} \beta_{i,k}\left[E(F_k) - R_f\right]$$

  - ✧ This is referred to as the APT.[13]
  - ✧ The expression $\left[E(F_k) - R_f\right]$ is the excess return of the k-th systematic risk factor over the risk-free rate, and as such it can be thought of as the "**price**" (or **risk premium**) for the k-th **systematic risk factor**.
  - ✧ For example, k-th systematic risk factor could be related to size risk factor and thus $E(F_k) = E(R_S) - E(R_L)$, where $E(R_S)$ and $E(R_L)$ are the expected portfolio returns for the small and large-firm returns. Consequently, risk premium for the **systematic risk factor of size effect is** $\left[E(F_k) - R_f\right]$.

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[12] We choose to discuss a simplified version of the APT. Specifically, we assume that the nonsystematic errors are independent. In this case, returns are said to have a strict factor structure. Generalizations to an approximate factor structure, where the covariance matrix satisfies $cov(\boldsymbol{\varepsilon}) = \boldsymbol{\Omega}$ possible, but technical. We also omit discussing approximate factor structures with infinitely many assets, which is the framework where the APT was originally established.

[13] Strictly speaking, this is not fully correct. In particular, the equality holds in the mean-squared sense, when the number of assets approaches infinity. That is, the APT states that in the absence of asymptotic arbitrage opportunities (see Huberman, 1982):

$$\lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{K} \left[E(R_i) - R_f - \sum_{i=k}^{K} \beta_{i,k}\left[E(F_k) - R_f\right]\right]^2 = 0$$

Huberman, G., 1982.A simple approach to arbitrage pricing theory. Journal of Economic Theory 28, 183-191.

- ✧ The APT asserts that investors want to be compensated for all the risk factors that systematically affect the return of a security. The compensation is the sum of the products of each risk factor's systematic risk $\beta_{i,k}$ and the risk premium assigned to it by the financial market, $[E(F_k) - R_f]$.

- CAPM versus APT
  - ✧ As a matter of fact, it turns out that the CAPM is actually a special case of the APT. If the only risk factor in the APT is market risk, the APT reduces to the CAPM.
  - ✧ Both (CAPM and APT) say that investors are compensated for accepting all systematic risk, but not nonsystematic risk. The CAPM states that systematic risk is market risk, while the APT does not specify what the systematic risk factors are.
  - ✧ How do the two different models differ? Supporters of the APT argue that it has several major advantages over the CAPM.
    1. First, it makes less restrictive assumptions about investor preferences toward risk and return.
    2. The CAPM is a market equilibrium model, whereas APT relies upon the no-arbitrage condition. **While a market equilibrium implies no-arbitrage, no-arbitrage does not necessarily imply that the market is in equilibrium**.
    3. APT is a "relative" pricing model, in that it prices securities on the basis of the prices of other securities. Conversely, CAPM is an "absolute" pricing model that relates returns on the securities to the fundamental source of risk inherent in the portfolio of total wealth.
    4. In the APT, no assumptions are made about the distribution of asset returns besides the factor structure.

5. Since the APT does not rely on the identification of **the true market portfolio**, **the theory is potentially testable**.

- In a paper by Roll (1976), he demonstrates that the CAPM is not testable unless (1) the exact composition of the "true" market portfolio is known; and (2) the only valid test of the CAPM is to observe whether the ex ante true market portfolio is mean-variance efficient.[14]
- As a result of his findings, Roll states that he does not believe there ever will be an unambiguous test of the CAPM. He does not say that the CAPM is invalid, but rather that there is likely to be no unambiguous way to test the CAPM and its implications due to the nonobservability of the true market portfolio and its characteristics.

[14] Roll. R., 1976. A critique of the asset pricing theory's tests Part I: On past and potential testability of the theory. Journal of Financial Economics 4, 129-176

5. Factor Models

The APT provides theoretical support for an asset pricing model in which there is more than one risk factor. Consequently, models of this type are referred to as **multifactor risk models**. **Multifactor risk models** are applied to equity portfolio management, they provide the tools for measuring the risk profile of a portfolio relative to a benchmark, for constructing a portfolio relative to a benchmark, and controlling risk.

- Below, we provide a brief overview of the three different types of multifactor risk models used in equity portfolio management: statistical factor models, macroeconomic factor models, and fundamental factor models, and explain briefly how factor models can be estimated using linear regression.

5.1. Statistical Factor Models

- In a statistical factor model, historical (time-series and cross-sectional data) on stock returns are tossed into a statistical model. The goal of the statistical model is to explain best the observed stock returns with "factors" that are linear return combinations and uncorrelated with each other.
- This is typically accomplished by principal component analysis (PCA), three-pass filer regression (TPRF), among others.
- For example, suppose that monthly returns for 5,000 companies for ten years are computed. The goal of the statistical analysis is to produce factors that explain best the variance of the observed stock returns (e.g., PCA).
- In statistical factor models, the number of factors is normally much smaller than in macroeconomic and fundamental factor models **because the main purpose of statistical factor models is to reduce the dimensionality of obsessed risk factors.**
- Because of the problem of interpretation, it is difficult to use the factors from a statistical factor model for valuation, portfolio construction, and risk control.

5.2. Macroeconomic Factor Models

- In a macroeconomic factor model, the inputs to the model are historical stock returns and observable macroeconomic variables.
- The goal is to determine which macroeconomic variables are persistent in explaining historical stock returns. Those variables that consistently explain the returns then become the factors and are included in the model.
- An example of a proprietary macroeconomic factor model is the Burmeister, Ibbotson, Roll, and Ross (BIRR) model. In this model, there are five macroeconomic factors that **reflect unexpected (unanticipated) changes** in the following macroeconomic variables: **investor confidence** (confidence risk); **interest rates** (time horizon risk); **inflation** (inflation risk); **real business activity** (business cycle risk); and **a market index** (market risk).

5.3. Fundamental Factor Models

- One of the most well-known fundamental factor models is the Fama-French three-factor model. Besides the market portfolio, the other two factors are the "size factor" and the "book-to-market" factor.[15,16,17]

$$E(R_{i,t}) - R_{f,t} = \alpha_i + \beta_i[E(R_{M,t}) - R_{f,t}] + \gamma_i SMB_t + \rho_i HML_t + \varepsilon_{i,t}$$

- The Fama/French factors are constructed using the 6 value-weight portfolios formed on size and book-to-market. (See the description of the 6 size/book-to-market portfolios.)
- The size factor is the return on a zero-cost portfolio that is long on small-cap stocks and short on large-cap stocks (SMB_t). SMB (Small Minus Big) is the average return on the three small portfolios minus the average return on the three big portfolios,

$$SMB = 1/3\ (\text{Small Value} + \text{Small Neutral} + \text{Small Growth})$$
$$-1/3\ (\text{Big Value} + \text{Big Neutral} + \text{Big Growth}).$$

- The book-to-market (B/M) factor is the return on a zero-cost portfolio that is long on high B/M stocks and short on low B/M stocks (HML_t). HML (High Minus Low) is the average return on the two value portfolios minus the average return on the two growth portfolios,

$$HML = 1/2\ (\text{Small Value} + \text{Big Value})$$
$$-\ 1/2\ (\text{Small Growth} + \text{Big Growth})$$

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[15] Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33, pp 3–56.

[16] Fama and French (1996) sort firms into size deciles based on their market value at the end of June in year t. Based on the portfolio constructed at the end of June in year t, the portfolio returns for July of year t to June of t+1 can be used to calculate SMB. BE/ME portfolios are formed on BE/ME at the end of each June in year t. The BE used in June of year t is the book equity for the last fiscal year end in t-1. ME is price times shares outstanding at the end of December of t-1. Based on the portfolio formed at the end of each June in year t, the portfolio returns for July of year t to June of t+1 can be used to calculate HML.

[17] It turns out that the model explains the cross-sectional variation in stock returns fairly well.

- Besides the three Fama-French factors, typical fundamental factor models include other factors in the model. These factors are price/earnings ratios, estimated economic growth, trading activity, and liquidity. Other technical factors, such as volatility of total return and momentum, are also often included in modern factor models.

Chapter 5: Introduction in R Package "portsort"

- Portfolio sorts have been a key part of empirical financial research since the 1970s. Sorting procedures are extensively used to identify and explore relationships between expected returns and asset class characteristics.
- The portfolio sorting approach has become widely used and is currently the dominant approach in empirical finance to test for and establish **cross-sectional relationships** between expected asset returns and asset characteristics.
- The "portsort" R package includes functionality for both a conditional sort and an unconditional sort up to the two dimension and the third dimension will be included soon.
- After the sorting procedure has been conducted, there is further functionality to analyze **the turnover**, **relative frequency of the assets** in each sub-portfolio and the **mean sub-portfolio size**.
- This approach in empirical finance is to first sort assets based on some factor into multiple sub-portfolios at some formation point and then examine the out-of-sample performance of **these sub-portfolios**.
- Often, **long-short zero-cost portfolios** are constructed by initiating positions in the top and bottom sub-portfolios (in the case of a one-way sort).
- The output of the core sorting functions includes the out-of-sample sub-portfolio returns and **a list of the assets in each sub-portfolio**.
- From this, you can create trading strategies (long-short portfolios), pass the sub-portfolios to a regression analysis or conduct further analysis on the sub-portfolios themselves.
- The primary goal of the package is to offer academic researchers, students and practitioners an easy way to conduct the portfolio sort procedure. Nevertheless, the package only provides the returns of the equal-weighted portfolio.

- **Conditional Sorting Procedure**
  - Assume an asset manager has identified two stock factors which he/she thinks are drivers of expected stock returns due to prior statistical analysis.
  - A conditional double sort strongly depends on which factor the asset manager decides to sort the stocks by first.
  - Let factor one (Fa) be market capitalization and factor two (Fb) be the book-to-market ratio. Let us assume that the manager thinks that **market capitalization** has a stronger effect on expected returns.

1. At time t, stocks are first sorted into n sub-portfolios based on market capitalization. Instead of using quantile break-points, the manager decides to form 3 sub-portfolios (tercile portfolios). The result is 3 sub-portfolios sorted from low to high based on market capitalization (a basic univariate sort).
2. Now, within these 3 sub-portfolios, the manager decides to sort each sub-portfolio into a further m sub-portfolios based on the book-to-market ratio. Let us assume the manager decides on terciles for ease of interpretation. This means that the stocks within each market capitalization sub-portfolio are further sorted from low to high resulting in 3 book-to-market sorted sub-portfolios within the market capitalization sorted sub-portfolio tercile.
3. Overall, 9 sub-portfolios are formed (there are 3 book-to-market sorted portfolios within each market capitalization sorted tercile).

- **Unconditional Sorting Procedure**
  - In an unconditional sort, **the order of the sort does not matter**, e.g., the manager will get the same results regardless of the sort order.

1. Using the example above, Fa and Fb are sorted independently (at the same time) into terciles.
2. This results in "Sort 1" which is the stock universe sorted by market capitalization from low to high. These same stocks (regardless of the first sort) are sorted into book-to-market terciles called 'Sort 2'.
3. We now have two univariate sorts stored in tercile portfolios which are sorted independently of each other.
4. This is the crucial difference between a conditional and unconditional sort - the double or triple sorting procedure is based on the intersection between each sub-portfolio (in this case terciles).
5. For example, if stock 1, 23 and 43 is in the 1st market capitalization tercile, while stock 1, 20 and 43 is in the 1st book-to-market tercile, the 'L1-L1' sub-portfolio would only contain stock 1 and stock 43.

- **Impact on Sub-Portfolios**

The key difference between the two sorting procedures is that in a conditional sort, the factor which is sorted on first impacts the second sort. This means that the sub-portfolios formed within the portfolios formed by the first sort are "conditional" on the first factor. This may seem obvious, but will result in large differences between sub-portfolios formed in the unconditional sorting process. In the case of the conditional sort, the factor that is sorted on first will have a much greater influence on forward returns than the other factors that are subsequently sorted conditional on the first factor sort.

Chapter 6: Liquidity Portfolios

- As recent events in financial markets have demonstrated, illiquidity appears to be one of the most important market frictions that influence asset prices.
- On the theoretical side, the seminal studies of Constantinindes (1986), Heaton and Lucas (1996), Vayanos (1998), and Lo et al. (2004) address the impact of transaction costs on asset prices.[18]
- On the empirical side, research in asset pricing has shown that liquidity risk generates substantially higher cost of capital for the least liquid stocks. In particular, Amihud and Mendelson (1986) use the bid-ask spread to proxy for illiquidity and find a positive relation between illiquidity and stock returns. Following Amihud (2002), several formal asset pricing tests have shown that illiquidity is a priced factor (see for example, Pastor and Stambaugh, 2003; Acharya and Pedersen, 2005; Liu, 2006; Sadka, 2006; Korajczyk and Sadka, 2008).[19]
- The most commonly used measure in the recent empirical literature is the price impact ratio of Amihud (2002), which is defined as the average monthly ratio of daily absolute returns to daily trading volume in monetary terms (henceforth, $RtoV$) (see e.g. Acharya and Pedersen, 2005; Korajczyk and Sadka, 2008; among

[18] Constantinindes, G., 1986. Capital market equilibrium with transactions costs. Journal of Political Economy 94, 842–862. Heaton, J., Lucas, D., 1996. Evaluating the effects of incomplete markets on risk sharing and asset pricing. Journal of Political Economy 104, 443–497. Lo, A., Mamaysky, H., Wang, J., 2004. Asset prices and trading volume under fixed transactions costs. Journal of Political Economy 112, 1054–1090. Vayanos, D., 1998. Transaction costs and asset prices: a dynamic equilibrium model. Review of Financial Studies 11, 1–58.

[19] Acharya, V., Pedersen, L.H., 2005. Asset pricing with liquidity risk. Journal of Financial Economics 77, 375–410. Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. Journal of Financial Markets 5, 31–56. Amihud, Y., Mendelson, H., 1986. Asset pricing and the bid-ask spread. Journal of Financial Economics 17, 223–249. Liu, W., 2006. A liquidity-augmented capital asset pricing model. Journal of Financial Economics 82, 631–671. Korajczyk, R.A., Sadka, R., 2008. Pricing the commonality across alternative measures of liquidity. Journal of Financial Economics 87, 45–72. Pastor, L., Stambaugh, R.F., 2003. Liquidity risk and expected stock returns. Journal of Political Economy 111, 642–685. Sadka, R., 2006. Momentum and post-earnings-announcement drift anomalies: the role of liquidity risk. Journal of Financial Economics 80, 309–349.

others).

- Despite its popularity, Amihud's price impact ratio is not free of limitations. In the cross-sectional dimension, $RtoV$ ratio builds in a size effect because trading volume in monetary terms is by construction positively correlated with market capitalization and, as a result, it is by no means comparable across stocks with different market values. This shortcoming has been firstly highlighted by Cochrane (2005),[20] who argues that the $RtoV$ ratio is expected to be much higher for small capitalization stocks, forcing the conclusion that small capitalization stocks are more illiquid than big capitalization stocks. Another important limitation of $RtoV$ is that it neglects the trading frequency dimension of liquidity; this may be inappropriate given that actual stocks' trading frequency exhibits considerable time-series as well as cross-sectional variation (Datar et al., 1998).[21]
- In order to reduce the potential size and trading frequency effects on the proxy of illiquidity ratio, Florackis et al. (2011) contribute to the literature by proposing an alternative, more appropriate, price impact ratio defined as the average monthly ratio of daily absolute stock return to its turnover ratio (henceforth $RtoTR$).[22]
- This new measure is free of any size bias. This is because turnover does not necessarily exhibit an inherent size-related pattern (Brown et al., 2009). Therefore, the proposed measure is more suitable to insulate the price impact from the size effect. Furthermore, the role of trading frequency, which can be effectively approximated by turnover rate, is highlighted by the fundamental theoretical result of prior literature.

[20] Cochrane, J.H., 2005. Asset Pricing Program Review: Liquidity, Trading and Asset Prices. NBER Reporter.
[21] Datar, V.T., Naik, N.Y., Radcliffe, R., 1998. Liquidity and stock returns: an alternative test. Journal of Financial Markets 1, 205–219.
[22] Florackis, C., Gregoriou, A., Kostakis, A., 2011. Trading frequency and asset pricing on the London Stock Exchange: Evidence from a new price impact ratio. Journal of Banking and Finance 35, pp. 3335–3350.

- **Amihud's Price Impact Ratio**

  - The ratio proposed by Amihud (2002) is widely accepted as one of the most appropriate and straightforward price impact measures to construct. Formally, it is calculated by the following ratio:

$$RtoV_i = \frac{1}{D_{i,t}} \sum_{d=1}^{D_{i,t}} \frac{|R_{i,t,d}|}{V_{i,t,d}}$$

where $R_{i,t,d}$ and $V_{i,t,d}$ are, respectively, the return and monetary volume of stock i on day d at month t and $D_{i,t}$ is the number of valid observation days in month t for stock i.

  - Stocks with high stock price reaction to a dollar of trading volume yield higher returns. **Stocks with high $RtoV$ command high abnormal returns** because of illiquidity risk premium.

  - Amihud's ratio does not come without its shortcomings. Focusing on the properties of this measure from a cross-sectional asset pricing perspective, two major issues arise:

1. $RtoV$ is by no means comparable across stocks with different market capitalization and, therefore, carries a significant size bias. Small cap stocks are bound to exhibit lower trading volume (in monetary terms) than big cap stocks even when they exhibit the same turnover ratio. In other words, based on the $RtoV$ ratio, small cap stocks are automatically characterized as "illiquid" only due to their size.

2. $RtoV$ neglects investors' stock holding horizons. More specifically, even though this ratio attempts to proxy the cost of transacting in a rather intuitive way, it is uninformative with respect to the frequency at which this cost is incurred. As a result, Amihud's ratio implicitly assumes that trading frequency is similar across stocks, and it should not affect liquidity premia.

Table 1
Performance and characteristics of decile portfolios constructed on the basis of the Return-to-Volume (*RtoV*) price impact ratio. This table reports the characteristics of portfolios constructed on the basis of the Return-to-Volume (*RtoV*) price impact ratio. All stocks listed on the London Stock Exchange during the period January 1991–December 2008 are sorted at month *t-1* in ascending order according to their *RtoV* ratio and they are assigned to 10 portfolios. P1 is the decile portfolio containing the stocks with the lowest *RtoV* ratios and P10 is the decile portfolio containing the stocks with the highest *RtoV* ratios. The excess returns of these portfolios at month *t* are calculated (i.e. post-ranking returns). P10–P1 stands for the spread between P10 and P1. Portfolios are rebalanced on a monthly basis. EW returns correspond to the annualized average monthly returns of equal weighted portfolios. VW returns refer to the annualized average monthly returns of value weighted portfolios. MV is the average market value of the stocks in each portfolio (in £m), calculated as the share price multiplied by the number of outstanding ordinary shares. Price-to-book is the average ratio of the share price divided by the book value per share for the stocks in each portfolio. CAPM beta is the average stocks' beta in each portfolio calculated using a 36-month rolling window. The last column reports values for *t*-tests referring to the null hypothesis of no difference in means between portfolios' P10 and P1 characteristics.

	Decile portfolio											
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P10–P1	*t*-Test
Panel A. February 1991–December 2008												
EW returns (% p.a.)	0.852	0.260	1.670	1.421	3.277	5.247	5.735	6.905	7.344	14.242	13.390	3.093
VW returns (% p.a.)	−0.735	−0.564	0.359	−0.418	0.408	1.593	0.127	1.046	0.665	5.479	6.215	1.517
RtoV ratio ($\times 10^6$)	5.97E−04	2.19E−03	6.28E−03	1.63E−02	3.77E−02	7.48E−02	1.37E−01	2.54E−01	5.09E−01	2.50	2.50	15.015
MV (£m)	12005.541	2216.778	1028.539	588.152	366.601	258.182	187.943	136.944	101.645	64.412	−11941.13	−47.977
Price-to-book	3.414	3.300	3.437	3.458	3.368	3.343	3.203	2.902	2.813	2.542	−0.872	−14.784
CAPM beta	1.005	1.095	1.062	1.088	1.073	1.045	1.028	1.023	1.021	1.011	0.006	0.290
Panel B. February 1991–December 1999												
EW returns (% p.a.)	7.094	4.563	5.628	9.242	9.016	9.050	12.080	13.727	14.641	20.499	13.405	2.151
VW returns (% p.a.)	6.754	5.470	4.459	7.019	7.078	6.889	8.071	7.477	8.341	16.490	9.736	1.717
RtoV ratio ($\times 10^6$)	7.99E−04	2.32E−03	5.34E−03	1.11E−02	2.14E−02	3.97E−02	7.26E−02	1.32E−01	2.60E−01	1.15	1.15	16.46
MV (£m)	9514.635	2573.908	1300.101	776.216	496.561	351.341	255.295	187.895	140.512	86.815	−9427.820	−33.393
Price-to-book	3.202	3.146	3.232	3.263	3.369	3.494	3.336	2.998	2.907	2.389	−0.813	−10.097
CAPM beta	1.000	1.043	1.047	1.028	1.020	1.020	0.995	1.000	1.005	1.030	0.030	0.964
Panel C. January 2000–December 2008												
EW returns (% p.a.)	−5.332	−4.003	−2.251	−6.328	−2.408	1.479	−0.552	0.146	0.115	8.043	13.375	2.213
VW returns (% p.a.)	−8.155	−6.542	−3.703	−7.786	−6.201	−3.654	−7.744	−5.325	−6.940	−5.429	2.726	0.461
RtoV ratio ($\times 10^6$)	3.97E−04	2.07E−03	7.21E−03	2.13E−02	5.39E−02	1.09E−01	2.02E−01	3.74E−01	7.56E−01	3.84	3.84	14.37
MV (£m)	14473.384	1862.954	759.491	401.829	237.845	165.885	121.214	86.464	63.137	42.217	−14431.16	−63.304
Price-to-book	3.625	3.452	3.640	3.651	3.367	3.192	3.071	2.807	2.719	2.694	−0.931	−10.799
CAPM beta	1.009	1.146	1.077	1.148	1.125	1.069	1.061	1.046	1.037	0.992	−0.017	−0.585

- **A New Price Impact Ratio**

  - The shortcomings of Amihud's ratio motivate us to introduce a new price impact ratio. This is formally given by the following expression:

$$RtoTR_i = \frac{1}{D_{i,t}} \sum_{d=1}^{D_{i,t}} \frac{|R_{i,t,d}|}{TR_{i,t,d}}$$

where $RtoTR_{i,t}$ is the Turnover Ratio of stock i at day d and $D_{i,t}$ and $R_{i,t,d}$ are as previously defined.

  - Comparing our measure to $RtoV$, we essentially replace trading volume with turnover ratio in the denominator.
  - $RtoTR$ has a similar intuitive interpretation showing how much stock return responds to one percent of turnover rate. Most importantly, it is free of any size bias because there is no mechanical reason why big capitalization stocks should exhibit by construction higher turnover ratios. This fact renders it more appropriate for cross-sectional asset pricing studies and it enables us to **disentangle** price impact and size effects.
  - **Stocks with very high turnover ratios command higher abnormal returns because investors require the compensation on risk premiums for trading frequency. Hence, very low $RtoTR$ values, command high abnormal returns even if the price impact of this trading activity is relatively low.** These findings support the view that the **trading frequency effect overwhelmingly dominates the transaction cost effect in the determination of the corresponding premium**, leading to the result that low $RtoTR$ stocks outperform high $RtoTR$ stocks.

Table 2
Performance and characteristics of decile portfolios constructed on the basis of the Return-to-Turnover Rate (*RtoTR*) price impact ratio. This table reports the characteristics of portfolios constructed on the basis of the Return-to-Turnover Rate (*RtoTR*) price impact ratio. All stocks listed on the London Stock Exchange over the period January 1991–December 2008 are sorted at month *t-1* in ascending order according to their *RtoTR* ratio and they are assigned to 10 portfolios. P1 is the decile portfolio containing the stocks with the lowest *RtoTR* ratios and P10 is the decile portfolio containing the stocks with the highest *RtoTR* ratios. The excess returns of these portfolios at month *t* are calculated (i.e. post-ranking returns). P1–P10 stands for the spread between P1 and P10. Portfolios are rebalanced on a monthly basis. EW returns refer to the annualized average monthly returns of equal weighted portfolios. VW returns refer to the annualized average monthly returns of value weighted portfolios. MV is the average market value of the stocks in each portfolio (in £m), calculated as the share price multiplied by the number of outstanding ordinary shares. Price-to-book is the average ratio of the share price divided by the book value per share for the stocks in each portfolio. CAPM beta is the average stocks' beta in each portfolio calculated using a 36-month rolling window. The last column reports values for *t*-tests referring to the null hypothesis of no difference in means between portfolios' P1 and P10 characteristics.

	Decile portfolio											
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P1–P10	*t*-Test
Panel A. February 1991–December 2008												
EW returns (% p.a.)	13.902	6.357	8.220	2.886	6.032	4.659	3.080	0.645	2.556	−1.493	15.395	5.156
VW returns (% p.a.)	6.551	1.842	1.939	−2.538	−3.239	−1.684	−5.087	−6.487	−7.814	−5.918	12.469	3.896
RtoTR ratio	1.441	2.899	4.131	5.598	7.522	10.203	14.200	20.968	35.425	120.436	−118.996	−42.730
MV (£m)	2719.871	3499.827	3345.892	2313.492	1789.928	1231.784	835.937	579.423	386.894	253.512	2466.359	31.029
Price-to-book	3.660	3.315	3.219	3.246	3.281	3.203	3.083	2.948	2.956	2.875	0.785	9.966
CAPM beta	0.952	1.023	1.053	1.081	1.066	1.070	1.060	1.047	1.034	1.073	−0.122	−6.986
Panel B. February 1991–December 1999												
EW returns (% p.a.)	22.920	13.347	11.892	5.688	12.028	7.488	9.373	7.606	9.705	5.345	17.575	4.361
VW returns (% p.a.)	14.922	10.565	7.103	5.612	6.898	2.775	1.752	2.388	1.996	3.873	11.050	2.957
RtoTR ratio	1.435	3.104	4.467	5.838	7.447	9.648	12.951	18.734	31.132	103.487	−102.053	−33.174
MV (£m)	2109.883	2242.825	2292.215	2114.469	2246.136	1808.805	1224.806	773.567	530.643	342.754	1767.130	24.756
Price-to-book	3.935	3.231	3.156	3.076	3.181	3.137	3.083	2.935	2.919	2.759	1.175	14.296
CAPM beta	0.965	0.981	1.006	1.031	1.031	1.016	1.030	1.027	1.032	1.071	−0.106	−5.249
Panel C. January 2000–December 2008												
EW returns (% p.a.)	4.968	−0.568	4.582	0.110	0.093	1.856	−3.155	−6.251	−4.527	−8.267	13.235	2.999
VW returns (% p.a.)	−1.743	−6.800	−3.178	−10.612	−13.282	−6.103	−11.862	−15.280	−17.533	−15.617	13.875	2.668
RtoTR ratio	1.447	2.696	3.797	5.360	7.597	10.752	15.438	23.181	39.678	137.228	−135.781	−33.633
MV (£m)	3324.211	4745.190	4389.812	2510.672	1337.945	660.105	450.668	387.078	244.477	165.096	3159.115	29.928
Price-to-book	3.388	3.398	3.281	3.414	3.379	3.268	3.082	2.962	2.992	2.990	0.398	3.222
CAPM beta	0.938	1.065	1.099	1.132	1.101	1.123	1.089	1.067	1.036	1.075	−0.137	−4.835

- **Asset Pricing Tests: Time-Series Evidence**
  - We estimate the abnormal performance of the portfolios constructed on the basis of the $RtoV$ and $RtoTR$ ratios correspondingly using three asset pricing models. Firstly, we estimate Jensen's alpha from the CAPM:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,MKT} MKT_t + \varepsilon_{i,t}$$

where $r_{i,t}$ is the return of portfolio i in month t, $r_{f,t}$ is the risk-free rate for month t and MKT_t is the excess market portfolio return, $(r_{m,t} - r_{f,t})$, in month t.

  - Second, we compute Fama–French alpha, i.e. the intercept of the three-factor Fama and French (1993) model:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,MKT} MKT_t + \beta_{i,SMB} SMB_t + \beta_{i,HML} HML_t + \varepsilon_{i,t}$$

where SMB_t and HML_t stand for the size and value risk factors respectively.

  - Third, we estimate Carhart's alpha, i.e. the intercept of the four-factor Carhart (1997) model:[23]

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,MKT} MKT_t + \beta_{i,SMB} SMB_t + \beta_{i,HML} HML_t + \beta_{i,MON} MOM_t + \varepsilon_{i,t}$$

where MOM_t stands for the momentum risk factor.

[23] Carhart, M., 1997. On persistence in mutual fund performance. Journal of Finance 52, 57–82.

Table 3
Alphas of value-weighted portfolios sorted by the Return-to-Volume (*RtoV*) price impact ratio. This table reports the abnormal performance of 10 value-weighted portfolios constructed on the basis of the Return-to-Volume (*RtoV*) price impact ratio. All stocks listed on the London Stock Exchange during the period January 1991 to December 2008 are sorted in ascending order according to their *RtoV* ratio and they are assigned to 10 portfolios. P1 is the decile portfolio containing the stocks with the lowest *RtoV* ratios and P10 is the decile portfolio containing the stocks with the highest *RtoV* ratios. P10–P1 stands for the spread between P10 and P1. CAPM alpha is the annualized alpha estimate derived from the Capital Asset Pricing Model. Fama–French alpha is the annualized alpha estimate derived from the Fama–French three-factor model. Carhart alpha is the annualized alpha estimate from the Carhart four-factor model. *t*-Statistics are reported in parentheses. The last column reports the chi-square (χ^2) statistic of the Wald test referring to the null hypothesis that the 10 portfolios' alphas are jointly equal to zero; *p*-values are reported below the statistic.

	Decile portfolio											
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P10–P1	Chi-sq.
Panel A. February 1991–December 2008												
CAPM alpha (% p.a.)	0.90	1.37	2.27	1.64	2.51	3.54	2.05	2.85	2.60	7.49	6.58	19.29
	(1.27)	(0.85)	(1.16)	(0.72)	(1.04)	(1.34)	(0.78)	(0.91)	(0.84)	(2.10)**	(1.64)	(0.037)
Fama–French alpha (% p.a.)	1.49	0.18	0.54	−0.19	0.70	1.62	0.33	0.47	0.58	5.33	3.83	20.12
	(3.19)***	(0.15)	(0.45)	(−0.12)	(0.39)	(0.71)	(0.14)	(0.20)	(0.20)	(1.62)	(1.13)	(0.028)
Carhart alpha (% p.a.)	1.54	0.05	−0.02	0.08	0.93	1.22	0.36	0.61	1.99	5.38	3.84	17.82
	(3.19)***	(0.04)	(−0.02)	(0.04)	(0.50)	(0.50)	(0.15)	(0.24)	(0.65)	(1.49)	(1.02)	(0.058)
Panel B. February 1991–December 1999												
CAPM alpha (% p.a.)	1.58	−0.19	−1.38	0.84	0.47	0.93	1.97	1.99	2.25	10.46	8.88	22.73
	(1.33)	(−0.11)	(−0.63)	(0.30)	(0.18)	(0.27)	(0.61)	(0.50)	(0.57)	(2.18)**	(1.57)	(0.012)
Fama–French alpha (% p.a.)	1.23	0.60	−0.76	1.62	1.15	1.82	2.76	2.99	3.27	11.41	10.19	24.08
	(2.20)**	(0.05)	(−0.49)	(0.76)	(0.79)	(0.54)	(1.03)	(0.98)	(1.09)	(2.73)***	(2.33)**	(0.007)
Carhart alpha (% p.a.)	1.19	−1.07	−0.71	2.94	2.43	1.76	3.36	3.88	4.94	12.95	11.76	26.17
	(1.67)*	(−0.73)	(−0.45)	(1.25)	(1.64)	(0.49)	(1.22)	(1.14)	(1.60)	(2.70)***	(2.30)**	(0.003)
Panel C. January 2000–December 2008												
CAPM alpha (% p.a.)	0.33	4.17	6.65	3.38	4.82	6.86	2.19	4.40	3.18	5.27	4.94	23.38
	(0.47)	(1.56)	(2.16)**	(0.95)	(1.23)	(1.66)	(0.52)	(0.92)	(0.67)	(1.02)	(0.88)	(0.009)
Fama–French alpha (% p.a.)	1.42	−0.60	1.07	−1.81	−0.76	1.95	−2.54	−2.44	−2.28	−0.85	−2.27	19.24
	(3.10)***	(−0.47)	(0.62)	(−0.84)	(−0.24)	(0.62)	(−0.71)	(−0.68)	(−0.47)	(−0.21)	(−0.54)	(0.037)
Carhart alpha (% p.a.)	1.54	−0.29	0.58	−1.92	−0.78	1.50	−2.68	−2.51	−1.30	−1.26	−2.80	19.29
	(3.49)***	(−0.23)	(0.34)	(−0.90)	(−0.24)	(0.47)	(−0.71)	(−0.70)	(−0.26)	(−0.29)	(−0.63)	(0.037)

* The corresponding alpha coefficient is statistically significant at the 10% level.
** The corresponding alpha coefficient is statistically significant at the 5% level.
*** The corresponding alpha coefficient is statistically significant at the 1% level.

Table 4
Alphas of value-weighted portfolios sorted by the Return-to-Turnover Rate (*RtoTR*) price impact ratio. This table reports the abnormal performance of 10 value-weighted portfolios constructed on the basis of the Return-to-Turnover Rate (*RtoTR*) price impact ratio. All stocks listed on the London Stock Exchange during the period January 1991–December 2008 are sorted in ascending order according to their *RtoTR* ratio and they are assigned to 10 portfolios. P1 is the decile portfolio containing the stocks with the lowest *RtoTR* ratios and P10 is the decile portfolio containing the stocks with the highest *RtoTR* ratios. P1–P10 stands for the spread between P1 and P10. CAPM alpha is the annualized alpha estimate derived from the Capital Asset Pricing Model. Fama–French alpha is the annualized alpha estimate derived from the Fama–French three-factor model. Carhart alpha is the annualized alpha estimate from the Carhart four-factor model. *t*-Statistics are reported in parentheses. The last column reports the chi-square (χ^2) statistic of the Wald test referring to the null hypothesis that the 10 portfolios' alphas are jointly equal to zero; *p*-values are reported below the statistic.

	Decile portfolio											
	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P1–P10	Chi-sq.
Panel A. February 1991–December 2008												
CAPM alpha (% p.a.)	8.30	3.48	3.66	−0.70	−1.46	0.25	−3.08	−4.53	−5.60	−3.89	12.20	42.23
	(4.46)***	(2.31)**	(1.98)**	(−0.45)	(−0.59)	(0.10)	(−1.07)	(−1.53)	(−1.44)	(−1.32)	(3.65)***	(0.00)
Fama–French alpha (% p.a.)	7.79	3.10	3.87	−0.72	−1.73	−0.31	−3.82	−5.80	−7.74	−5.78	13.58	42.30
	(4.39)***	(2.03)*	(2.12)**	(−0.44)	(−0.71)	(−0.12)	(−1.52)	(−2.08)**	(−2.45)**	(−2.25)**	(4.22)***	(0.00)
Carhart alpha (% p.a.)	6.53	3.07	3.41	−0.99	−1.75	0.88	−4.22	−5.97	−6.85	−6.85	13.38	33.22
	(3.93)***	(1.91)*	(2.03)**	(−0.63)	(−0.72)	(0.34)	(−1.62)	(−1.96)**	(−2.16)**	(−2.26)**	(3.77)***	(0.00)
Panel B. February 1991–December 1999												
CAPM alpha (% p.a.)	8.87	5.55	1.22	0.01	1.85	2.66	−4.16	−2.62	−4.33	−2.67	11.54	68.31
	(6.17)***	(2.34)**	(0.68)	(−0.01)	(0.93)	(−1.20)	(−2.03)**	(−0.98)	(−0.84)	(−0.75)	(3.19)***	(0.00)
Fama–French alpha (% p.a.)	8.92	5.71	1.26	0.16	1.60	−2.67	−4.29	−2.42	−3.45	−1.95	10.87	65.07
	(6.16)***	(2.37)**	(0.73)	(0.10)	(0.87)	(−1.21)	(−2.11)**	(−0.98)	(−0.95)	(−0.74)	(3.82)***	(0.00)
Carhart alpha (% p.a.)	6.67	6.21	−0.49	−0.32	2.34	−1.82	−3.84	−1.49	−2.39	−2.66	9.33	39.99
	(4.83)***	(2.44)**	(−0.33)	(−0.20)	(1.24)	(−0.86)	(−1.78)*	(−0.52)	(−0.63)	(−1.01)	(3.11)***	(0.00)
Panel C. January 2000–December 2008												
CAPM alpha (% p.a.)	6.65	1.83	5.28	−0.83	−3.42	4.94	−5.36	−3.65	−5.18	−5.36	12.01	20.92
	(2.04)**	(0.95)	(2.02)**	(−0.30)	(−0.87)	(1.22)	(−0.16)	(−0.74)	(−0.83)	(−1.04)	(2.12)**	(0.02)
Fama–French alpha (% p.a.)	3.82	−0.14	6.86	−0.20	−6.03	2.87	−4.38	−8.48	−12.02	−11.16	14.98	27.26
	(1.36)	(−0.08)	(3.09)***	(−0.06)	(−1.65)	(0.72)	(−1.12)	(−1.82)*	(−2.42)**	(−2.46)**	(2.58)***	(0.00)
Carhart alpha (% p.a.)	3.46	−0.20	6.82	−0.46	−6.13	3.87	−4.79	−8.94	−11.34	−11.88	15.34	26.25
	(1.26)	(−0.11)	(3.06)***	(−0.16)	(−1.64)	(0.96)	(−1.15)	(−1.91)*	(−2.36)**	(−2.42)***	(2.61)***	(0.00)

* The corresponding alpha coefficient is statistically significant at the 10% level.
** The corresponding alpha coefficient is statistically significant at the 5% level.
*** The corresponding alpha coefficient is statistically significant at the 1% level.

- **Asset Pricing Tests: Cross-Sectional Evidence**

  - Given the time-series evidence for the abnormal performance of portfolios constructed on the basis of the $RtoTR$ price impact ratio (presented in the previous section), we are motivated to examine whether a price impact (PI) factor, once added to the traditional asset pricing model, helps explain the cross-sectional variation in these portfolios' returns. We define the PI factor as the spread return of value weighted portfolios P1–P10, constructed on the basis of the $RtoV$ ratio.

  - As a first step, we perform the traditional two-stage Fama and MacBeth (1973) regressions using information from all of the 10 $RtoTR$-sorted portfolios. This exercise will shed light on whether the risk premium associated with this factor is positive and statistically significant. The first stage of the Fama–MacBeth approach involves the estimation of betas from time-series regressions of excess portfolio returns on a set of risk factors. The most general among the model specifications we consider is the augmented Carhart model, given by:

$$r_{p,t} - r_{f,t} = \alpha_p + \beta_{p,MKT} MKT_t + \beta_{p,SMB} SMB_t + \beta_{p,HML} HML_t + \beta_{p,MON} MOM_t + \beta_{p,PI} PI_t + e_{p,t}$$

  where $r_{p,t}$ is the return of portfolio p at time t, $r_{f,t}$ is the monthly risk free rate for month t, and $MKT, SMB, HML, MOM\ and\ PI$ refer to the market, size, value, momentum and price impact factors, as previously defined. As it is common practice in the literature, in the first stage we estimate a time-series of factor loadings for each of the 10 portfolios using rolling windows of 60 monthly observations. However, for the early sample periods we start computing these betas when there are at least 36 observations.

- The second stage involves the estimation of monthly cross-sectional regressions of the 10 excess portfolios' returns on the corresponding betas that were estimated in the first-stage. The cross-sectional regression model is:

$$r_{p,t} - r_{f,t} = \lambda_0 + \lambda_{MKT}\hat{\beta}_{p,MKT} + \lambda_{SMB}\hat{\beta}_{p,SMB} + \lambda_{HML}\hat{\beta}_{p,HML} + \lambda_{MOM}\hat{\beta}_{p,MOM} + \lambda_{PI}\hat{\beta}_{p,PI} + w_{p,t}$$

  where now lambdas represent the risk premium coefficients associated with each beta. The main hypothesis we test is whether the time-series average of the estimated coefficients λ_{PI} is positive and statistically significant, which is perceived as evidence that the PI factor is priced.

- Table 6 presents the estimated lambda coefficients for the 10 value-weighted portfolios sorted according to the $RtoTR$ ratio. For comparability purposes, we report results for the three asset pricing models that are most commonly used (CAPM, Fama–French and Carhart models), once they have been augmented by the PI factor. Panel A reports the results for the unrestricted model where λ_0 can take any value. The results support the existence of a large positive and statistically significant premium (λ_{PI}).

- Moreover, the fact that none of the models' intercept (λ_0) is found to be statistically significant points to the conclusion that the PI augmented models are not misspecified.

Table 6
Cross-sectional asset pricing tests. This table reports the mean of the risk premium coefficients λ_i estimated from the second stage of the Fama and MacBeth (1973) methodology. This involves monthly cross-sectional regressions of 10 value-weighted excess portfolio returns on the corresponding risk factor loadings estimated in the first stage of the procedure (see Section 4.2 for details). These 10 portfolios have been constructed by sorting all listed stocks in ascending order according to their Return-to-Turnover Rate (*RtoTR*) ratio. The sample period is February 1991–December 2008. The risk factor loadings used as regressors in the second stage are betas estimated from time-series regressions of excess portfolio returns on several risk factors using rolling windows of 60 monthly observations. However, for early sample periods we start computing betas when there are at least 36 observations. The reported results refer to augmented versions of the Capital Asset Pricing Model, the Fama–French model and the Carhart model (see Eq. (8)). The additional factor is the price impact (PI) factor that goes long the value-weighted decile portfolio P1 containing the stocks with the lowest *RtoTR* ratios and goes short the value-weighted decile portfolio P10 containing the stocks with the highest *RtoTR* ratios. In Panel A we report results from the unrestricted model, while in Panel B we report results from the restricted model where λ_0 is set equal to its zero theoretical value. Fama–MacBeth *t*-statistics are reported in parentheses. We also report in brackets *t*-statistics using Shanken (1992)-corrected standard errors to mitigate the error-in-variables bias. The pre-last column in Panel A reports the adjusted *R*-squared coefficient for each model, while the last column reports the increase in the adjusted *R*-squared coefficient due to the addition of the PI factor to the plain version of the model.

	λ_0	λ_{MKT}	λ_{SMB}	λ_{HML}	λ_{MOM}	λ_{PI}	Adj. R^2	ΔR^2
Panel A. Unrestricted model								
1. CAPM & PI	0.99	−1.24				0.81		
	(1.15)	(−1.24)	–	–	–	(2.34)**	0.14	0.06
	[1.08]	[−1.18]				[2.30]**		
2. Fama–French & PI	2.02	−2.41	0.21	−0.32		0.97		
	(1.63)	(−1.70)*	(1.24)	(−0.44)	–	(2.96)***	0.20	0.05
	[1.33]	[−1.40]	[1.06]	[−0.36]		[2.82]***		
3. Carhart & PI	1.74	−2.16	0.29	−0.59	0.25	0.94		
	(1.19)	(−1.32)	(1.02)	(−0.76)	(0.68)	(2.94)***	0.25	0.05
	[0.96]	[−1.07]	[0.84]	[−0.62]	[0.56]	[2.82]***		
Panel B. Restricted model: λ_0 is set equal to zero								
4. CAPM & PI	Set to 0	−0.19				0.81		
		(−0.62)	–	–	–	(2.25)**	–	–
		[−0.62]				[2.23]**		
5. Fama–French & PI	Set to 0	−0.21	0.08	0.29		0.93		
		(−0.68)	(0.57)	(0.50)	–	(2.79)***	–	–
		[−0.67]	[0.55]	[0.48]		[2.75]***		
6. Carhart & PI	Set to 0	−0.19	−0.13	0.18	0.63	0.92		
		(−0.62)	(−0.75)	(0.32)	(2.22)**	(2.83)***	–	–
		[−0.61]	[−0.68]	[0.28]	[1.97]**	[2.74]***		

* The coefficient is statistically significant at the 10% level.
** The coefficient is statistically significant at the 5% level.
*** The coefficient is statistically significant at the 1% level.

- **Preliminary Evidence of the Taiwan Stock Market**

  - Results for $RtoTR$

	RT1	RT2	RT3	RT4
Annualized Return	0.1701	0.1408	0.1060	0.0986
Annualized Std Dev	0.2010	0.1307	0.1214	0.1301
Annualized Sharpe (Rf=0%)	0.8463	1.0770	0.8730	0.7581

  - Results for $RtoV$

	PI1	PI2	PI3	PI4
Annualized Return	0.0880	0.0917	0.1285	0.2088
Annualized Std Dev	0.1551	0.1337	0.1380	0.1573
Annualized Sharpe (Rf=0%)	0.5674	0.6857	0.9310	1.3275

- References

Florackis, C., Gregoriou, A., Kostakis, A., 2011. Trading frequency and asset pricing on the London Stock Exchange: Evidence from a new price impact ratio. Journal of Banking and Finance 35, pp. 3335–3350.

Chapter 7: Return Seasonalities

- Heston and Sadka (2008) present a new pattern in the cross-section of expected stock returns. Stocks tend to have relatively high (or low) returns every year in the same calendar month. This pattern explains an economically and statistically significant magnitude of the cross-sectional variation in average stock returns. The main evidence presented by Heston and Sadka (2008) is shown as follows:

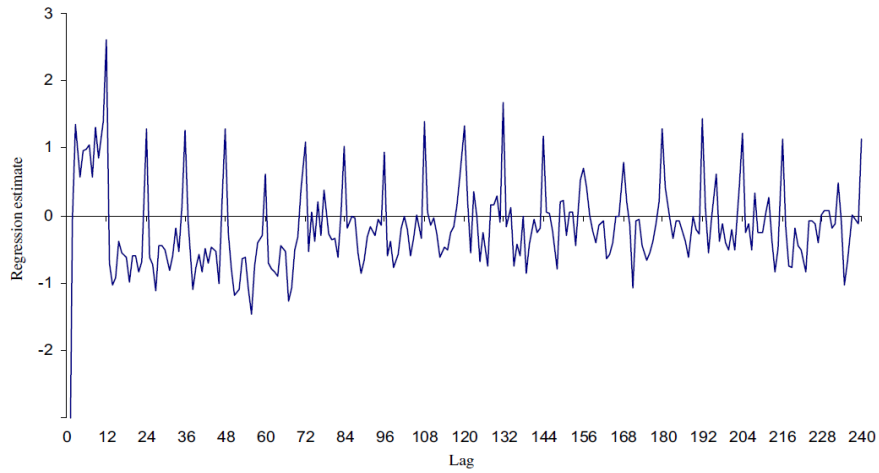


Table 2
Returns of strategies based on past performance

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2–5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The average monthly returns of the various trading strategies for the period January 1965 through December 2002 (456 months) are reported below (in percent), as well as the corresponding *t*-statistics (in brackets). The analysis uses NYSE- and AMEX-listed stocks.

Strategy	1 (Losers)	2	3	4	5	6	7	8	9	10 (Winners)	10-1
Year 1											
All	0.64	0.90	1.01	1.08	1.17	1.25	1.36	1.52	1.73	2.10	1.46
	[1.65]	[3.12]	[3.97]	[4.54]	[5.11]	[5.50]	[5.77]	[6.05]	[6.31]	[6.39]	[5.58]
Annual	0.82	0.97	1.06	1.11	1.21	1.27	1.38	1.48	1.60	1.97	1.15
	[2.53]	[3.66]	[4.36]	[4.77]	[5.20]	[5.56]	[5.78]	[5.82]	[5.77]	[5.80]	[7.60]
Nonannual	0.80	1.05	1.11	1.09	1.19	1.20	1.28	1.44	1.65	1.97	1.17
	[1.99]	[3.59]	[4.23]	[4.53]	[5.23]	[5.26]	[5.48]	[5.88]	[6.08]	[6.13]	[4.20]
Years 2–5											
All	1.91	1.45	1.35	1.27	1.29	1.26	1.28	1.16	1.12	0.84	−1.07
	[5.28]	[5.25]	[5.50]	[5.59]	[5.68]	[5.39]	[5.25]	[4.50]	[3.99]	[2.57]	[−5.02]
Annual	1.10	1.03	1.05	1.09	1.26	1.24	1.37	1.49	1.57	1.77	0.67
	[3.48]	[3.85]	[4.37]	[4.74]	[5.44]	[5.27]	[5.74]	[5.83]	[5.71]	[5.44]	[5.35]
Nonannual	1.99	1.59	1.41	1.30	1.29	1.26	1.20	1.14	1.01	0.74	−1.25
	[5.42]	[5.67]	[5.78]	[5.74]	[5.71]	[5.36]	[4.99]	[4.44]	[3.59]	[2.28]	[−5.60]
Years 6–10											
All	1.56	1.47	1.38	1.31	1.21	1.21	1.11	1.11	1.18	1.17	−0.39
	[5.17]	[5.76]	[5.97]	[5.82]	[5.40]	[5.27]	[4.71]	[4.46]	[4.31]	[3.68]	[−3.32]
Annual	1.04	0.93	1.01	1.12	1.17	1.14	1.35	1.44	1.55	1.72	0.68
	[3.44]	[3.68]	[4.27]	[4.90]	[5.24]	[5.10]	[5.90]	[6.02]	[5.92]	[5.60]	[6.15]
Nonannual	1.67	1.56	1.46	1.35	1.26	1.14	1.10	1.02	1.05	1.12	−0.55
	[5.47]	[6.14]	[6.23]	[5.96]	[5.61]	[5.02]	[4.64]	[4.09]	[3.91]	[3.51]	[−4.62]
Years 11–15											
All	1.35	1.16	1.10	1.20	1.20	1.22	1.23	1.26	1.27	1.33	−0.02
	[4.67]	[4.76]	[4.87]	[5.46]	[5.60]	[5.51]	[5.54]	[5.33]	[5.01]	[4.69]	[−0.17]
Annual	1.01	1.04	1.02	1.11	1.12	1.29	1.30	1.34	1.45	1.66	0.66
	[3.61]	[4.25]	[4.52]	[5.13]	[5.24]	[5.87]	[5.83]	[5.77]	[5.81]	[5.97]	[6.43]
Nonannual	1.44	1.22	1.22	1.20	1.22	1.15	1.24	1.21	1.19	1.25	−0.19
	[4.98]	[5.02]	[5.28]	[5.37]	[5.61]	[5.37]	[5.56]	[5.07]	[4.71]	[4.45]	[−1.77]
Years 16–20											
All	1.50	1.17	1.16	1.10	1.13	1.20	1.17	1.13	1.23	1.19	−0.31
	[5.48]	[4.89]	[5.21]	[5.11]	[5.29]	[5.54]	[5.43]	[5.01]	[5.14]	[4.65]	[−2.82]
Annual	1.00	1.00	1.03	1.11	1.12	1.15	1.22	1.31	1.37	1.53	0.52
	[3.78]	[4.24]	[4.61]	[5.02]	[5.14]	[5.36]	[5.64]	[5.95]	[5.86]	[5.84]	[4.58]
Nonannual	1.54	1.28	1.15	1.22	1.15	1.11	1.08	1.20	1.09	1.15	−0.39
	[5.57]	[5.38]	[5.16]	[5.60]	[5.37]	[5.17]	[4.95]	[5.37]	[4.61]	[4.49]	[−3.35]

- Based on the Evidence by Heston and Sadka (2008), Keloharju et al. (2016) extend the sample period of the US stocks to re-examine the effect of return seasonalities. The following figure shown by Keloharju et al. (2016) plots the average coefficients from cross-sectional regressions of monthly stock returns against one-month returns of the same stock at different lags.

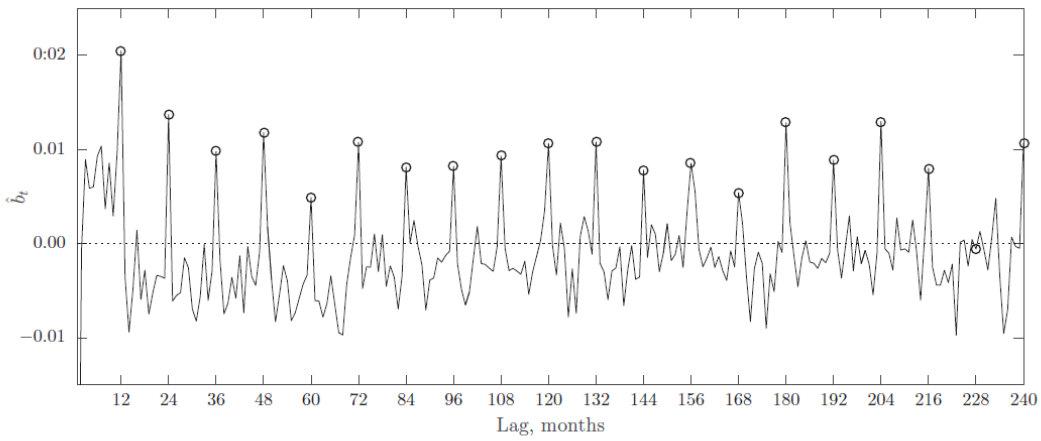


Figure 1. Seasonalities in individual stock returns. This figure plots slope coefficients from univariate Fama and MacBeth (1973) regressions of month t returns against month $t-k$ returns, $r_{i,t} = a_t + b_t\, r_{i,t-k} + e_{i,t}$, with k ranging from 1 to 240 months. The circles denote estimates at annual lags. The regressions use monthly data from January 1963 through December 2011 for NYSE, Amex, and NASDAQ stocks.

- The figure is an updated version of that in Heston and Sadka (2008). The figure shows that the pattern is not the momentum up to the one-year mark or the long-term reversals that follow, but rather the positive peaks can be found for every annual lag.
- More importantly, return seasonalities are not confined to individual stocks or to the monthly frequency. Keloharju et al. (2016) show that seasonality strategies that trade well-diversified portfolios formed by characteristics such as size and industry are about as profitable as those that trade individual stocks. Seasonalities also exist in the returns of commodities and country portfolios and at the daily frequency.

- The empirical findings of raw returns and FF-Alpha for the US stocks and US factors are presented as follows:

Table I

Seasonalities in Individual-Stock and Portfolio Returns

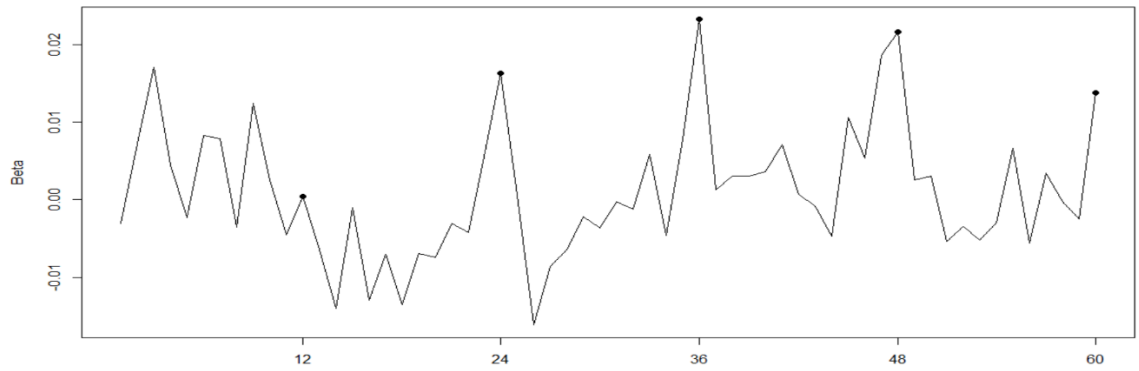
We form 10 portfolios by sorting on size, book-to-market, momentum, and gross profitability; 11 portfolios by sorting on dividend-to-price and earnings-to-price with zero-dividend and negative-earnings stocks in their own portfolios; and 17 portfolios for Fama-French industries. Momentum is the prior one-year return skipping a month and gross profitability is the ratio of revenue minus cost of goods sold to total assets. Except for momentum, we rebalance the portfolios every June and compute value-weighted returns from July of year t to June of year $t+1$; momentum portfolios are rebalanced monthly. The row "Composite" combines the 58 size, value, momentum, dividend-to-price, and industry portfolios. Same-month sorts are based on average same-calendar-month returns over the prior 20 years. Other-month sorts are based on average other-calendar-month returns over the same period, skipping months $t-11$ through $t-1$. The individual stock strategy trades the top and bottom deciles. Strategies based on size, value, momentum, profitability, dividend-to-price, and earnings-to-price trade the top and bottom portfolios. The industry strategy trades the top two and bottom two portfolios. The composite strategy trades the top six and bottom six portfolios. FF3 α is the Fama and French (1993) three-factor model alpha. We use monthly data from January 1963 through December 2011 except for the gross profitability and earnings-to-price portfolios, for which the data begin in January 1973.

	Monthly Returns and Alphas (%)				t-Values			
	Sort by				Sort by			
	Same-Month Return		Other-Month Return	Same − Other	Same-Month Return		Other-Month Return	Same−Other
Set of Assets	Avg	FF3 α			Avg	FF3 α		
All Months								
Individual stocks	1.19	1.22	−0.96	2.16	6.27	6.45	−4.12	7.94
Portfolios								
Size	1.35	1.27	−0.94	2.29	6.64	6.33	−3.94	6.53
Value	0.47	0.24	0.23	0.24	2.76	1.66	1.27	1.05
Momentum	1.83	1.97	1.89	−0.07	5.77	6.15	5.39	−0.20
Profitability	0.20	0.32	0.33	−0.13	1.21	1.91	1.91	−0.64
Dividend-to-price	0.48	0.59	−0.50	0.99	3.12	3.76	−3.08	3.97
Earnings-to-price	0.57	0.39	0.12	0.45	2.93	2.06	0.50	1.35
Industry	0.70	0.80	−0.81	1.51	3.79	4.22	−4.32	5.40
Composite	1.30	1.30	0.01	1.29	8.65	8.77	0.06	6.31
Excluding January								
Individual stocks	0.99	1.07	−0.56	1.56	5.45	6.14	−2.51	6.28
Portfolios								
Size	0.73	0.77	−0.55	1.28	3.88	4.07	−2.15	3.61
Value	0.23	0.17	0.16	0.07	1.55	1.17	0.85	0.33
Momentum	1.82	2.01	2.30	−0.48	6.08	6.61	5.81	−1.77
Profitability	0.21	0.34	0.36	−0.15	1.31	2.14	2.03	−0.74
Dividend-to-price	0.34	0.47	−0.31	0.65	2.09	2.98	−1.88	2.60
Earnings-to-price	0.30	0.27	0.18	0.12	1.73	1.51	0.68	0.33
Industry	0.59	0.72	−0.69	1.28	3.11	3.76	−3.66	4.74
Composite	0.90	1.02	0.28	0.62	6.95	7.72	1.81	3.68

- Based on the findings by Heston and Sadka (2008), Heston and Sadka (2010) extend the results to international evidence of the stock markets. As suggested by Heston and Sadka (2010), stocks that outperform the domestic market in a particular month continue to outperform the domestic market in that same calendar

month for up to 5 years. The pattern appears in Canada, Japan, and 12 European countries. Global trading strategies based on seasonal predictability outperform similar nonseasonal strategies by over 1% per month.

- However, the data used in Heston and Sadka (2010) are from developed markets. The question of whether emerging markets exhibit similar return patterns has not yet been thoroughly investigated, and the few existing studies on international markets focus on stock market indexes rather than individual stocks.
- Lin et al. (2018) investigate whether the same-month return pattern exists in emerging markets and whether this pattern differs from that observed in advanced markets. Specifically, Lin et al. (2018) extend the studies of stock return seasonality by Heston and Sadka (2008, 2010) to a comprehensive sample of 42 international markets, including 21 advanced markets and 21 emerging markets.
- The empirical results by Lin et al. (2018) show a large variation in stock seasonality across markets and suggest that this phenomenon exists primarily in advanced markets. A winner–loser portfolio approach shows that return seasonality is economically significant in advanced markets but not in emerging markets (see Table 4).
- Particularly, Lin et al. (2018) show that the effect of return seasonalities is insignificant in the Taiwan stock market.
- Lin et al. (2018) only use 454 firms to explore the effect of return seasonalities in the Taiwan stock market. The insignificant effect of return seasonalities for the Taiwan stock market might be related to the survivorship bias in sample selection.
- We collect a comprehensive data (listing and delisting) for 961 firms in the Taiwan stock market and re-examine the effect of return seasonalities. The results are presented as follows.



Same-Month

	P1	P2	P3	P4	P5	P6	P7
Annualized Return	-0.0132	0.0531	0.0451	0.0603	0.0786	0.0918	0.1085
Annualized Std Dev	0.1917	0.1670	0.1568	0.1437	0.1353	0.1354	0.1382
Annualized Sharpe (Rf=0%)	-0.0686	0.3181	0.2878	0.4198	0.5811	0.6781	0.7852

	P8	P9	P10
Annualized Return	0.1189	0.0947	0.1287
Annualized Std Dev	0.1397	0.1474	0.1739
Annualized Sharpe (Rf=0%)	0.8511	0.6423	0.7402

Other-Month

	P1	P2	P3	P4	P5	P6	P7
Annualized Return	0.0641	0.0872	0.0785	0.0854	0.0879	0.0922	0.0935
Annualized Std Dev	0.1794	0.1628	0.1590	0.1515	0.1381	0.1424	0.1407
Annualized Sharpe (Rf=0%)	0.3572	0.5356	0.4934	0.5638	0.6368	0.6480	0.6645

	P8	P9	P10
Annualized Return	0.0799	0.0775	0.0461
Annualized Std Dev	0.1340	0.1485	0.1553
Annualized Sharpe (Rf=0%)	0.5964	0.5220	0.2969

- Hirshleifer et al. (2020) and Keloharju et al. (2021) provide the possible behavioral biases to explain the return seasonalities. Can we provide other reasons to explain the same month effects? E.g., Peak season effect for the sale growth.

References:

Heston, S.L., Sadka, R., 2008. Seasonality in the cross-section of stock returns. Journal of Financial Economics 87, 418–445.

Heston, S.L., Sadka, R., 2010. Seasonality in the cross section of stock returns: The international evidence. Journal of Financial and Quantitative Analysis 45, 1133-1160.

Hirshleifer, D., Jiang, D., DiGiovanni, Y.M, 2020. Mood beta and seasonalities in stock returns. Journal of Financial Economics 137, 272–295.

Keloharju, M. , Linnainmaa, J.T. , Nyberg, P. , 2016. Return seasonalities. Journal of Finance 71, 1557–1590 .

Keloharju, M., Linnainmaa, J.T., Nyberg, P., 2021.Are return seasonalities due to risk or mispricing? Journal of Financial Economics 139 (2021) 138–161.

Li, F., Zhang, H., Zheng, D., 2018. Seasonality in the cross section of stock returns: Advanced markets versus emerging markets. Journal of Empirical Finance 49, 263–281.

Chapter 8: The Global Latent Factor and International Index Futures Returns Predictability

- A voluminous literature has explored aggregate stock market return predictability (e.g., Lo & MacKinlay, 1990; Rapach, Strauss, & Zhou, 2013; Baetje, 2018).[24]
- Rapach et al. (2013) find lagged U.S. returns can be used to forecast returns in numerous non-U.S. industrialized nations, but that lagged non-U.S. returns have limited forecasting capacity to U.S. returns.
- Rather than using the stock market indices, Tse (2018)[25] uses the index futures markets to re-examine the lead-lag relationship among industrialized countries. Specifically, Tse (2018) shows the forecasting capability for lagged U.S. returns for the returns of other countries, and further documents illustrating that some countries may bring influence on others in comparison to that from the United States.
- The issue of the estimation of the latent factor has attracted considerable attention in the literature. Ahn and Perez (2010)[26] propose a new version of generalized method of moments (GMM) estimators for latent factors within a linear factor model. Applying the estimators to international stock markets, Ahn and Perez (2010) find that one strong global factor explains about international stock returns.
- He, Zhu, and Zhu (2015)[27] indicate that the Fama–French pricing model with dynamic latent factors extracted by Kalman filter improves cross-sectional

[24] Lo, A., & MacKinlay, C. (1990). When are contrarian profits due to stock market overreaction? Review of Financial Studies, 3, 175–205. Rapach, D. E., Strauss, J. K., & Zhou, G. (2013). International stock return predictability: What is the role of the United States. Journal of Finance, 68, 1633–1662. Baetje, F. (2018). Does a lot help a lot? Forecasting stock returns with pooling strategies in a data-rich environment. Journal of Forecasting, 37, 37–63.

[25] Tse, Y. (2018). Return predictability and contrarian profits of international index futures. Journal of Futures Markets, 38, 788–803.

[26] Ahn, S.C., & Perez, M.F. (2010). GMM estimation of the number of latent factors: With application to international stock markets. Journal of Empirical Finance, 17, 783–802.

[27] He, Z., Zhu, J., Zhu, X. (2015). Dynamic factors and asset pricing: International and further U.S. evidence. Pacific-Basin Finance Journal, 32, 21–39

explanatory capacity of the Fama-French three factor model, and further point out that such the Fama–French pricing model has a more accurate capacity than other models for predicting portfolio returns.

- Chen (2018)[28] uses a Bayesian dynamic latent factor model for categorizing global, regional, and idiosyncratic country-specific factors out of stock market returns. Empirical works point out that variations of stock market returns from global market based on international stock market co-movements show a significant global factor. Lin (2018) employs the $TPRF$ suggested by Kelly and Pruitt (2015) to extract a latent factor from technical indicators and then uses the latent factor to predict aggregate stock returns.
- Lin (2018)[29] finds that the latent factor obtained from the $TPRF$ displays vital forecasting capacity of in-sample and out-of-sample criteria in both statistic and economic formats, outperforming other competing models, such as the principal component analysis (PCA).
- Motivated by Tse (2018) and Lin (2018), this paper examines the forecasting performance of international index futures returns based on the global latent factor extracted by the $TPRF$ with lagged international index futures returns. The results by Tse (2018) present that lagged monthly returns for several countries have return predictability comparable to those in the U.S. for the 1988–2016 period. This finding implies a specific global latent factor for each index futures in predicting its excess returns.
- As indicated by Lin (2018), econometrically, the $TPRF$ method works efficiently as it can identify latent factors that influence the future stock returns while

[28] Chen, P. (2018). Understanding international stock market comovements: A comparison of developed and emerging markets. International *Review of Economics and Finance*, 56, 451–464.
[29] Lin, Q. (2018). Technical analysis and stock return predictability: An aligned approach. Journal of Financial Markets, 38, 103–123.

discarding idiosyncratic error components that are irrelevant for forecasting (see also Kelly & Pruitt, 2015). This paper employs the $TPRF$ to obtain a specific global latent factor from lagged international index futures returns for each index futures and then uses the specific global latent factor to predict its corresponding index futures returns. For comparison purposes, this paper also uses the average of cross-sectional weekly excess returns of the index futures and the PCA to proxy the global latent factor. Based on our best knowledge and research, this shall be the first paper employing the $TPRF$ to explore the predictive power with international index futures data.

- **The $TPRF$ regression**

Following the variable definition and estimation procedure suggested by Kelly and Pruitt (2015), the $TPRF$ approach is introduced as follows. Let $ER_{i,w} = R_{i,w} - RF_w$, where $R_{i,w}$ is calculated as $ln\,(P_{i,w}) - ln\,(P_{i,w-1})$, $P_{i,w}$ is the futures prices on week w for country i, and RF_w is the U.S. weekly risk-free interest rate on week w. Suppose that $ER_{i,w}$ is a target variable to be forecast, where $ER_{i,w}$ is the excess returns on week w for country i $(i = 1, \ldots, 9)$. For each individual predictor in $\{ER_{i,w-1}\}_{i=1}^{9}$, the first pass of the $TPRF$ is to run a time series regression of $ER_{i,w-1}$ on $ER_{i,w}$, for nine countries,

$$ER_{i,w-1} = \pi_{i,0,TPRF} + \pi_{i,1,TPRF} ER_{i,w} + e_{i,w} \tag{1}$$

where $e_{i,w}$ is an error term, yielding $\hat{\pi}_{i,1,TPRF}$. The second pass for the cross-sectional regression of $ER_{i,w-1}$ on $\hat{\pi}_{i,1,TPRF}$ is estimated for each week as follows:

$$ER_{i,w-1} = \tau_{0,w-1} + GLF_{TPRF_i,w-1} \hat{\pi}_{i,1,TPRF} + \varepsilon_{i,w-1} \tag{2}$$

where $\varepsilon_{i,w-1}$ is an error term and the regression coefficient $GLF_{TPRF_i,w-1}$ is the latent factor extracted from the nine individual predictors using the $TPRF$. Finally,

in the third stage, the estimated latent factor ($\widehat{GLF}_{TPRF_{i,w-1}}$) is used to obtain the factor loading as follows:

$$ER_{i,w} = \beta_{0,i,TPRF} + \beta_{1,i,TPRF}\widehat{GLF}_{TPRF_{i,w-1}} + \eta_{i,w} \quad (3)$$

where $\eta_{i,w}$ is an error term and $\widehat{GLF}_{TPRF_{i,w-1}}$ is the estimated regression coefficient of Eq. (2), yielding $\hat{\beta}_{0,i,TPRF}$ and $\hat{\beta}_{1,i,TPRF}$.

Based on estimated coefficients, $\hat{\beta}_{0,i,TPRF}$ and $\hat{\beta}_{1,i,TPRF}$, from Eq. (3), the out-of-sample forecast value of $\widehat{ER}_{i,w+1}$ can be obtained as follows:

$$\widehat{ER}_{i,w+1} = \hat{\beta}_{0,i,TPRF} + \hat{\beta}_{1,i,TPRF} \times \widehat{GLF}_{TPRF_{i,w}} \quad (4)$$

where the new global latent factor on week w ($\widehat{GLF}_{TPRF_{i,w}}$) is estimated by the cross-sectional regression for $ER_{i,w}$ ($i = 1, \ldots, 9$) on week w regressing on $\hat{\pi}_{i,1,TPRF}$. $\hat{\pi}_{i,1,TPRF}$ is obtained from Eq. (1) and $\hat{\beta}_{0,i,TPRF}$ as well as $\hat{\beta}_{1,i,TPRF}$ are estimated from Eq. (3).

- **The competing models**

This paper introduces a host of competing models, including *prevailing mean* (PM) *forecasts*, the AR(1) model, predictive regression models with using lagged US returns, and the predictive regression models estimated *via* the global latent factor measured by the average of cross-sectional weekly excess returns and principal component (PC) approach, among others.

➢ *Prevailing mean* (PM)

This paper uses *prevailing mean* (PM) *forecasts* as the baseline model (see Goyal & Welch 2008; Rapach, Strauss, Tu, and Zhou, 2019) and compares the forecasting performance with that of predictive regression models.

➢ *AR* (1) model

The *AR* (1) model is expressed as follows:

$$ER_{i,w} = \beta_{0,i,AR} + \beta_{1,i,AR}ER_{i,w-1} + \eta_{i,w} \quad (5)$$

After obtaining $\hat{\beta}_{0,i,AR}$ and $\hat{\beta}_{1,i,AR}$, the out-sample forecast value for $\widehat{ER}_{i,w+1}$ based on the AR (1) model is calculated as follows:

$$\widehat{ER}_{i,w+1} = \hat{\beta}_{0,i,AR} + \hat{\beta}_{1,i,AR} ER_{i,w} + \eta_{i,w} \tag{6}$$

- US return as a predictor

The predictive regression model based on the lagged US returns is presented as follows:

$$ER_{i,w} = \beta_{0,i,US} + \beta_{1,i,US} ER_{US,w-1} + \eta_{i,w} \tag{7}$$

where $ER_{US,w-1}$ is the US index futures return. After estimating Eq. (7), $\hat{\beta}_{0,i,US}$ and $\hat{\beta}_{1,i,US}$ can be obtained and the out-of-sample forecast value for $\widehat{ER}_{i,w+1}$ based on the US returns is calculated as follows:

$$\widehat{ER}_{i,w+1} = \hat{\beta}_{0,i,US} + \hat{\beta}_{1,i,US} \times ER_{US,w} \tag{8}$$

- The average of cross-sectional weekly excess returns

The average of cross-sectional weekly excess returns of the index futures across nine countries is used as the proxy of the global latent factor and the predictive model is estimated as follows:

$$ER_{i,w} = \beta_{0,i,Avg} + \beta_{1,i,Avg} \widehat{GLF}_{Avg,w-1} + \eta_{i,w} \tag{9}$$

where $\widehat{GLF}_{Avg,w-1}$ is the average of cross-sectional weekly excess returns of the index futures across nine countries on week $w-1$. After obtaining $\hat{\beta}_{0,i,Avg}$ and $\hat{\beta}_{1,i,Avg}$, the out-of-sample forecast value for $\widehat{ER}_{i,w+1}$ is calculated as follows:

$$\widehat{ER}_{i,w+1} = \hat{\beta}_{0,i,Avg} + \hat{\beta}_{1,i,Avg} \times \widehat{GLF}_{Avg,w} \tag{10}$$

- The principal component approach (PCA)

The PCA is used to compute the factors common to the weekly excess returns of the index futures of the nine countries. This paper employs the first principal component as a proxy for the global latent factor. The predictive model based on principal components (PC) is present as follows.

Step 1: For the excess returns $ER_{1,w-1}, \dots, ER_{i,w-1}$ ($i = 1, \dots, 9$ and $w = 1, \dots, w-1$), the first principal component ($\widehat{GLF}_{PCA,w-1}$) and factor loading are estimated by the principal component analysis.

Step 2: Then, the predictive regression based on the first principal component from step 1 is estimated as follows:

$$ER_{i,w} = \beta_{0,i,PCA} + \beta_{1,i,PCA}\widehat{GLF}_{PCA,w-1} + \eta_{i,w} \tag{11}$$

where $\widehat{GLF}_{PCA,w-1}$ is the first principal component on week $w-1$, yielding $\hat{\beta}_{0,i,PCA}$ and $\hat{\beta}_{1,i,PCA}$.

Step 3: In order to estimate the out-of-sample forecast value for $\widehat{ER}_{i,w+1}$, the $ER_{1,w}, \dots, ER_{i,w}$ ($i = 1, \dots, 9$) on week w and factor loading estimated from Step 1 are used to calculate $\widehat{GLF}_{PCA,w}$, and then the out-of-sample forecast value for $\widehat{ER}_{i,w+1}$ is obtained as follows:

$$\widehat{ER}_{i,w+1} = \hat{\beta}_{0,i,PCA} + \hat{\beta}_{1,i,PCA} \times \widehat{GLF}_{PCA,w} \tag{12}$$

- **Out-of-Forecasting**

  - A 52-week initial window of the recursive estimation based on the approaches mentioned above is used to obtain the forecasts of the excess returns for the nine countries. Following Rapach et al. (2019), the long-short trading strategies based on out-of-sample forecasts are used in this research to evaluate the forecasting performance for the out-of-sample return predictability. Specifically, the paper sorts the index futures of the nine countries in ascending order according to the excess return forecasts and forms three equal-weighted portfolios. The out-of-sample excess return forecasts are based on *prevailing mean* (PM) *forecasts*, the $AR(1)$ model, predictive regression models with using lagged US returns ($USLag$), and the predictive regression models estimated *via* the global latent factor measured by the average of cross-sectional weekly excess returns (Avg), principal component approach (PCA), and three-pass regression filter ($TPRF$).

Then, for each long-short portfolio calculated by the PM, AR, $USLag$, Avg, PCA, or $TPRF$ forecasts, we create a zero-investment portfolio that goes long (short) in the three highest (lowest) excess returns of the index futures. The long-short portfolio returns calculated by the PM, AR, $USLag$, Avg, PCA, or $TPRF$ forecasts are denoted as $LSR_{PM,w}$, $LSR_{AR,w}$, $LSR_{USLag,w}$, $LSR_{Avg,w}$, $LSR_{PCA,w}$, and $LSR_{TPRF,w}$, respectively. The annualized mean, volatility, and Sharpe ratio, annualized downside risk and the Sortino ratio, maximum drawdown, and annualized manipulation-proof performance measures (MPPM, Goetzmann et al., 2007)[30] are used to evaluate the out-of-sample forecasting performance of the long-short portfolios.

- This paper expects that the $TPRF$ has a better performance in forecasting subsequent excess returns than other competing models. The reason is presented as follows. The PM and $AR(1)$ approaches only include the country's own lagged returns for predicting subsequent excess returns. However, the $TPRF$ captures the global latent factor and thus contains more information in predicting subsequent excess returns. Furthermore, while the US, Avg (average of cross-sectional), and PCA approaches obtain a (or same) global latent factor for nine index futures returns, the $TPRF$ approach extracts the specific global latent factor for each excess returns of the index futures i. As such, it is reasonable that the global latent factor extracted from $TPRF$ has a better performance in predicting subsequent excess returns of the index futures than that of US, Avg, and PCA approach.

[30] Goetzmann, W., Ingersoll, J., Spiegel, M., & Welch, I. (2007). Portfolio Performance Manipulation and Manipulation-Proof Performance Measures. Review of Financial Studies, 20, 1503–1546.

TABLE 5

Long-short portfolio returns.

	Ann. Mean	Ann. Vol	Ann. Sharpe ratio	Ann. Downside risk	Ann. Sortino ratio	Max. drawdown	Ann. MPPM
$LSR_{PM,w}$	0.007	0.103	0.063	0.073	0.089	0.514	-0.015
$LSR_{AR,w}$	0.090	0.109	0.827	0.068	1.331	0.194	0.066
$LSR_{US,w}$	0.066	0.100	0.659	0.062	1.058	0.190	0.046
$LSR_{Avg,w}$	0.073	0.113	0.645	0.069	1.063	0.265	0.048
$LSR_{PCA,w}$	0.069	0.115	0.604	0.071	0.977	0.239	0.043
$LSR_{TPRF,w}$	0.153	0.123	1.247	0.071	2.165	0.191	0.123

This table reports the performance measures for long-short index futures rotation portfolios across nine countries. This paper uses a 52-week initial window of the recursive estimation to obtain the forecasts of excess returns for nine countries. Then, this paper sorts the index futures of nine countries in ascending order according to the excess return forecasts and forms three equal-weighted portfolios. Specifically, the out-of-sample excess return forecasts are based on *prevailing mean* (PM) *forecasts*, the $AR(1)$ model, predictive regression models with using lagged US returns (US), and the predictive regression models estimated *via* the global latent factor measured by the average of cross-sectional weekly excess returns (Avg), principal component approach (PCA), and three-pass regression filter ($TPRF$). For each long-short portfolio based on the PM, AR, US, Avg, PCA, or $TPRF$ forecasts, we create a zero-investment portfolio that goes long (short) in the three highest (lowest) excess returns of the index futures. The long-short portfolio returns based on forecasts of the PM, AR, US, Avg, PCA, or $TPRF$ are $LSR_{PM,w}$, $LSR_{AR,w}$, $LSR_{US,w}$, $LSR_{Avg,w}$, $LSR_{PCA,w}$, and $LSR_{TPRF,w}$. The downside risk is measured as the semi-standard deviation. MPPM is the Goetzmann et al. (2007) manipulation-proof performance measure for a relative risk aversion coefficient of four. All futures prices of futures contracts are converted to U.S. dollars and thus all long-short returns are given in U.S. dollars. The sample period is from May 2001 to September 2020.

TABLE 6

Long-short portfolio returns under different market conditions.

Forecast	Ann. Mean	Ann. Vol	Ann. Sharpe ratio	Ann. Downside risk	Ann. Sortino ratio	Max. drawdown	Ann. MPPM
Panel A: Bull regimes							
$LSR_{PM,w}$	-0.012	0.088	-0.138	0.064	-0.189	0.521	-0.027
$LSR_{AR,w}$	0.022	0.083	0.265	0.058	0.379	0.321	0.008
$LSR_{US,w}$	0.023	0.083	0.275	0.058	0.393	0.236	0.009
$LSR_{Avg,w}$	0.008	0.089	0.093	0.063	0.130	0.360	-0.008
$LSR_{PCA,w}$	0.009	0.090	0.101	0.064	0.140	0.388	-0.007
$LSR_{TPRF,w}$	0.079	0.095	0.831	0.062	1.271	0.195	0.061
Panel B: Bear regimes							
$LSR_{PM,w}$	0.054	0.135	0.402	0.092	0.588	0.142	0.018
$LSR_{AR,w}$	0.265	0.155	1.713	0.088	3.013	0.184	0.218
$LSR_{US,w}$	0.177	0.133	1.330	0.072	2.462	0.190	0.143
$LSR_{Avg,w}$	0.239	0.157	1.521	0.080	2.989	0.154	0.191
$LSR_{PCA,w}$	0.224	0.160	1.400	0.085	2.628	0.155	0.174
$LSR_{TPRF,w}$	0.343	0.172	1.996	0.089	3.862	0.108	0.285

This table reports the performance measures for long-short index futures rotation portfolios across nine countries under different market conditions. Panels A and B report the long-short performance for the bull and bear regimes, respectively. The market conditions are determined by a simple Markov-switching model. Let $S_t = 0$ indicate the bull market and $S_t = 1$ represent the bear market. $p^{00} = \mathrm{P}(S_t = 0|S_{t-1} = 0)$ is the probability of the bull regime and $p^{11} = \mathrm{P}(S_t = 1|S_{t-1} = 1)$ is the probability of the bear regime. The regime is classified as the bull regime ($Bull$) when the smooth probability for $S_t = 0$ is higher than 0.5. The regime is classified as the bear regime ($Bear$) when the smooth probability for $S_t = 1$ is higher than 0.5. This paper uses a 52-week initial window of the recursive estimation to obtain the forecasts of excess returns for nine countries. Then, this paper sorts the index futures of nine countries in ascending order according to the excess return forecasts and forms three equal-weighted portfolios. Specifically, the out-of-sample excess return forecasts are based on *prevailing mean* (PM) *forecasts*, the $AR(1)$ model, predictive regression models with using lagged US returns (US), and the predictive regression models estimated *via* the global latent factor measured by the average of cross-sectional weekly excess returns (Avg), principal component approach (PCA), and three-pass regression filter

($TPRF$). For each long-short portfolio based on the PM, AR, US, Avg, PCA, or $TPRF$ forecasts, we create a zero-investment portfolio that goes long (short) in the three highest (lowest) excess returns of the index futures. The long-short portfolio returns based on forecasts of the PM, AR, US, Avg, PCA, or $TPRF$ are $LSR_{PM,w}$, $LSR_{AR,w}$, $LSR_{US,w}$, $LSR_{Avg,w}$, $LSR_{PCA,w}$, and $LSR_{TPRF,w}$. The downside risk is measured as the semi-standard deviation. MPPM is the Goetzmann et al. (2007) manipulation-proof performance measure for a relative risk aversion coefficient of four. All futures prices of futures contracts are converted to U.S. dollars and thus all long-short returns are given in U.S. dollars. The sample period is from May 2001 to September 2020.

TABLE 7

Long-short portfolio returns for the periods of non-extreme and extreme fear in the global stock markets.

Forecast	Ann. Mean	Ann. Vol	Ann. Sharpe ratio	Ann. Downside risk	Ann. Sortino ratio	Max. drawdown	Ann. MPPM
Panel A: Period of the non-extreme fear in the global stock markets							
$LSR_{PM,w}$	-0.007	0.091	-0.081	0.065	-0.113	0.519	-0.024
$LSR_{AR,w}$	0.053	0.091	0.587	0.059	0.897	0.200	0.037
$LSR_{US,w}$	0.040	0.087	0.458	0.059	0.680	0.151	0.025
$LSR_{Avg,w}$	0.027	0.092	0.290	0.063	0.425	0.255	0.010
$LSR_{PCA,w}$	0.026	0.093	0.278	0.064	0.405	0.269	0.009
$LSR_{TPRF,w}$	0.099	0.099	1.003	0.062	1.604	0.191	0.080
Panel B: Period of the extreme fear in the global stock markets							
$LSR_{PM,w}$	0.061	0.141	0.435	0.098	0.628	0.165	0.021
$LSR_{AR,w}$	0.236	0.161	1.469	0.093	2.530	0.188	0.185
$LSR_{US,w}$	0.170	0.140	1.216	0.075	2.258	0.152	0.132
$LSR_{Avg,w}$	0.257	0.172	1.495	0.089	2.906	0.154	0.200
$LSR_{PCA,w}$	0.242	0.174	1.391	0.093	2.603	0.165	0.183
$LSR_{TPRF,w}$	0.366	0.187	1.962	0.098	3.744	0.103	0.298

This table reports the performance measures for long-short index futures rotation portfolios across nine countries under the periods of the non-extreme and extreme fear in the global stock markets. Panels A and B report the long-short performance for the periods of the non-extreme and extreme fear in the global stock markets, respectively. The US VIX (the Chicago Board Option Exchange's S&P500 volatility index) is employed to identify the periods of the non-extreme and extreme fear in the global stock markets (see Da et al., 2015; Fatum and Yamamoto, 2016). The regime is classified as the extreme fear in the stock market when the value of the VIX index is higher than the 80% percentile of its corresponding index value. The other sample period is defined as the non-extreme fear period. This paper uses a 52-week initial window of the recursive estimation to obtain the forecasts of excess returns for nine countries. Then, this paper sorts the index futures of nine countries in ascending order according to the excess return forecasts and forms three equal-weighted portfolios. Specifically, the out-of-sample excess return forecasts are based on *prevailing mean* (PM) *forecasts*, the $AR(1)$ model, predictive regression models with using lagged US returns (US), and the predictive regression models estimated *via* the global latent factor measured by the average of cross-sectional weekly excess returns (Avg), principal component

approach (PCA), and three-pass regression filter ($TPRF$). For each long-short portfolio based on the PM, AR, US, Avg, PCA, or $TPRF$ forecasts, we create a zero-investment portfolio that goes long (short) in the three highest (lowest) excess returns of the index futures. The long-short portfolio returns based on forecasts of the PM, AR, US, Avg, PCA, or $TPRF$ are $LSR_{PM,w}$, $LSR_{AR,w}$, $LSR_{US,w}$, $LSR_{Avg,w}$, $LSR_{PCA,w}$, and $LSR_{TPRF,w}$. The downside risk is measured as the semi-standard deviation. MPPM is the Goetzmann et al. (2007) manipulation-proof performance measure for a relative risk aversion coefficient of four. All futures prices of futures contracts are converted to U.S. dollars and thus all long-short returns are given in U.S. dollars. The sample period is from May 2001 to September 2020.

Chapter 9: Machine Learning and Long-Short Portfolios

- A variety of recent studies provide a skeptical view on the predictability of stock returns. Empirical evidence shows that most prediction models suffer from a loss of information, model uncertainty, and structural instability by relying on lowdimensional information sets.
- On the other hand, a voluminous literature investigates aggregate stock market return predictability. In contrast, a relatively limited number of studies examine stock return predictability along industry lines, even though analyst reports and asset allocations are often industry based.
- Studies that analyze industry return predictability typically rely on popular predictor variables from the literature on aggregate market return predictability, such as the aggregate dividend yield, nominal yields, and yield spreads (e.g., Ferson and Harvey 1991, 1999; Ferson and Korajczyk 1995; Avramov 2004).[31]
- In this paper, we examine industry return predictability using a different information set - namely, lagged industry returns from across the entire economy.[32] This study is the first to directly analyze industry return predictability based on lagged industry returns from across the economy.
- The lack of previous studies investigating this topic is perhaps due to the statistical challenges inherent in estimating regression models with a large number of predictors.
- Our use of lagged industry returns to forecast individual industry returns is

[31] Ferson, W. E., Harvey, C. R., 1991. The Variation of economic risk premiums. Journal of Political Economy 99, 385-415. Ferson, W. E., Harvey. C. R., 1999. Conditioning variables and the cross section of stock returns. Journal of Finance 54, 1325-1360. Ferson, W. E., Korajczyk R. A., 1995. Do arbitrage pricing models explain the predictability of stock returns? Journal of Business 68, 309-349. Avramov, D., 2004. Stock return predictability and asset pricing models. Review of Financial Studies 17, 699-738.

[32] The data and code are used from the paper by Rapach, D.E., Strauss, J.K., Tu, J., Zhou,G., 2019. Industry return predictability: A machine learning approach. Journal of Financial Data Science 1(3), 9-28.

motivated by the theoretical model in Hong et al. (2007),[33] who introduce information frictions into an economy with multiple linked industries. Due to the industry links, cash flow shocks originating in one industry can affect expected cash flows in related industries. In a frictionless rational-expectations equilibrium, investors readily recognize all of the inter-industry implications of a cash ow shock in a particular industry. Consequently, equity prices across all relevant industries immediately adjust to fully impound the interindustry effects of the cash flow shock, and lagged industry returns do not have predictive power.

- However, incorporating insights from Merton (1987) and Hong and Stein (1999),[34] Hong et al. (2007) posit that investors with limited information-processing capabilities specialize in specific market segments. In this environment, when a cash flow shock arises in a particular industry, information-processing limitations prevent investors specializing in related industries from quickly working out the full implications of the shock. Information thus diffuses gradually across industries, and the resulting delayed adjustment in equity prices gives rise to industry return predictability on the basis of lagged industry returns.
- We analyze the predictive ability of lagged industry returns in a general predictive regression framework that allows each industry's return to respond to the lagged returns for all industries, thereby accommodating a wide array of industry links, both direct and indirect.
- Given the plethora of potential predictors in the predictive regression models, conventional ordinary least squares (OLS) estimation has potential drawbacks.

    1. If all lagged returns are used, the abundance of predictors makes OLS

[33] Hong, H., Torous W., Valkanov. R., 2007. Do industries lead stock markets? Journal of Financial Economics 83, 367-396.

[34] Merton, R. C. 1987. A simple model of capital market equilibrium with incomplete information. Journal of Finance 42, 483-510. Hong, H., Stein. J. C., 1999. A unified theory of underreaction, momentum trading, and overreaction in asset markets." Journal of Finance 54, 2143-2184.

estimation to overfitting.

2. If only a few lagged returns are chosen, it is difficult to know in advance which are the most important.

- The paper employs the least absolute shrinkage and selection operator (LASSO, Tibshirani 1996), a powerful and popular technique in machine learning, to overcome the disadvantages from the OLS.[35]
- For the in-sample analysis, we estimate predictive regression models via OLS post-LASSO using monthly return data spanning 1960 to 2016 for 30 industry portfolios from Kenneth French's Data Library. The LASSO selects at least one lagged industry return as a predictor for 29 of the individual industries, while multiple lagged industry returns are selected for 22 of the individual industries. Furthermore, the OLS post-LASSO estimation results indicate that the lagged industry returns selected by the LASSO are often statistically significant predictors of industry returns.
- The out-of-sample analysis simulates the situation of an investor in real time and measures the economic value of industry return predictability. We first compute out-of-sample forecasts of monthly industry returns based on predictive regressions estimated via OLS post-LASSO. We then sort the 30 industries according to their forecasted returns over the next month and construct a zero-investment portfolio that goes long (short) the top (bottom) quintile of sorted industries. For the 1970:01 to 2016:12 out-of-sample period, the industry-rotation portfolio generates an average return of 7.33% per annum and annualized Sharpe (Sortino) ratio of 0.65 (1.16).

[35] Tibshirani, R., 1996. Regression shrinkage and selection via the LASSO." Journal of the Royal Statistical Society. Series B (Methodological) 58, 267-288.

EXHIBIT 1
Summary Statistics, Industry Portfolio Excess Returns, 1959:12–2016:12

(1) Industry portfolio	(2) Ann. mean (%)	(3) Ann. vol. (%)	(4) Min. (%)	(5) Max. (%)	(6) Ann. Sharpe ratio	(7) Industry portfolio	(8) Ann. mean (%)	(9) Ann. vol. (%)	(10) Min. (%)	(11) Max. (%)	(12) Ann. Sharpe ratio
Food	8.29	15.03	−18.15	19.89	0.55	Carry	8.62	21.80	−31.10	23.39	0.40
Beer	8.53	17.63	−20.19	25.51	0.48	Mines	6.58	25.82	−34.55	35.15	0.26
Smoke	11.79	21.00	−25.32	32.38	0.56	Coal	8.83	35.31	−38.09	45.55	0.25
Games	8.42	24.88	−33.40	34.52	0.34	Oil	7.81	18.53	−18.96	23.70	0.42
Books	6.34	20.12	−26.56	33.13	0.32	Util	5.80	13.83	−12.94	18.26	0.42
Hshld	6.65	16.49	−22.24	18.22	0.40	Telcm	6.26	16.04	−16.44	21.22	0.39
Clths	8.03	22.12	−31.50	31.79	0.36	Servs	8.20	22.61	−28.67	23.38	0.36
Hlth	7.81	17.07	−21.06	29.01	0.46	BusEq	6.97	23.34	−32.07	24.66	0.30
Chems	6.24	19.12	−28.60	21.68	0.33	Paper	6.12	17.51	−27.74	21.00	0.35
Txtls	8.01	24.33	−33.11	59.03	0.33	Trans	7.02	19.90	−28.50	18.50	0.35
Cnstr	6.28	20.75	−28.74	25.02	0.30	Whlsl	7.46	19.42	−29.25	17.53	0.38
Steel	3.49	25.26	−32.99	30.30	0.14	Rtail	7.97	18.53	−29.74	26.49	0.43
FabPr	6.75	21.18	−31.63	22.91	0.32	Meals	8.47	21.15	−31.89	27.38	0.40
ElcEq	8.45	21.52	−32.80	23.21	0.39	Fin	7.31	18.75	−22.53	20.59	0.39
Autos	5.45	23.17	−36.49	49.56	0.24	Other	4.64	20.09	−28.11	19.96	0.23

Notes: The exhibit reports summary statistics for excess returns for 30 value-weighted industry portfolios from Kenneth French's Data Library. Excess returns are computed relative to the one-month Treasury bill return. The industry abbreviations are as follows: Food *= Food Products;* Beer *= Beer and Liquor;* Smoke *= Tobacco Products;* Games *= Recreation;* Books *= Printing and Publishing;* Hshld *= Consumer Goods;* Clths *= Apparel;* Hlth *= Healthcare, Medical Equipment, and Pharmaceutical Products;* Chems *= Chemicals;* Txtls *= Textiles;* Cnstr *= Construction and Construction Materials;* Steel *= Steel Works, Etc.;* FabPr *= Fabricated Products and Machinery;* ElcEq *= Electrical Equipment;* Autos *= Automobiles and Trucks;* Carry *= Aircraft, Ships, and Railroad Equipment;* Mines *= Precious Metals, Non-Metallic, and Industrial Metal Mining;* Coal *= Coal;* Oil *= Petroleum and Natural Gas;* Util *= Utilities;* Telcm *= Communication;* Servs *= Personal and Business Services;* BusEq *= Business Equipment;* Paper *= Business Supplies and Shipping Containers;* Trans *= Transportation;* Whlsl *= Wholesale;* Rtail *= Retail;* Meals *= Restaurants, Hotels, and Motels;* Fin *= Banking, Insurance, Real Estate, and Trading;* Other *= Everything Else.*

EXHIBIT 4
Industry-Rotation Portfolio Performance, 1970:01–2016:12

(1) Forecasts	(2) Ann. mean (%)	(3) Ann. vol. (%)	(4) Ann. Sharpe ratio	(5) Ann. downside risk (%)	(6) Ann. Sortino ratio	(7) Max. drawdown (%)	(8) Ann. MPPM (%)
Prevailing mean	−2.22	10.98	−0.20	8.34	−0.27	73.97	−4.73
OLS	5.52	11.64	0.47	7.17	0.77	29.39	2.84
OLS post-LASSO	7.33	11.29	0.65	6.31	1.16	25.65	4.84

Notes: The exhibit reports performance measures for long-short industry-rotation portfolios. At the end of each month, we sort 30 industry portfolios according to out-of-sample forecasts of their excess returns for the subsequent month. The out-of-sample industry excess return forecasts are based on predictive regression models estimated via OLS or OLS post-LASSO, as well as prevailing mean forecasts, as indicated in the first column. We then form equal-weighted quintile portfolios based on the sorts, and each long-short industry-rotation portfolio is a zero-investment portfolio that goes long (short) the top (bottom) quintile portfolio. Downside risk is measured as the semi-standard deviation. MPPM is the Goetzmann et al. (2007) manipulation-proof performance measure for a relative risk aversion coefficient of four.

EXHIBIT 6
Annualized Average Returns During Good and Bad States, 1970:01–2016:12

(1)	(2)	(3)	(4)	(5)	(6)	(7)
	NBER business-cycle phases			CFNAI observations		
Forecasts	Expansion	Recession	Difference	Upper 80%	Lower 20%	Difference
Prevailing mean	−2.54%	−0.57%	1.97%	−3.49%	2.72%	6.22%
	[−1.45]	[−0.14]	[0.45]	[−1.95]*	[0.77]	[1.57]
OLS	4.20%	12.54%	8.34%	5.56%	5.34%	−0.22%
	[2.27]**	[2.94]***	[1.79]*	[2.92]***	[1.42]	[−0.05]
OLS post-LASSO	4.62%	21.75%	17.13%	5.07%	16.14%	11.08%
	[2.61]***	[5.31]***	[3.84]***	[2.76]***	[4.45]***	[2.72]***

*Notes: The exhibit reports annualized average returns for long-short industry-rotation portfolios during good and bad macroeconomic states. At the end of each month, we sort 30 industry portfolios according to out-of-sample forecasts of their excess returns for the subsequent month. The out-of-sample industry excess return forecasts are based on predictive regression models estimated via OLS or OLS post-LASSO, as well as prevailing mean forecasts, as indicated in the first column. We then form equal-weighted quintile portfolios based on the sorts, and each long-short industry-rotation portfolio is a zero-investment portfolio that goes long (short) the top (bottom) quintile portfolio. Good (bad) states are months when the economy is in an expansion (recession) according to NBER-dated cyclical turning points or months in the upper 80% (lower 20%) of Chicago Fed national activity index (CFNAI) observations. Difference is the difference in average returns during bad versus good states. Brackets report t-statistics; *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.*

EXHIBIT 7
Multifactor Model Estimation Results, 1970:01–2016:12

(1)	(2)	(3)	(4)	(5)	(6)	(7)
	PM forecasts		OLS forecasts		OLS post-LASSO forecasts	
Factor	Carhart four-factor model	HXZ q-factor model	Carhart four-factor model	HXZ q-factor model	Carhart four-factor model	HXZ q-factor model
Ann. α	−1.84%	−2.91%	6.64%	5.57%	8.78%	8.04%
	[−1.29]	[−1.84]*	[3.78]***	[3.04]***	[5.19]***	[4.55]***
Market	−0.03	−0.02	−0.05	−0.02	−0.13	−0.10
	[−1.17]	[−0.63]	[−1.62]	[−0.56]	[−3.87]***	[−3.09]***
Size	−0.12	−0.04	−0.12	−0.13	−0.07	−0.07
	[−3.19]***	[−0.85]	[−2.60]***	[−2.56]**	[−1.61]	[−1.54]
Value	−0.38		−0.10		−0.12	
	[−9.05]***		[−1.87]*		[−2.35]**	
Momentum	0.24		−0.01		0.004	
	[8.98]***		[−0.37]		[0.12]	
Investment		−0.37		0.13		0.002
		[−5.37]***		[1.64]		[0.02]
Profitability		0.45		−0.04		0.02
		[8.98]***		[−0.61]		[0.38]
R^2	27.80%	18.19%	2.22%	2.30%	4.06%	3.28%

*Notes: The exhibit reports Carhart (1997) four-factor and Hou et al. (2015, HXZ) q-factor model estimation results for long-short industry-rotation portfolios. At the end of each month, we sort 30 industry portfolios according to out-of-sample forecasts of their excess returns for the subsequent month. The out-of-sample industry excess return forecasts are based on predictive regression models estimated via OLS or OLS post-LASSO, as well as prevailing mean (PM) forecasts, as indicated at the top of the exhibit. We then form equal-weighted quintile portfolios based on the sorts, and each long-short industry-rotation portfolio is a zero-investment portfolio that goes long (short) the top (bottom) quintile portfolio. The regressand for the multiple regression model is the return for the portfolio indicated at the top of the exhibit. Brackets report t-statistics; *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.*

Chapter 10: Index Tracking

- Fund managers follow two basic investment strategies: active and passive.
- In **active management strategies**, the fund managers assume that the markets are not perfectly efficient and through their expertise and superior prediction methods they hope to add value by choosing high performing assets.
- On the contrary, **the passive management** strategies are based on the assumption that the market cannot be beaten in the long run. The passive managers have less flexibility and their role is to conform to a closely defined set of criteria.
- Index tracking, also known as index replication, is one of the most popular passive portfolio management strategies. It refers to the problem of reproducing the performance of a market index. A straightforward way to do that is to purchase all the assets that compose an index in appropriate quantities.
- However, to simplify the execution, avoid small and illiquid positions, and large transaction costs, **it is desired that the tracking portfolio consists of a small number of assets, i.e., we wish to create a sparse portfolio**.
- In this chapter, we present the code which is used to replicate stock index.
- More details in math, please see:

  https://palomar.home.ece.ust.hk/papers/2018/BenidisFengPalomar-FnT2018.pdf